# Syllabus 9758 Pure Math

Example Questions

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# 1 Systems of Linear Equations

**1.1** Three different blends of coffee beans – Blend A, Blend B and Blend C – are sold at both M Café and I Café. The usual selling prices of the blends are the same at both cafés. The total cost of buying one package each of Blend A, Blend B and Blend C is \$176. During a holiday sale the two cafés offered the following discounts:

| **Café** | **Discounts given for each package** | | | **Total price after the discount** |
|---|---|---|---|---|
| | **Blend A** | **Blend B** | **Blend C** | |
| **M Café** | 15% | 10% | 5% | \$161.80 |
| **I Café** | 7% | 5% | 2% | \$169.56 |

**(i)** Find the usual selling price of each package of Blend A, Blend B and Blend C coffee beans respectively. [3]

**[MI 2025 Prelim P1 Q4]**

| **Solution** | | **Comments** |
|---|---|---|
| **(i)**<br>**[3]** | Let the usual prices be \$$a$, \$$b$, \$$c$.<br>$a+b+c=176 \quad ---(1)$<br>$0.85a+0.90b+0.95c=161.80 \quad ---(2)$<br>$0.93a+0.95b+0.98c=169.56 \quad ---(3)$<br>Using GC, $a=32,\ b=44,\ c=100$<br>$\therefore$ Blend A: \$32, Blend B: \$44, Blend C: \$100 | |

**1.2** A function f is defined by $\mathrm{f}(x)=ax^3+bx^2+cx+d$ where $a$, $b$, $c$ and $d$ are constants. The graph of $y=\mathrm{f}(x)$ has a turning point at $x=-1$ and passes through the points $(2, 10)$ and $(-2, -2)$. Given also that $\int_{-2}^{2} \mathrm{f}(x)\,\mathrm{d}x=16$, find the values of $a$, $b$, $c$ and $d$. [4]

**[SAJC 2025 Prelim P1 Q1]**

| **Solution** | | **Comments** |
|---|---|---|
| **[4]** | $\mathrm{f}'(x)=3ax^2+2bx+c$<br>$\mathrm{f}'(-1)=0$<br>$3a-2b+c=0 \quad ---(1)$<br>$\mathrm{f}(2)=10$<br>$8a+4b+2c+d=10 \quad ---(2)$<br>$\mathrm{f}(-2)=-2$<br>$-8a+4b-2c+d=-2 \quad ---(3)$<br>$\int_{-2}^{2} (ax^3+bx^2+cx+d)\,\mathrm{d}x=16$<br>$\left[\frac{a}{4}x^4+\frac{b}{3}x^3+\frac{c}{2}x^2+dx\right]_{-2}^{2}=16$<br>$(4a+\frac{8}{3}b+2c+2d)-(4a-\frac{8}{3}b+2c-2d)=16$<br>$\frac{16}{3}b+4d=16 \quad ---(4)$<br>Using GC, $a=3,\ b=0,\ c=-9,\ d=4$ | |

Two linear equations are independent if and only if neither equation is a linear combination of the other.

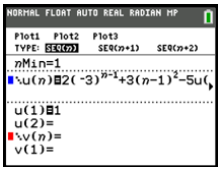
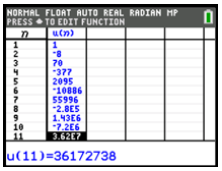
**1.3** A sequence of real numbers, $x_n$, satisfies the recurrence relation

$$x_{n+1} = a(-3)^n + bn^2 + cx_n, \text{ for } n, a, b, c \in \mathbb{Z} \text{ and } n \geqslant 1.$$

Given that $x_1 = 1$, $x_2 = -8$, $x_3 = 70$, and $x_4 = -377$, find the values of $a$, $b$ and $c$. [3]

Hence, find the values of $x_{10}$ and $x_{11}$. [2]

**[HCI 2025 Prelim P1 Q2]**

| **Solution** | | **Comments** |
|---|---|---|
| **[2]** | $n = 1$<br>$x_2 = a(-3)^1 + b(1)^2 + cx_1$<br>$-8 = -3a + b + c \quad ---(1)$<br>$n = 2$<br>$x_3 = a(-3)^2 + b(2)^2 + cx_2$<br>$70 = 9a + 4b - 8c \quad ---(2)$<br>$n = 3$<br>$x_4 = a(-3)^3 + b(3)^2 + cx_3$<br>$-377 = -27a + 9b + 70c \quad ---(3)$<br>Using GC, $a = 2,\ b = 3,\ c = -5$<br>$x_{n+1} = 2(-3)^n + 3n^2 - 5x_n$<br>Using GC,<br>$x_{10} = -7\,210\,868$<br>$x_{11} = 36\,172\,738$<br>  <br><br>  <br> | |

**1.4** The lines $l_1$ and $l_2$ have equations

$$l_1: \mathbf{r} = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix},$$

$$l_2: \mathbf{r} = \begin{pmatrix} 0 \\ 5 \\ -8 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ -2 \\ 3 \end{pmatrix},$$

where $\lambda$ and $\mu$ are parameters.

**(i)** Show that $l_1$ and $l_2$ are skew lines. [2]

**[SAJC 2025 Prelim P1 Q3]**

| **Solution** | | **Comments** |
|---|---|---|
| **(i)** **[2]** | $1-2\lambda = 2\mu$ $---(1)$<br>$2+\lambda = 5-2\mu$ $---(2)$<br>$2\lambda = -8+3\mu$ $---(3)$<br>Solve (1) and (2): $\lambda = -2,\ \mu = \frac{5}{2}$<br>Sub into (3):<br>$\text{LHS} = 2(-2) = -4$<br>$\text{RHS} = -8+3\left(\frac{5}{2}\right) = -\frac{1}{2}$<br>$\text{LHS} \neq \text{RHS},$ no solution<br>$\therefore l_1$ and $l_2$ do not intersect<br>$\because \begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix} \neq k \begin{pmatrix} 2 \\ -2 \\ 3 \end{pmatrix}$ for any $k \in \mathbb{R}$<br>$\therefore l_1$ and $l_2$ are not parallel<br>$\therefore l_1$ and $l_2$ are skew (shown) | |

**1.5** A movie theatre supports a charity event by providing discounted movie tickets at the following prices based on age group.

| Age group | Discounted ticket price |
|---|---|
| $\leqslant 12$ years | \$6 |
| 13 to 49 years | \$9 |
| $\geqslant 50$ years (senior citizen) | \$4 |

The event organiser plans to spend \$700 to sponsor 100 participants of which at least 35 of them are senior citizens. Find the possible number of participants for each age group. [4]

**[TJC 2022 Prelim P2 Q1]**

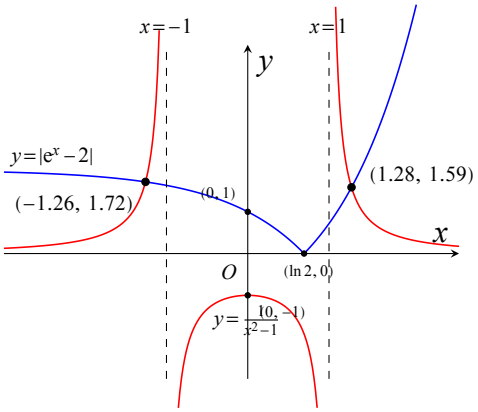
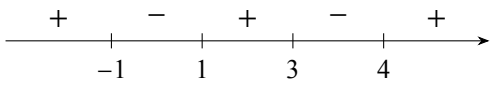
| **Solution** | | **Comments** |
|---|---|---|
| **[4]** | Let $x$, $y$, $z$ be the number of participants aged $\leqslant 12$, 13 to 49 and $\geqslant 50$ years respectively.<br>$\left.\begin{aligned} x+y+z&=100 \quad ---(1)\\ 6x+9y+4z&=700 \quad ---(2)\end{aligned}\right\} \Rightarrow$ Using GC, $x=\frac{200}{3}-\frac{5}{3}z,\ y=\frac{100}{3}+\frac{2}{3}z$<br>$z \geqslant 35$<br>$\because x \geqslant 0$<br>$\frac{200}{3}-\frac{5}{3}z \geqslant 0$<br>$z \leqslant 40$<br>$\therefore 35 \leqslant z \leqslant 40$<br>$x$, $y$, $z$ are non-negative integers. Using GC table:<br>$z=37 \Rightarrow x=5,\ y=58$<br>$z=40 \Rightarrow x=0,\ y=60$<br>$\therefore (x,y,z)=(5,58,37)$ or $(0,60,40)$ | |

# 2 Inequalities

**2.1** **(a)** Solve the inequality $|e^x - 2| > \dfrac{1}{x^2 - 1}$ using a graphical approach. [3]

**(b)** Without the use of a graphing calculator, solve the inequality $\dfrac{x+2}{x-1} \geqslant \dfrac{2}{x-3}$. [3]

**[RVHS 2026 Prelim P1 Q2]**

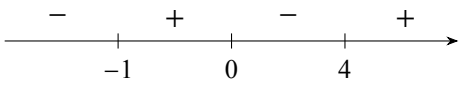
| Solution | Comments |
|---|---|
| **(a)** **[3]** |  <br>Using GC, intersections at $x=-1.2581$, $x=1.2770$<br>$\therefore x<-1.26$ or $-1<x<1$ or $x>1.28$ | |
| **(b)** **[3]** | $\dfrac{x+2}{x-1}-\dfrac{2}{x-3}\geqslant 0$<br>$\dfrac{(x+2)(x-3)-2(x-1)}{(x-1)(x-3)}\geqslant 0$<br>$\dfrac{x^2-3x-4}{(x-1)(x-3)}\geqslant 0$<br>$\dfrac{(x-4)(x+1)}{(x-1)(x-3)}\geqslant 0$<br>  <br>$\therefore x\leqslant -1$ or $1<x<3$ or $x\geqslant 4$ | |

**2.2** Use an algebraic method to solve the inequalities

$$\frac{x(5x+9)}{x^2+2x+5} > x \geqslant 3x-6.$$

[4]

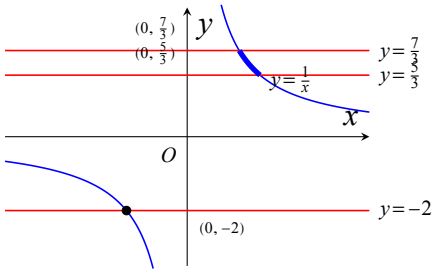
**[ACJC 2026 Prelim P1 Q1]**

| **Solution** | | **Comments** |
|---|---|---|
| **[4]** | $x \geqslant 3x-6$<br>$x \leqslant 3 \quad ---(1)$<br>$x^2+2x+5=(x+1)^2+4>0$ for all real $x$<br>$\frac{x(5x+9)}{x^2+2x+5} > x$<br>$5x^2+9x > x(x^2+2x+5) \quad (\because x^2+2x+5>0)$<br>$x^3-3x^2-4x<0$<br>$x(x-4)(x+1)<0$<br>  <br><br>$x<-1$ or $0<x<4 \quad ---(2)$<br>$(1)\cap(2)$ :<br>$\therefore x<-1$ or $0<x\leqslant 3$ | |

**2.3** **(i)** Solve the inequality $3|x^2-4| \leqslant |x+2|$. [4]

**(ii)** Hence, solve $\frac{3}{x^2}|1-4x^2| \leqslant \left|\frac{1+2x}{x}\right|$. [2]

**[SAJC 2025 Prelim P2 Q1]**

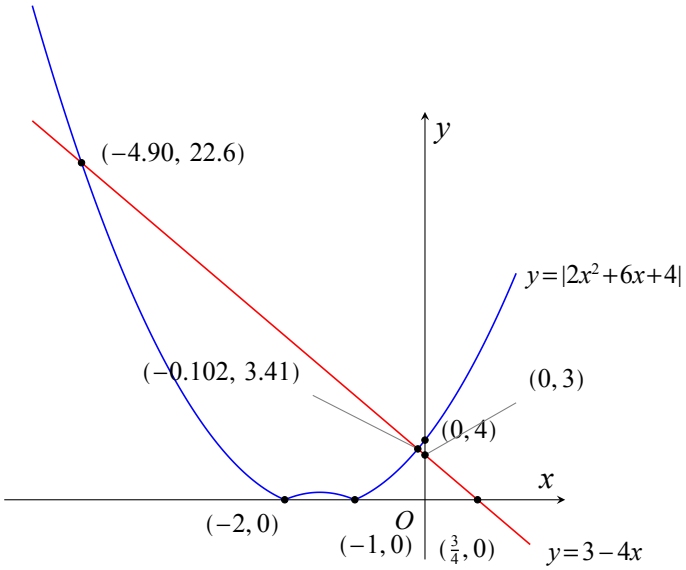
| Solution | | Comments |
|---|---|---|
| **(i)** **[4]** | $3\|x-2\|\|x+2\| \leqslant \|x+2\|$<br>$x=-2:\quad 0 \leqslant 0$ (true)<br>$x \neq -2:\ \|x+2\| > 0$<br>$3\|x-2\| \leqslant 1$<br>$-\frac{1}{3} \leqslant x-2 \leqslant \frac{1}{3}$<br>$\frac{5}{3} \leqslant x \leqslant \frac{7}{3}$<br>$\therefore x=-2$ or $\frac{5}{3} \leqslant x \leqslant \frac{7}{3}$ | |
| **(ii)** **[2]** | $\frac{3}{x^2}\|1-4x^2\| = 3\left\|\frac{1}{x^2}-4\right\| \quad (\because x^2>0)$<br>$\left\|\frac{1+2x}{x}\right\| = \left\|\frac{1}{x}+2\right\|$<br>$3\left\|\left(\frac{1}{x}\right)^2-4\right\| \leqslant \left\|\frac{1}{x}+2\right\| \quad (x \to \frac{1}{x})$<br>$\frac{1}{x}=-2$ or $\frac{5}{3} \leqslant \frac{1}{x} \leqslant \frac{7}{3}$<br>  <br><br>$\therefore x=-\frac{1}{2}$ or $\frac{3}{7} \leqslant x \leqslant \frac{3}{5}$ | |

**2.4** **(i)** On the same axes, sketch the curves with equations $y=|2x^2+6x+4|$ and $y=3-4x$, indicating any intercepts with the axes and points of intersection. Hence solve the inequality $3-4x<|2x^2+6x+4|$. [4]

**(ii)** Find the exact area bounded by the graphs of $y=3-4x$, $y=|2x^2+6x+4|$, $x=-3$ and $x=-1$. [3]

**[SAJC 2019 Prelim P1 Q2]**

| Solution | Comments |
|---|---|
| **(i)** **[4]** | |




$$3-4x=2x^2+6x+4$$

$$2x^2+10x+1=0$$

$$x=\frac{-5\pm\sqrt{23}}{2}$$

$$\therefore x<\frac{-5-\sqrt{23}}{2} \text{ or } x>\frac{-5+\sqrt{23}}{2}$$

**(ii)** **[3]**

$$\text{Area}=\int_{-3}^{-1}(3-4x-|2x^2+6x+4|)\,\mathrm{d}x$$

$$=\int_{-3}^{-2}(3-4x-2x^2-6x-4)\,\mathrm{d}x+\int_{-2}^{-1}(3-4x+2x^2+6x+4)\,\mathrm{d}x$$

$$=\int_{-3}^{-2}(-2x^2-10x-1)\,\mathrm{d}x+\int_{-2}^{-1}(2x^2+2x+7)\,\mathrm{d}x$$

$$=\left[-\tfrac{2}{3}x^3-5x^2-x\right]_{-3}^{-2}+\left[\tfrac{2}{3}x^3+x^2+7x\right]_{-2}^{-1}$$

$$=\left(\tfrac{16}{3}-20+2\right)-(18-45+3)+\left(-\tfrac{2}{3}+1-7\right)-\left(-\tfrac{16}{3}+4-14\right)$$

$$=20$$

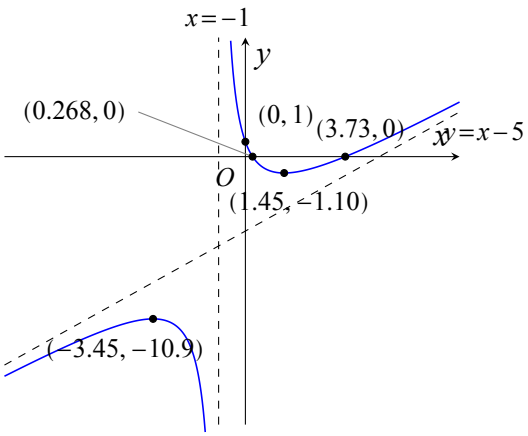
# 3 Graphing Techniques

**3.1** A curve $C$ has equation $y=\dfrac{x^2-4x+1}{x+1}$, $x\neq -1$.

**(a)** Prove, using an algebraic method, that $y=\dfrac{x^2-4x+1}{x+1}$ cannot lie between two certain values, to be determined in exact form. [3]

**(b)** Sketch $C$, stating the equations of any asymptotes, and the coordinates of any turning points and any axial intercepts. [4]

**[VJC 2025 Promo Q9(a), (b)]**

| Solution | | Comments |
|---|---|---|
| **(a)**<br>**[3]** | $y=\dfrac{x^2-4x+1}{x+1}$<br>$yx+y=x^2-4x+1$<br>$x^2+(-y-4)x+(1-y)=0$<br>No real solution for $x$: $(-y-4)^2-4(1-y)<0$<br>$y^2+8y+16-4+4y<0$<br>$y^2+12y+12<0$<br>$(y+6)^2-24<0$<br>$\left(y+6-\sqrt{24}\right)\left(y+6+\sqrt{24}\right)<0$<br>$-6-2\sqrt{6}<y<-6+2\sqrt{6}$<br>$\therefore y$ cannot lie between $-6-2\sqrt{6}$ and $-6+2\sqrt{6}$ | |
| **(b)**<br>**[4]** | $y=x-5+\dfrac{6}{x+1}$<br>Asymptotes: $x=-1,\ y=x-5$<br>$x=0:\ y=1$<br>$y=0:\ x^2-4x+1=0 \Rightarrow x=0.268,\ 3.73$<br>Turning points: $(1.45,-1.10),\ (-3.45,-10.9)$<br>  <br> | |

**3.2** A curve $C$ has equation $y=ax+b+\dfrac{2b-a}{x+3}$, where $a$ and $b$ are constants such that $a>0$, $b\neq\dfrac{1}{2}a$ and $x\neq -3$.

**(a)** If $C$ has no stationary points, use differentiation to find the relationship between $a$ and $b$. [3]

**(b)** It is now given that $b=a$.

**(i)** Find the $x$-coordinates of the turning points. [2]

**(ii)** Hence sketch $C$, stating the equations of any asymptotes and the coordinates of the axial intercepts and turning points. [3]

**(iii)** Verify that the point of intersection of the two asymptotes of $C$ lies on the line $y=kx+3k-2a$.

Hence, using the graph in part (b)(ii), find the range of values of $k$ in terms of $a$, such that the equation $x(k-a)+3k-2a-b-\dfrac{2b-a}{x+3}=0$ has real roots. [2]

**[SAJC 2025 Promo Q7]**

| Solution | | Comments |
|---|---|---|
| **(a)** **[3]** | $\dfrac{\mathrm{d}y}{\mathrm{d}x}=a-\dfrac{2b-a}{(x+3)^2}$<br>$\dfrac{\mathrm{d}y}{\mathrm{d}x}=0:\quad a-\dfrac{2b-a}{(x+3)^2}=0$<br>$(x+3)^2=\dfrac{2b-a}{a}\qquad ---(1)$<br>No stationary points $\Rightarrow\ \dfrac{2b-a}{a}<0$<br>Since $a>0,\quad 2b-a<0$<br>$\therefore\ a>2b$ | |
| **(b)(i)** **[2]** | $b=a:\quad (x+3)^2=\dfrac{2a-a}{a}=1$<br>$x+3=1$ or $-1$<br>$x=-2$ or $-4$ | |

**(b)(ii)**
**[3]**

$$y = ax + a + \frac{a}{x+3}$$

Asymptotes: $x = -3,\ y = ax + a$

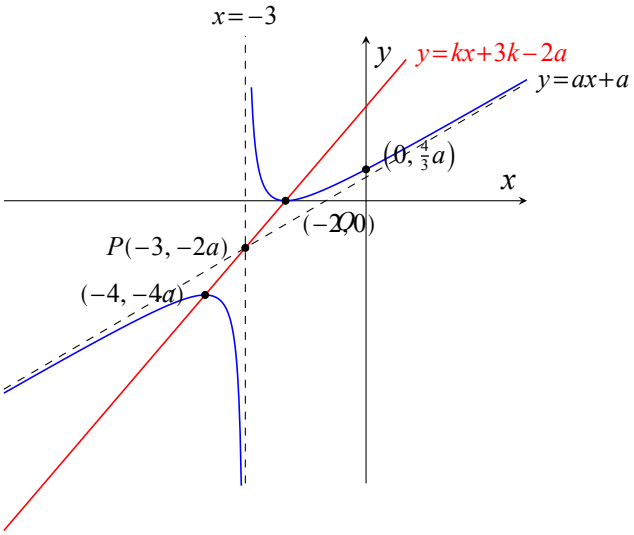
$x = -2 : y = 0; \quad x = -4 : y = -4a$

Turning points: $(-2, 0),\ (-4, -4a)$

$$x = 0 : y = \frac{4}{3}a$$

$$y = 0 : ax + a + \frac{a}{x+3} = 0 \Rightarrow (x+1)(x+3) + 1 = 0$$

$$x^2 + 4x + 4 = 0 \Rightarrow (x+2)^2 = 0 \Rightarrow x = -2$$




**(b)(iii)**
**[2]**

$x = -3$ and $y = ax + a$: $\quad y = -3a + a = -2a$

Intersection of asymptotes: $P(-3, -2a)$

$x = -3$: $\quad \text{RHS} = k(-3) + 3k - 2a = -2a = \text{LHS}$

$\therefore P$ lies on the line $y = kx + 3k - 2a$

$$x(k-a) + 3k - 2a - b - \frac{2b-a}{x+3} = 0$$

$$kx + 3k - 2a = ax + b + \frac{2b-a}{x+3}$$

Line through $P$ meets $C \Rightarrow k > a$

**3.3** A curve $C$ has equation $\mathrm{f}(x)=\dfrac{3x^2+6x+k}{x+2}$, where $x>-2$ and $k$ is a non-zero real constant.
It is given that the function f is strictly increasing.

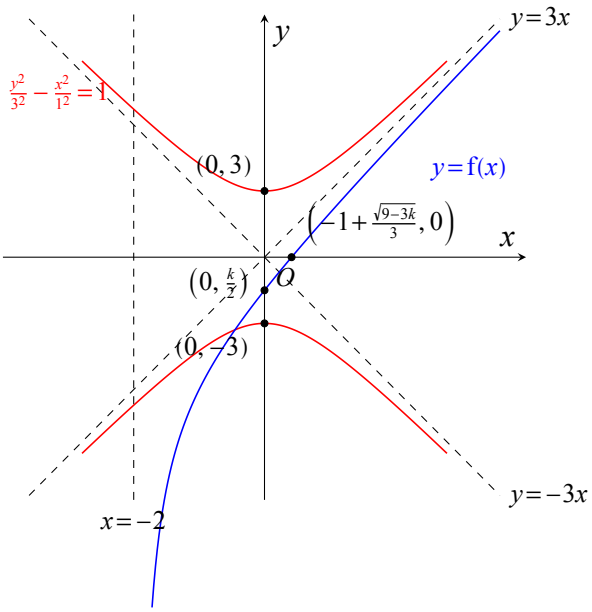
**(a)** Find the range of values of $k$. [3]

**(b)** Sketch the curve $C$, giving the coordinates of any points where $C$ crosses the $x$- and $y$-axes and the equations of any asymptotes. [3]

**(c)** By adding a suitable curve to the graph of the curve $C$ in (b), deduce the number of distinct real root(s) of the equation $(3x^2+6x+k)^2=(x^2+1)\,(9x^2+36x+36)$. [3]

**[DHS 2025 Promo Q8]**

| Solution | | Comments |
|---|---|---|
| **(a)** **[3]** | $\mathrm{f}(x)=\dfrac{3x^2+6x+k}{x+2}=3x+\dfrac{k}{x+2}$ | |
| | $\mathrm{f}'(x)=3-\dfrac{k}{(x+2)^2}=\dfrac{3x^2+12x+12-k}{(x+2)^2}$ | |
| | f strictly increasing $\Rightarrow$ $\mathrm{f}'(x)>0$ | |
| | $3x^2+12x+12-k>0$ $\quad\because (x+2)^2>0$ for $x>-2$ | |
| | $3(x+2)^2-k>0$ | |
| | Since $3(x+2)^2-k>-k$ for all $x>-2$, $\quad -k\geqslant 0$ | |
| | $k\neq 0 \Rightarrow k<0$ | |
| **(b)** **[3]** | $y=3x+\dfrac{k}{x+2}$ | |
| | Asymptotes: $y=3x,\ x=-2$ | |
| | $x=0:\ y=\dfrac{k}{2}$ | |
| | $y=0:\ 3x^2+6x+k=0 \Rightarrow x=\dfrac{-6\pm\sqrt{36-12k}}{6}=-1\pm\dfrac{\sqrt{9-3k}}{3}$ | |
| | $x>-2 \Rightarrow x=-1+\dfrac{\sqrt{9-3k}}{3}$ | |
| **(c)** **[3]** | $(3x^2+6x+k)^2=(x^2+1)\,(9x^2+36x+36)\quad ---(*)$ | |
| | $(3x^2+6x+k)^2=9\,(x^2+1)\,(x+2)^2$ | |
| | $\left(\dfrac{3x^2+6x+k}{x+2}\right)^2-9x^2=9$ | |
| | Draw the hyperbola $\dfrac{y^2}{3^2}-\dfrac{x^2}{1^2}=1$ | |
| | Asymptotes $y=\pm 3x$, vertices $(0,\pm 3)$ | |




$y=\mathrm{f}(x)$ and $\dfrac{y^2}{3^2}-\dfrac{x^2}{1^2}=1$ have one intersection point

$\therefore$ $(*)$ has one real root

**3.4** The curve $C$ is defined by the parametric equations $x=6t-5$ and $y=2t^2+1$, where $t\geqslant\dfrac{5}{6}$.

**(a)** Sketch the curve $C$, labelling the exact coordinates of any axial intercept(s). [1]

**(b)** Find the equation of the normal to $C$ at the point $P$ where $t=1$. [2]

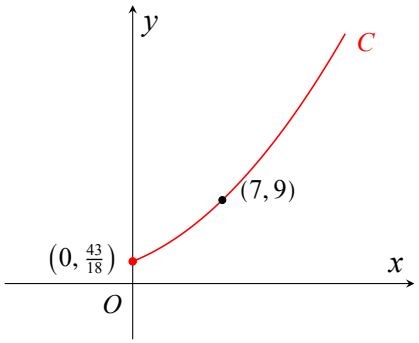
**(c)** Find the acute angle between the normal to $C$ at the point $P$ where $t=1$ and the tangent to $C$ at the point $M$ where $t=3$. [2]

**(d)** The curve $C$ is translated 5 units in the positive $x$-direction and translated 3 units in the negative $y$-direction, to form the curve $D$. Find the equation of $D$ in parametric form. [2]

The curve $E$ is defined by the parametric equations $x=7u$ and $y=\dfrac{9}{u}$, where $u\neq 0$.

**(e)** Show that at the point of intersection of the curves $C$ and $E$, $6t^3-5t^2+3t-34=0$. Deduce that there is only one point of intersection and find the coordinates of this point. [4]

**[RVHS 2025 Prelim P1 Q12]**

| Solution | | Comments |
|---|---|---|
| **(a)**<br>**[1]** | $t=\frac{5}{6}$: $\quad x=6\left(\frac{5}{6}\right)-5=0, \quad y=2\left(\frac{5}{6}\right)^2+1=\frac{43}{18}$<br>Endpoint ($y$-intercept): $\left(0, \frac{43}{18}\right)$<br>  <br> | |

**(b)** **[2]**

$$\frac{dx}{dt}=6, \quad \frac{dy}{dt}=4t$$

$$\frac{dy}{dx}=\frac{4t}{6}=\frac{2}{3}t$$

$t=1$: $\quad \frac{dy}{dx}=\frac{2}{3} \Rightarrow$ gradient of normal $=-\frac{3}{2}$

$t=1$: $\quad x=1,\ y=3$

$$y-3=-\frac{3}{2}(x-1)$$

$$y=-\frac{3}{2}x+\frac{9}{2}$$

**(c)** **[2]**

$t=3$: $\quad$ gradient of tangent $=\frac{2}{3}(3)=2$

$$\theta_1+\theta_2=\tan^{-1}(2)-\tan^{-1}\left(-\frac{3}{2}\right)=2.089942 \text{ or } 119.7^\circ$$

$$\text{Acute angle}=\pi-2.089942 \text{ or } 180^\circ-119.7^\circ$$

$$=1.05 \text{ or } 60.3^\circ$$

**(d)** **[2]**

$x-5=6t-5 \Rightarrow x=6t$

$y+3=2t^2+1 \Rightarrow y=2t^2-2$

$D$: $\quad x=6t,\ y=2t^2-2$

**(e)** **[4]**

$$6t-5=7u \Rightarrow u=\frac{6t-5}{7} \quad ---(1)$$

$$2t^2+1=\frac{9}{u} \quad ---(2)$$

Sub (1) into (2): $2t^2+1=\frac{63}{6t-5}$

$$(2t^2+1)(6t-5)=63$$

$$12t^3-10t^2+6t-68=0$$

$$6t^3-5t^2+3t-34=0 \quad \text{(shown)}$$

$$(t-2)(6t^2+7t+17)=0$$

$6t^2+7t+17$: $b^2-4ac=7^2-4(6)(17)=-359<0 \Rightarrow$ no real roots

$\therefore t=2$ only $\Rightarrow$ only one point of intersection

$t=2$: $\quad x=6(2)-5=7,\ y=2(2)^2+1=9$

Point of intersection: $(7, 9)$

**3.5** The parametric equations of the curve $C$ are

$$x=\tan\theta-1 \quad \text{and} \quad y=2\sec\theta, \quad \text{where } -\frac{\pi}{2}<\theta<\frac{\pi}{2}.$$

**(a)** Find $\frac{\mathrm{d}y}{\mathrm{d}x}$ in terms of $\theta$, leaving your answer in a single trigonometric function. [2]

**(b)** Find the equation of the tangent to $C$ at the point where it cuts the $y$-axis, leaving your answer in exact form. [3]

**(c)** Find the Cartesian equation of $C$. [1]

**(d)** Find the equations of the asymptotes and state the coordinates of any turning point(s). [3]

**(e)** The region bounded by the tangent found in part (b), the curve $C$ and the line $x=-1$ is rotated completely about the $x$-axis. Calculate the volume obtained. [3]

**[HCI 2024 Promo Q12]**

| **Solution** | | **Comments** |
|---|---|---|
| **(a)** **[2]** | $\frac{\mathrm{d}y}{\mathrm{d}x}=\frac{\mathrm{d}y}{\mathrm{d}\theta}\div\frac{\mathrm{d}x}{\mathrm{d}\theta}=\frac{2\sec\theta\tan\theta}{\sec^2\theta}=2\sin\theta$ | |
| **(b)** **[3]** | $x=0$: $\tan\theta-1=0 \Rightarrow \tan\theta=1$<br>$\theta=\frac{\pi}{4}$ since $-\frac{\pi}{2}<\theta<\frac{\pi}{2}$<br>$\theta=\frac{\pi}{4}$: $y=\frac{2}{\cos\frac{\pi}{4}}=2\sqrt{2}$, $\frac{\mathrm{d}y}{\mathrm{d}x}=2\sin\frac{\pi}{4}=\sqrt{2}$<br>Tangent: $y=\sqrt{2}x+2\sqrt{2}$ | |
| **(c)** **[1]** | $\tan^2\theta+1=\sec^2\theta$<br>$(x+1)^2+1=\left(\frac{y}{2}\right)^2$<br>$-\frac{\pi}{2}<\theta<\frac{\pi}{2} \Rightarrow 0<\cos\theta\leqslant 1 \Rightarrow \sec\theta\geqslant 1$<br>$\therefore (x+1)^2+1=\left(\frac{y}{2}\right)^2, \quad y\geqslant 2$ | |

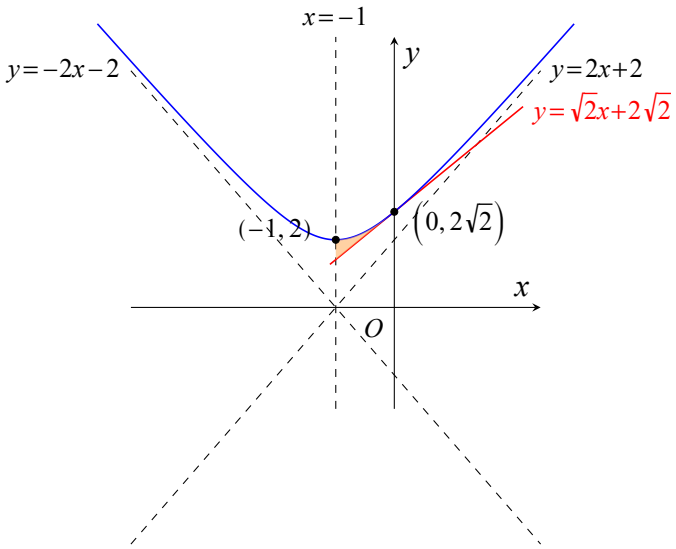
**(d)**
**[3]**

Asymptotes: $\left(\frac{y}{2}\right)^2 = (x+1)^2$

$$\frac{y}{2} = \pm(x+1)$$

$y = 2x+2$ or $y = -2x-2$

Centre $(-1, 0)$, $y \geqslant 2 \Rightarrow$ turning point $(-1, 2)$




**(e)**
**[3]**

$$\frac{y^2}{4} = (x+1)^2 + 1$$

$$y^2 = 4(x+1)^2 + 4$$

$$\text{Volume} = \pi \int_{-1}^{0} \left[ 4\left((x+1)^2+1\right) - \left(\sqrt{2}x + 2\sqrt{2}\right)^2 \right] dx$$

$$= 2.09 \text{ units}^3$$

# 4 Graphing Transformations

**4.1** A curve with equation $y=\mathrm{h}(x)$ undergoes the following sequence of transformations.

**A**: A translation of 2 units in the positive direction of the $x$-axis.

**B**: A scaling parallel to the $x$-axis by factor of 3.

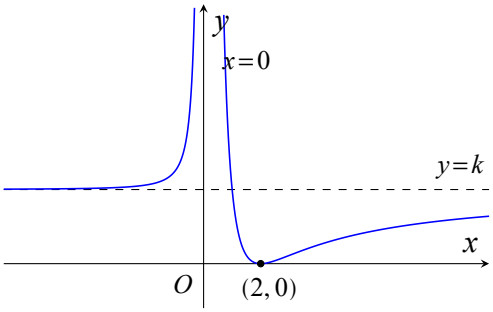
**C**: A reflection in the $y$-axis.

The resultant curve has equation $y=-\dfrac{x^2}{6x+45}$. Find $\mathrm{h}(x)$. [3]

**[NJC 2024 Promo Q4(b)]**

| Solution | | Comments |
|---|---|---|
| **[3]**<br><br>**(b)** | $y=-\dfrac{x^2}{6x+45}$<br>$\mathrm{C}^{-1}$: reflection in the $y$-axis $(x\to -x)$<br>$y=-\dfrac{(-x)^2}{6(-x)+45}=-\dfrac{x^2}{-6x+45}$<br>$\mathrm{B}^{-1}$: scaling parallel to the $x$-axis by factor $\frac{1}{3}$ $(x\to 3x)$<br>$y=-\dfrac{(3x)^2}{-6(3x)+45}=-\dfrac{9x^2}{-18x+45}=-\dfrac{x^2}{-2x+5}$<br>$\mathrm{A}^{-1}$: translation of 2 units in the negative $x$-direction $(x\to x+2)$<br>$y=-\dfrac{(x+2)^2}{-2(x+2)+5}=-\dfrac{(x+2)^2}{-2x+1}$<br>$\mathrm{h}(x)=\dfrac{(x+2)^2}{2x-1}$ | |

**4.2** The diagram below shows the graph of $y=\mathrm{f}(2x)$. The graph has a turning point at $(2,0)$, and asymptotes with equations $x=0$ and $y=k$.



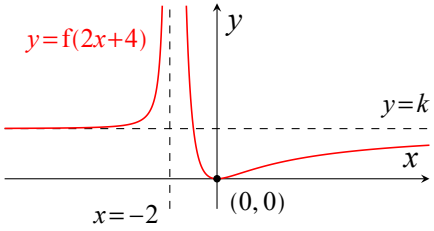
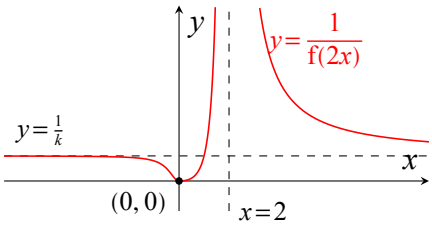
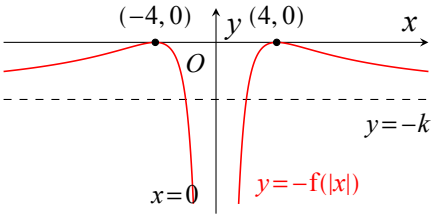

**(a)** State a single transformation that will transform the graph of $y=\mathrm{f}(2x)$ onto the graph of $y=\mathrm{f}(2x+4)$. Hence sketch the graph of $y=\mathrm{f}(2x+4)$. [3]

On separate clearly labelled diagrams, sketch the graphs of

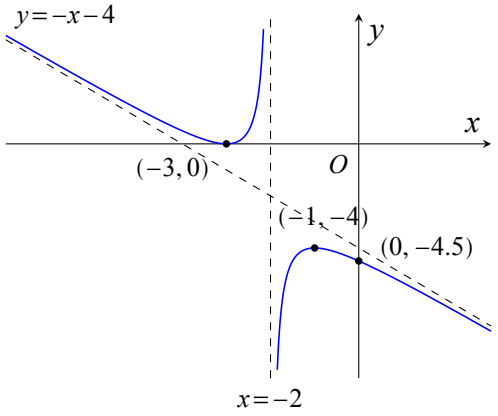
**(b)** $y=\dfrac{1}{\mathrm{f}(2x)}$, [3]

**(c)** $y=-\mathrm{f}(|x|)$. [3]

**[EJC 2025 Prelim P1 Q4]**

| Solution | | Comments |
|---|---|---|
| **(a)**<br>**[3]** |  | |
| **(b)**<br>**[3]** |  | |
| **(c)**<br>**[3]** |  | |

**4.3**




The diagram shows the curve $y = \mathrm{f}(x)$. The curve crosses the $x$-axis at its minimum point $(-3, 0)$. It has a maximum turning point $(-1, -4)$ and crosses the $y$-axis at the point $(0, -4.5)$. The asymptotes of the curve are $x = -2$ and $y = -x - 4$.

On separate diagrams, sketch the graphs of

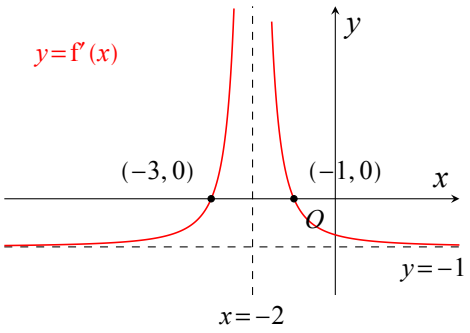
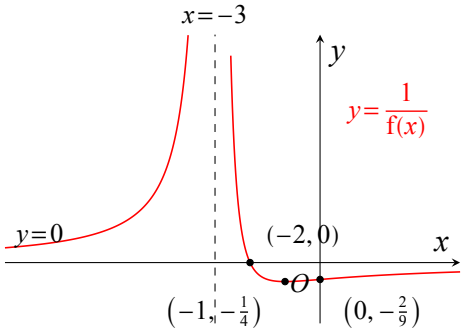
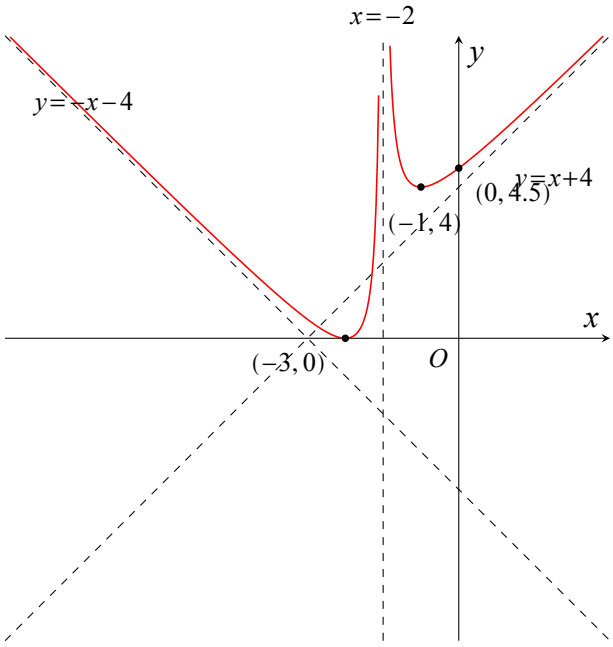
**(a)** $y = \mathrm{f}'(x)$, [3]

**(b)** $y = \dfrac{1}{\mathrm{f}(x)}$, [3]

**(c)** $y = |\mathrm{f}(x)|$, [3]

indicating the equations of any asymptotes, turning points and intersections with the axes.

**[CJC 2025 Promo Q8]**

| Solution | Comments |
|---|---|
| **(a)** **[3]**   | |
| **(b)** **[3]**   | |
| **(c)** **[3]**   | |

**4.4** Express $\frac{9x+5}{3x+2}$ in the form $a-\frac{b}{3x+2}$ where $a$ and $b$ are positive constants.

Hence, describe a sequence of transformations which transform the graph of $y=\frac{9x+5}{3x+2}$ to the graph of $y=\frac{1}{x}$. [5]

**[RVHS 2026 Prelim P1 Q10(b)]**

| Solution | | Comments |
|---|---|---|
| **[5]**<br><br>**(b)** | $\frac{9x+5}{3x+2}=3-\frac{1}{3x+2} \quad (a=3,\ b=1)$<br>$y=3-\frac{1}{3x+2} \xrightarrow{y\to y+3} y=-\frac{1}{3x+2}$<br>$\xrightarrow{y\to -y} y=\frac{1}{3x+2}$<br>$\xrightarrow{x\to \frac{x}{3}} y=\frac{1}{x+2}$<br>$\xrightarrow{x\to x-2} y=\frac{1}{x}$<br><br>1. Translate 3 units in the negative $y$-direction<br>2. Reflect about the $x$-axis<br>3. Scale by factor 3 parallel to the $x$-axis<br>4. Translate 2 units in the positive $x$-direction | |

# 5 Functions

**5.1** The function f is defined by

$$\mathrm{f}: x \to \left|\frac{2x+4}{3-x}\right|, \qquad x \in \mathbb{R}, \qquad x \neq 3.$$

**(a)** Sketch the graph of $y = \mathrm{f}(x)$, giving the equations of any asymptotes and the coordinates of the points where the curve meets the axes. [3]

**(b)** Hence state the range of f. [1]

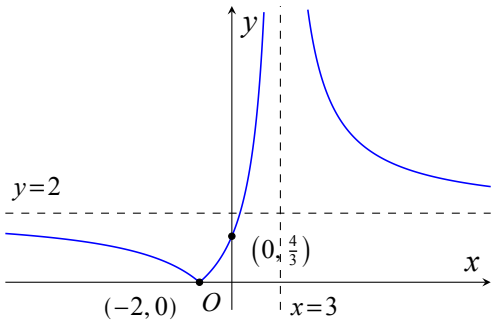
**(c)** Explain why the function $\mathrm{f}^2$ does not exist. [1]

The domain of f is further restricted to $x \leqslant a$, where $a$ is a constant.

**(d)** State the greatest value of $a$ such that the function $\mathrm{f}^{-1}$ exists. [1]

**(e)** Hence find $\mathrm{f}^{-1}(x)$ and state its domain. [4]

**[2023 A-Level P1 Q7]**

| Solution | | Comments |
|---|---|---|
| **(a)** **[3]** | $y = \left\|\frac{2x+4}{3-x}\right\| = \left\|-2+\frac{10}{3-x}\right\|$<br>Asymptotes: $y = \|-2\| = 2, \quad x = 3$<br>$x = 0: \quad y = \left\|\frac{4}{3}\right\| = \frac{4}{3}$<br>$y = 0: \quad x = -2$<br>  <br> | |
| **(b)** **[1]** | $\mathrm{R_f} = [0, \infty)$ | |

**(c)** **[1]**

$R_f = [0, \infty), \quad D_f = (-\infty, 3) \cup (3, \infty)$

$3 \in R_f,\ 3 \notin D_f \Rightarrow R_f \not\subseteq D_f$

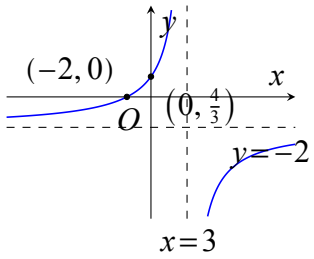
$\therefore$ $f^2$ does not exist

**(d)** **[1]**

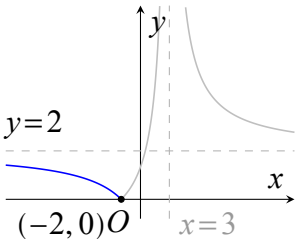
$a = -2$

**(e)** **[4]**

$x \leqslant -2: \quad f(x) = \left|\dfrac{2x+4}{3-x}\right| = -\dfrac{2x+4}{3-x}$




$\Rightarrow$




$y = f(x)$

$-3y + xy = 2x + 4$

$x(y-2) = 4 + 3y$

$x = \dfrac{4+3y}{y-2}$

$f^{-1}(x) = \dfrac{4+3x}{x-2}, \qquad D_{f^{-1}} = [0, 2)$

**5.2** The function f is defined by

$$\mathrm{f}: x \mapsto \frac{ax-3}{5x-4}, \text{ for } x \in \mathbb{R}, \ x \neq \frac{4}{5}, \text{ and } a \in \mathbb{R}.$$

**(a)** Find the range of f in terms of $a$. [1]

The function f is such that $\mathrm{f}(x) = \mathrm{f}^{-1}(x)$ for all $x$ in the domain of f.

**(b)** Find the value of $a$. [2]

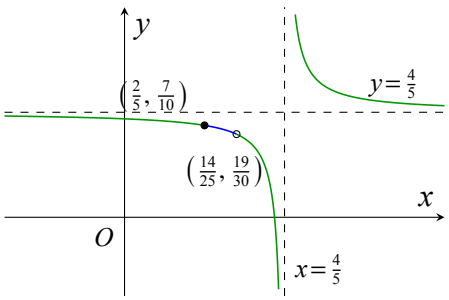
**(c)** Hence find $\mathrm{f}^{2025}(2)$. [2]

Another function g is defined by

$$\mathrm{g}: x \mapsto \left(x - \frac{1}{5}\right)^2 + \frac{2}{5}, \text{ for } x \in \mathbb{R}, \ 0 < x < \frac{3}{5}.$$

**(d)** Find the range of fg. [2]

**[HCI 2025 Prelim P1 Q8]**

| Solution | | Comments |
|---|---|---|
| **(a)** **[1]** | Horizontal asymptote: $y=\frac{a}{5}$<br>$\mathrm{R_f}=\mathbb{R}\setminus\left\{\frac{a}{5}\right\}$ | |
| **(b)** **[2]** | $y=\frac{ax-3}{5x-4}$<br>$5xy-4y=ax-3$<br>$5xy-ax=4y-3$<br>$x=\frac{4y-3}{5y-a}$<br>$\mathrm{f}^{-1}(x)=\frac{4x-3}{5x-a},\quad x\in\mathbb{R},\ x\neq\frac{a}{5}$<br>$\mathrm{f}(x)=\mathrm{f}^{-1}(x)\ \Rightarrow\ a=4$ | |
| **(c)** **[2]** | $\mathrm{f}^{-1}(x)=\mathrm{f}(x)\ \Rightarrow\ \mathrm{f}^2(x)=\mathrm{ff}^{-1}(x)=x$<br>$\mathrm{f}^{2025}(2)=\mathrm{f}^2\left(\mathrm{f}^{2023}(2)\right)$<br>$=\mathrm{f}^2\left(\mathrm{f}^{2021}(2)\right)$<br>$=\cdots$<br>$=\mathrm{f}(2)$<br>$=\frac{4(2)-3}{5(2)-4}=\frac{5}{6}$ | |
| **(d)** **[2]** | $\mathrm{R_g}=\left[\frac{2}{5},\frac{14}{25}\right)$<br>$\mathrm{f}(x)=\frac{4x-3}{5x-4}$<br>$\left(0,\frac{3}{5}\right)\xrightarrow{\mathrm{g}}\left[\frac{2}{5},\frac{14}{25}\right)\xrightarrow{\mathrm{f}}\left(\frac{19}{30},\frac{7}{10}\right]$<br>$\mathrm{R_{fg}}=\left(\frac{19}{30},\frac{7}{10}\right]$<br>  <br> | |

**5.3** The function f is given by

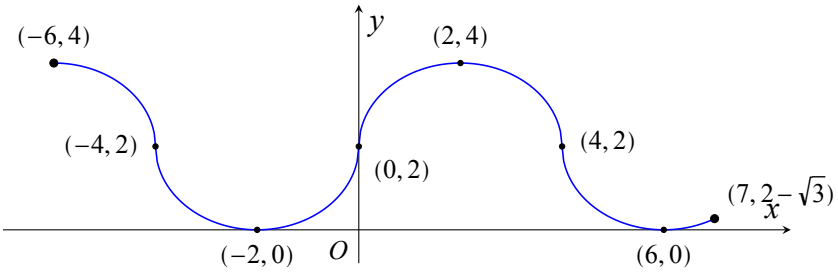
$$\mathrm{f}(x)=\begin{cases}2+\sqrt{2^2-(x-2)^2}, & 0<x\leqslant 4,\\ 2-\sqrt{2^2-(x-6)^2}, & 4<x\leqslant 8.\end{cases}$$

It is given that $\mathrm{f}(x)=\mathrm{f}(x+8)$ for all real values of $x$.

**(a)** On the diagram in the Printed Answer Book, sketch the graph of $y=\mathrm{f}(x)$ for $-6\leqslant x\leqslant 7$, indicating clearly the coordinates of the end points and the points where the graph cuts the axes. [3]

**(b)** Without integrating, write down the exact area of the region bounded by $y=\mathrm{f}(x)$, the line $x=4$, the $x$-axis and the $y$-axis. [1]

**[VJC 2025 Prelim P1 Q5(a), (b)]**

| Solution | | Comments |
|---|---|---|
| **(a)** **[3]** | $y=2+\sqrt{2^2-(x-2)^2}$<br>$y-2=\sqrt{2^2-(x-2)^2}$<br>$(x-2)^2+(y-2)^2=2^2$<br>Centre $(2,2)$, radius 2.<br>Since it is a positive square root, it is the top half of this circle.<br><br>$y=2-\sqrt{2^2-(x-6)^2}$<br>$y-2=-\sqrt{2^2-(x-6)^2}$<br>$(x-6)^2+(y-2)^2=2^2$<br>Centre $(6,2)$, radius 2.<br>Since it is a negative square root, it is the bottom half of this circle.<br><br>  <br> | |
| **(b)** **[1]** | $\text{Area}=2\times 4+\frac{1}{2}\pi(2^2)=8+2\pi$ | |

**5.4** The function f is defined by

$$\mathrm{f}(x) = 1 - ax,\ x \in \mathbb{R},\ \text{where } a \text{ is a real constant.}$$

The function g is defined by

$$\mathrm{g}(x) = \begin{cases} x^2 + 3 & \text{for} \quad 0 \leqslant x \leqslant 2, \\ 7 - x & \text{for} \quad 2 < x \leqslant 7. \end{cases}$$

**(i)** Find the set of possible values of $a$ such that $\mathrm{f}^{-1}$ exist. [1]

**(ii)** If $a = 2$, describe a sequence of transformations that transform the graph of $y = \mathrm{g}(x)$ onto the graph of $y = \mathrm{fg}(x)$. [3]

**(iii)** The function gg is defined by

$$\mathrm{gg}(x) = \begin{cases} \mathrm{h}_1(x) & \text{for} \quad 0 \leqslant x \leqslant 2, \\ \mathrm{h}_2(x) & \text{for} \quad 2 < x < 5, \\ \mathrm{h}_3(x) & \text{for} \quad 5 \leqslant x \leqslant 7. \end{cases}$$

Find $\mathrm{h}_1(x)$, $\mathrm{h}_2(x)$ and $\mathrm{h}_3(x)$. [3]

**[ACJC 2022 Prelim P2 Q1]**

| Solution | | Comments |
|---|---|---|
| **(i)** **[1]** | $a \in \mathbb{R} \setminus \{0\}$ | |
| **(ii)** **[3]** | $y = \text{fg}(x) = 1 - 2\text{g}(x)$<br>Reflection of the graph of $y = \text{g}(x)$ in the $x$-axis, followed by scaling by scale factor 2 parallel to the $y$-axis, followed by translation 1 unit in the positive $y$-direction.<br>**OR**<br>Translation of the graph of $y = \text{g}(x)$ by $\frac{1}{2}$ unit in the negative $y$-direction followed by reflection in the $x$-axis followed by scaling by scale factor 2 parallel to the $y$-axis.<br>**OR**<br>Translation of the graph of $y = \text{g}(x)$ by $\frac{1}{2}$ unit in the negative $y$-direction followed by scaling by scale factor 2 parallel to the $y$-axis followed by reflection in the $x$-axis. | |
| **(iii)** **[3]** | $\text{gg}(x) = \begin{cases} 7-(x^2+3) & \text{for} \quad 0 \leqslant x \leqslant 2 \\ 7-(7-x) & \text{for} \quad 2 < x < 5 \\ (7-x)^2+3 & \text{for} \quad 5 \leqslant x \leqslant 7 \end{cases}$<br>$\therefore \text{gg}(x) = \begin{cases} 4-x^2 & \text{for} \quad 0 \leqslant x \leqslant 2 \\ x & \text{for} \quad 2 < x < 5 \\ x^2-14x+52 & \text{for} \quad 5 \leqslant x \leqslant 7 \end{cases}$ | |

**5.5** Functions f and g are defined by

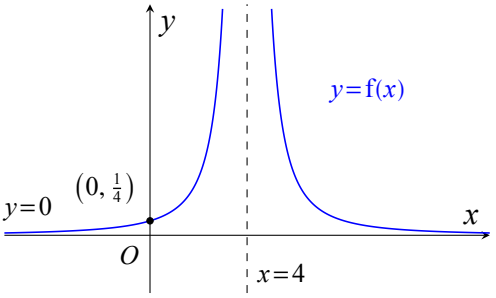
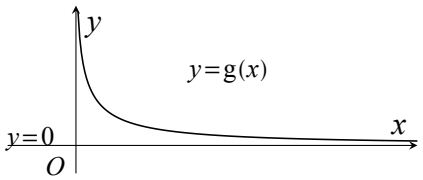
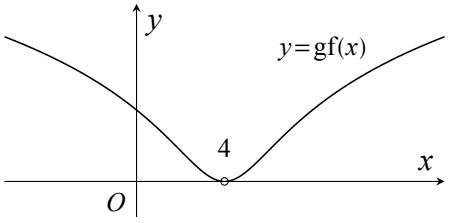
$$\mathrm{f}: x \to \frac{4}{(x-4)^2}, \quad x \in \mathbb{R},\ x \neq 4,$$

$$\mathrm{g}: x \to \ln\left(1+\frac{1}{x}\right), \quad x \in \mathbb{R},\ x > 0.$$

**(a)** Sketch the graph of $y = \mathrm{f}(x)$, stating the equations of any asymptotes, the coordinates of the points where it crosses the axes and the coordinates of the turning points, if any. [2]

**(b)** Show that gf exists. Hence find the rule, domain and range of gf. [4]

**[CJC 2025 Prelim P2 Q2(a), (b)]**

| Solution | | Comments |
|---|---|---|
| **(a)**<br>**[2]** |  | |
| **(b)**<br>**[4]** | For gf to exist, $R_f \subseteq D_g$,<br>$R_f = (0, \infty)$<br>$D_g = (0, \infty)$<br>Since $R_f \subseteq D_g$, gf exists (shown)<br>$\text{gf}(x) = \text{g}\left[\frac{4}{(x-4)^2}\right]$<br>$= \ln\left[1 + \left(\frac{1}{\frac{4}{(x-4)^2}}\right)\right]$<br>$= \ln\left[1 + \frac{(x-4)^2}{4}\right]$<br>$D_{gf} = D_f = (-\infty, 4) \cup (4, \infty)$<br>OR<br>$D_{gf} = D_f = \mathbb{R} \setminus \{4\}$<br>**Method 1: Mapping**<br>$D_f = (-\infty, 4) \cup (4, \infty) \xrightarrow{f} R_f = (0, \infty) \xrightarrow{g} R_{gf} = (0, \infty)$<br>  <br>**Method 2: Sketch $y = \text{gf}(x)$ for given domain**<br>  <br>$R_{gf} = (0, \infty)$ | |

**5.6** The function g, with domain the set of non-negative integers, is given by

$$\mathrm{g}(n)=\begin{cases}1 & \text{for } n=0,\\ 2+\mathrm{g}\left(\frac{1}{2}n\right) & \text{for } n \text{ even},\\ 1+\mathrm{g}(n-1) & \text{for } n \text{ odd}.\end{cases}$$

**(i)** Find g(4), g(7) and g(12). [3]

**(ii)** Does g have an inverse? Justify your answer. [2]

**[2016 A-Level P1 Q10(b)]**

| Solution | Comments |
|---|---|
| **(i)** **[3]** | $\mathrm{g}(4)=2+\mathrm{g}\left(\frac{1}{2}\times 4\right)=2+\mathrm{g}(2)=2+\left[2+\mathrm{g}\left(\frac{1}{2}\times 2\right)\right]=4+\mathrm{g}(1)$<br>$=4+[1+\mathrm{g}(1-1)]=5+\mathrm{g}(0)=5+1=6$<br>$\mathrm{g}(7)=1+\mathrm{g}(7-1)=1+\mathrm{g}(6)=1+\left[2+\mathrm{g}\left(\frac{1}{2}\times 6\right)\right]=3+\mathrm{g}(3)$<br>$=3+[1+\mathrm{g}(3-1)]=4+\mathrm{g}(2)=4+\left[2+\mathrm{g}\left(\frac{1}{2}\times 2\right)\right]=6+\mathrm{g}(1)$<br>$=6+[1+\mathrm{g}(1-1)]=7+\mathrm{g}(0)=7+1=8$<br>$\mathrm{g}(12)=2+\mathrm{g}\left(\frac{1}{2}\times 12\right)=2+\mathrm{g}(6)=2+\left[2+\mathrm{g}\left(\frac{1}{2}\times 6\right)\right]=4+\mathrm{g}(3)$<br>$=4+[1+\mathrm{g}(3-1)]=5+\mathrm{g}(2)=5+\left[2+\mathrm{g}\left(\frac{1}{2}\times 2\right)\right]=7+\mathrm{g}(1)$<br>$=7+[1+\mathrm{g}(1-1)]=8+\mathrm{g}(0)=8+1=9$<br>Solution source: Achevas (Jack Ng), "A-Level H2 Maths 2016 – Paper 1", www.achevas.com. | |
| **(ii)** **[2]** | $\mathrm{g}(5)=1+\mathrm{g}(4)=1+6=7$<br>$\mathrm{g}(6)=2+\mathrm{g}(3)=2+[1+\mathrm{g}(2)]=3+\mathrm{g}(2)$<br>$=3+[2+\mathrm{g}(1)]=5+\mathrm{g}(1)=5+[1+\mathrm{g}(0)]=6+1=7$<br>Since $\mathrm{g}(5)=\mathrm{g}(6)$, g does not have an inverse. | |

# 6 Sequences and Series

**6.1** It is given $\sum_{r=1}^{n} r^3 = \frac{n^2(n+1)^2}{4}$ and $u_r = 2r^3 + 5$.

**(i)** Find $\sum_{r=1}^{n} u_r$. [2]

**(ii)** Hence, or otherwise, find $\sum_{r=2}^{n} (2(r+2)^3 + 5)$. [3]

**(iii)** Explain why the series $\sum_{r=1}^{\infty} u_r$ does not converge. [1]

**[JPJC 2025 Prelim P1 Q8(b)]**

| Solution | | Comments |
|---|---|---|
| **(i)** **[2]** | $\sum_{r=1}^{n} (2r^3+5) = 2\sum_{r=1}^{n} r^3 + \sum_{r=1}^{n} 5$<br>$= 2\left(\frac{n^2(n+1)^2}{4}\right) + 5n$<br>$= \frac{n^2(n+1)^2}{2} + 5n$ | |
| **(ii)** **[3]** | $\sum_{r=2}^{n} (2(r+2)^3+5) = (2(2+2)^3+5) + \cdots + (2(n+2)^3+5)$<br>$= \sum_{r=4}^{n+2} (2r^3+5)$<br>$= \sum_{r=1}^{n+2} (2r^3+5) - \sum_{r=1}^{3} (2r^3+5)$<br>$= \frac{(n+2)^2(n+3)^2}{2} + 5(n+2) - \left[\left(\frac{3^2(4)^2}{2}\right) + 5(3)\right]$<br>$= \frac{(n+2)^2(n+3)^2}{2} + 5n - 77$ | |
| **(iii)** **[1]** | From (i) $\sum_{r=1}^{n} u_r = \frac{n^2(n+1)^2}{2} + 5n$<br>As $n \to \infty$, $\frac{n^2(n+1)^2}{2} \to \infty$, $5n \to \infty$<br>Hence, $\sum_{r=1}^{n} u_r = \frac{n^2(n+1)^2}{2} + 5n \to \infty$, series $\sum_{r=1}^{\infty} u_r$ does not converge. | |

**6.2** It is given that $\sum_{r=2}^{n} \frac{1}{r(r^2-1)} = \frac{1}{4} + \frac{1}{2(n+1)} - \frac{1}{2n}$.

**(a)** Show that $\sum_{r=2}^{n} \frac{1}{r(r^2-1)}$ is less than $\frac{1}{4}$. [2]

**(b)** Give a reason why the series $\sum_{r=2}^{\infty} \frac{1}{r(r^2-1)}$ converges, and write down its value. [2]

**(d)** Find $\sum_{r=m+1}^{N} \frac{1}{(r-m)(r-m+1)(r-m+2)}$, where $m$ and $N$ are integers with $N > m > 0$.
(There is no need to express your answer as a single algebraic fraction.) [2]

**[VJC 2025 Prelim P1 Q6(a), (b), (d)]**

| Solution | | Comments |
|---|---|---|
| **(a)** **[2]** | $\frac{1}{2(n+1)} - \frac{1}{2n} = \frac{n-(n+1)}{2n(n+1)} = \frac{-1}{2n(n+1)} < 0$ for all $n \geqslant 2$<br>$\sum_{r=2}^{n} \frac{1}{r(r^2-1)} = \frac{1}{4} + \frac{1}{2(n+1)} - \frac{1}{2n} < \frac{1}{4}$ | |
| **(b)** **[2]** | As $n \to \infty$, $\frac{1}{2(n+1)} \to 0$ and $\frac{1}{2n} \to 0$,<br>$\therefore \sum_{r=2}^{n} \frac{1}{r(r^2-1)} \to \frac{1}{4}$, a constant.<br>Hence $\sum_{r=2}^{\infty} \frac{1}{r(r^2-1)}$ converges and $\sum_{r=2}^{\infty} \frac{1}{r(r^2-1)} = \frac{1}{4}$. | |
| **(d)** **[2]** | $\sum_{r=m+1}^{N} \frac{1}{(r-m)(r-m+1)(r-m+2)}$<br>$= \frac{1}{1(2)(3)} + \frac{1}{2(3)(4)} + \cdots + \frac{1}{(N-m)(N-m+1)(N-m+2)}$<br>$= \sum_{r=2}^{N-m+1} \frac{1}{(r-1)(r)(r+1)}$<br>$= \sum_{r=2}^{N-m+1} \frac{1}{r(r^2-1)}$<br>$= \frac{1}{4} + \frac{1}{2(N-m+2)} - \frac{1}{2(N-m+1)}$ | |

**6.3** A sequence is such that $u_{n+1} = u_n + 1.5$, for $n = 1, 2, 3, \ldots$.

**(a)** Given that $u_{50} = 99u_1$, show that $u_1 = 0.75$. [2]

**(b)** Find the least value of $n$ such that $u_2 + u_4 + u_6 + \cdots + u_{2n}$ is more than 2025. [2]

**(c)** It is given that $u_{25}$, $u_k$ and $u_5$ are the first three terms of a geometric progression.

**(i)** Find $k$. [2]

**(ii)** Find the sum of the first 13 terms of the geometric series, giving your answer to 3 decimal places. [2]

**[NYJC 2025 Prelim P1 Q8]**

| Solution | | Comments |
|---|---|---|
| **(a)** **[2]** | Note that the sequence is an arithmetic progression with common difference 1.5<br>$u_{50} = u_1 + 49(1.5)$<br>$99u_1 = u_1 + 73.5$<br>$u_1 = 0.75$ | |
| **(b)** **[2]** | **Method 1**<br>$\sum_{k=1}^{n} u_{2k} = u_2 + u_4 + \cdots + u_{2n}$ is a sum of $n$ terms of an arithmetic sequence with first term $u_2 = 0.75 + 1.5 = 2.25$ and common difference $2(1.5) = 3$<br>$u_2 + u_4 + \cdots + u_{2n} = \frac{n}{2}(2(2.25) + (n-1)(3))$<br>$= \frac{n}{2}(1.5 + 3n)$<br>From GC, for $\frac{n}{2}(1.5+3n) > 2025$, least $n = 37$<br>**Method 2**<br>$\sum_{k=1}^{n} u_{2k} = \sum_{k=1}^{n} (2.25 + (2k-1)3) > 2025$<br>Using GC, least $n = 37$ | |
| **(c)(i)** **[2]** | $\frac{u_k}{u_{25}} = \frac{u_5}{u_k}$<br>$u_k^2 = u_5 u_{25}$<br>$[0.75 + (k-1)(1.5)]^2 = [0.75 + (5-1)(1.5)]\,[0.75 + (25-1)(1.5)]$<br>$0.75 + (k-1)(1.5) = \sqrt{6.75 \times 36.75} = 15.75$<br>$k = 11$ | |
| **(c)(ii)** **[2]** | **Method 1**<br>Common ratio $= \frac{15.75}{36.75} = \frac{3}{7}$<br>$S_{13} = \frac{36.75\left(1 - \left(\frac{3}{7}\right)^{13}\right)}{1 - \frac{3}{7}}$<br>$= 64.31144$<br>$= 64.311$<br>**Method 2**<br>Using GC, $\sum_{r=1}^{13} \left(36.75\left(\frac{3}{7}\right)^{r-1}\right) = 64.311$ | |

**6.4** A sequence $u_1, u_2, u_3, \ldots$ is defined by

$$u_{n+1} = 2u_n(1-u_n)$$

for all positive integers $n$.

**(a)** Find the possible limits of the sequence, if the sequence is convergent. [2]

**(b)** For each of the following, evaluate $u_2$ and $u_3$, and state whether the sequence is convergent:

- $u_1 = 0$,
- $u_1 = 0.01$,
- $u_1 = -0.01$. [6]

**[EJC 2025 Prelim P2 Q2]**

| Solution | | Comments |
|---|---|---|
| **(a)** **[2]** | Let $L$ be the limit of the sequence $u_{n+1} = 2u_n(1-u_n)$.<br>$L = 2L(1-L)$<br>$2L^2 - L = 0$<br>$L(2L-1) = 0$<br>$L = 0$ or $L = \frac{1}{2}$ | |
| **(b)** **[6]** | When $u_1 = 0$, $u_2 = 0$, $u_3 = 0$, sequence is convergent.<br>When $u_1 = 0.01$, $u_2 = 0.0198$ (exact), $u_3 = 0.0388$ (3 s.f.), sequence is convergent.<br>When $u_1 = -0.01$, $u_2 = -0.0202$ (exact), $u_3 = -0.0412$ (3 s.f.), sequence is not convergent. | |

# 7 Arithmetic and Geometric Progressions

**7.1** **(a)** **(i)** The ninth, fifth and second terms of an arithmetic progression are successive terms of a geometric progression with first term $a$ and common ratio $r$, where $r \neq 1$ and $a > 0$. Find the value of $r$ and deduce that the geometric series is convergent. [3]

**(ii)** Using the value of $r$ found in (i), find the least value of $n$ such that the sum of all the terms after the $n$th term of the geometric progression is less than 1% of its sum of the first $n$ terms. [3]

**(b)** The sum, $S_n$, of the first $n$ terms of another sequence is given by $S_n = n \ln 2^{q(n+1)}$, where $q$ is a constant. Prove that the sequence follows an arithmetic progression. [4]

**[ASRJC 2025 Prelim P1 Q8]**

| Solution | | Comments |
|---|---|---|
| **(a)(i)** **[3]** | Let the first term and common difference of the arithmetic progression be $b$ and $d$ respectively.<br>Since $b+8d$, $b+4d$ and $b+d$ are successive terms of a GP, then $\frac{b+4d}{b+8d} = \frac{b+d}{b+4d}$ (common ratio).<br>$(b+4d)^2 = (b+8d)(b+d)$<br>$8bd+16d^2 = 9bd+8d^2$<br>$d(8d-b) = 0$<br>$b = 8d \quad (d \neq 0)$<br>$\Rightarrow r = \frac{8d+4d}{8d+8d} = \frac{3}{4}$<br>Since $\lvert r \rvert = \left\lvert \frac{3}{4} \right\rvert < 1$, the series is convergent. | |

**(a)(ii)**
**[3]**

$S_\infty - S_n < 0.01S_n \Rightarrow S_\infty < 1.01S_n$

$$\frac{a}{1-\frac{3}{4}} < 1.01\left[\frac{a\left(1-\left(\frac{3}{4}\right)^n\right)}{1-\frac{3}{4}}\right]$$

$$1 < 1.01\left(1-\left(\frac{3}{4}\right)^n\right)$$

$$\left(\frac{3}{4}\right)^n < 0.009901$$

$$n\ln\left(\frac{3}{4}\right) < \ln(0.009901)$$

$$n > \frac{\ln(0.009901)}{\ln\left(\frac{3}{4}\right)} = 16.042$$

Hence least $n$ is 17.

**(b)**
**[4]**

$$T_n = S_n - S_{n-1} = n\ln 2^{q(n+1)} - (n-1)\ln 2^{q(n)}$$

$$= n(n+1)q\ln 2 - (n-1)(nq)\ln 2$$

$$= nq\ln 2 + nq\ln 2$$

$$= 2nq\ln 2$$

$$T_n - T_{n-1} = 2nq\ln 2 - 2(n-1)q\ln 2$$

$$= 2q\ln 2 \quad \text{(constant } \because q \text{ is a constant)}$$

$\therefore$ the sequence is an arithmetic progression (Proved).

**7.2** An arithmetic series has first term $a$ and common difference $d$, where $a$ and $d$ are non-zero real constants. A convergent geometric series has first term $b$ and common ratio $r$, where $b>0$ and $r$ is a non-zero real constant. The sum of the first $n$ terms of the geometric series is denoted by $S$.

It is given that the 7th, 10th and 11th terms of the arithmetic series are equal to the 3rd, 6th and 10th terms of the geometric series respectively.

**(a)** Show that $r$ satisfies the equation $3r^7 - 4r^3 + 1 = 0$. Solve this equation, giving your answer correct to 4 decimal places. [4]

**(b)** A new series is formed by taking the even numbered terms of the geometric series. Find the smallest value of $n$ for which $S$ differs from the sum to infinity of the new series by at least $1.78b$. [4]

**[TMJC 2025 Prelim P1 Q4]**

| Solution | | Comments |
|---|---|---|
| **(a)** **[4]** | $a+6d=br^2 \quad --(1)$<br>$a+9d=br^5 \quad --(2)$<br>$a+10d=br^9 \quad --(3)$<br>$(2)-(1):\ 3d=br^2(r^3-1) \quad --(4)$<br>$(3)-(2):\ d=br^5(r^4-1) \quad --(5)$<br>Substitute (5) into (4),<br>$3br^5(r^4-1)=br^2(r^3-1)$<br>$3r^3(r^4-1)=(r^3-1) \quad$ (since $b, r\neq 0$)<br>$3r^7-4r^3+1=0 \quad$ (shown)<br>By GC, since the series is convergent, $\lvert r\rvert<1$,<br>$r=0.6639$ (4 d.p.)<br>[GC SCREEN PLACEHOLDER] GC 1: [APPS] PlySmlt2, POLY ROOT FINDER, ORDER 7 [NEXT]<br>[GC SCREEN PLACEHOLDER] GC 2: coefficients $a_7=3,\ a_6=0,\ a_5=0,\ a_4=0,\ a_3=-4,\ a_2=0,\ a_1=0,\ a_0=1$ [SOLVE] | |
| **(b)** **[4]** | New Series: $br, br^3, br^5, \ldots$<br>$\left\lvert \frac{br}{1-r^2}-S\right\rvert \geqslant 1.78b$<br>$\left\lvert \frac{br}{1-r^2}-\frac{b}{1-r}(1-r^n)\right\rvert \geqslant 1.78b$<br>$\left\lvert \frac{0.6639b}{1-0.6639^2}-\frac{b}{1-0.6639}(1-0.6639^n)\right\rvert \geqslant 1.78b$<br>$\left\lvert \frac{0.6639}{1-0.6639^2}-\frac{1}{1-0.6639}(1-0.6639^n)\right\rvert \geqslant 1.78 \quad (\because b>0)$<br>Using GC,<br>When $n=14$, $\left\lvert \frac{0.6639}{1-0.6639^2}-\frac{1}{1-0.6639}(1-0.6639^n)\right\rvert = 1.7785 < 1.78$<br>When $n=15$, $\left\lvert \frac{0.6639}{1-0.6639^2}-\frac{1}{1-0.6639}(1-0.6639^n)\right\rvert = 1.7818 \geqslant 1.78$<br>Therefore, least $n=15$. | |

**7.3** An arithmetic progression has first term $a$ and common difference $d$, where $a$ and $d$ are non-zero. The 3rd, 9th and 11th terms of the arithmetic progression are three distinct consecutive terms of a geometric progression.

**(a)** Find $d$ in terms of $a$. [3]

**(b)** Let the sum of the first $n$ terms of the arithmetic progression be $S_n$ where $S_1 = 143$. Find the largest possible value of $S_n$. [3]

The first term of the geometric progression is $b$, where $b$ is positive.

**(c)** Determine whether the geometric series is convergent. [2]

**(d)** Show that the sum of all terms after the $n$th term of the geometric progression is at most $\frac{1}{2}b$, where $n$ is a positive integer. [2]

**[SAJC 2025 Prelim P1 Q5]**

| **Solution** | | **Comments** |
|---|---|---|
| **(a)**<br>**[3]** | $\frac{a+8d}{a+2d} = \frac{a+10d}{a+8d}$<br>$(a+8d)^2 = (a+2d)(a+10d)$<br>$a^2+16ad+64d^2 = a^2+12ad+20d^2$<br>$4ad+44d^2 = 0$<br>$4d(a+11d) = 0$<br>$d=0$ (reject since terms of AP, GP are distinct) or $a+11d=0$<br>$d = -\frac{1}{11}a$ | |

**(b)** **[3]**

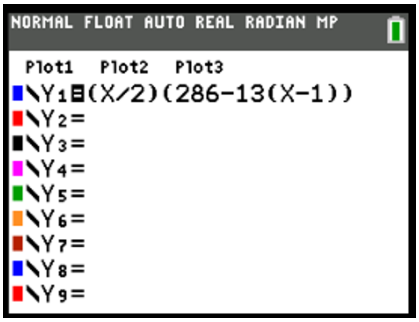
$S_1 = a = 143$

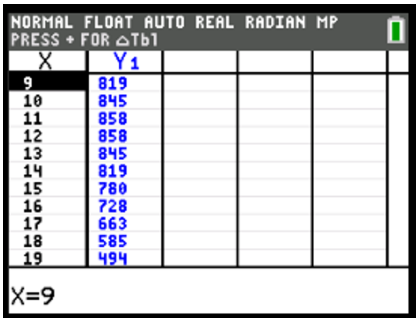
$$d = -\frac{1}{11}(143) = -13$$

$$S_n = \frac{n}{2}[2a + (n-1)d]$$

$$= \frac{n}{2}[286 - 13(n-1)]$$

From GC, largest $S_n = 858$




**(c)** **[2]**

$$r = \frac{a+8d}{a+2d} = \frac{-11d+8d}{-11d+2d} = \frac{-3d}{-9d} = \frac{1}{3}$$

Since $|r| = \frac{1}{3} < 1$, the geometric series is convergent.

**(d)** **[2]**

Let $u_n$ be the $n$th term of the geometric progression.

$u_{n+1} + u_{n+2} + \ldots$

$= S_\infty - S_n$

$$= \frac{b}{1-r} - \frac{b(1-r^n)}{1-r}$$

$$= \frac{br^n}{1-r}$$

$$= \frac{b\left(\frac{1}{3}\right)^n}{1-\frac{1}{3}}$$

$$= \frac{3}{2}\left(\frac{1}{3}\right)^n b$$

$$= \frac{b}{2}\left(\frac{1}{3}\right)^{n-1} \leqslant \frac{1}{2}b \quad \text{since } \left(\frac{1}{3}\right)^{n-1} \leqslant 1$$

Alternative:

$u_{n+1} + u_{n+2} + u_{n+3} + \ldots$

$$= b\left(\frac{1}{3}\right)^n + b\left(\frac{1}{3}\right)^{n+1} + b\left(\frac{1}{3}\right)^{n+2} + \ldots$$

$$= \frac{b\left(\frac{1}{3}\right)^n}{1-\frac{1}{3}}$$

$$= \frac{3}{2}b\left(\frac{1}{3}\right)^n$$

$$= \frac{b}{2}\left(\frac{1}{3}\right)^{n-1} \leqslant \frac{b}{2} \quad \text{since } \left(\frac{1}{3}\right)^{n-1} \leqslant 1$$

**7.4** Mabel and Janice decided to start a 5-year savings plan beginning in January 2026.

Mabel saves using a piggy bank. At the start of January 2026, she deposits \$101. Each subsequent month, she increases her deposit by \$1—so she deposits \$101 in January, \$102 in February, \$103 in March, and so on, until \$112 in December. At the start of each new year, she resets her monthly deposit to \$101 in January and repeats the same pattern through December. She continues this routine from 2026 to 2030, inclusive.

Janice, on the other hand, deposits \$100 at the start of every month into a bank account that earns 0.3% interest per month, with interest calculated and added into the account at the end of each month.

**(a)** Show that Janice will have more money in her savings account than Mabel has in her piggy bank at the end of December 2030. [5]

**(b)** Find the month and year when Janice's savings first exceed Mabel's savings. [4]

**[TJC 2025 Prelim P1 Q7]**

**Solution** | **Comments**

**(a)**
**[5]**

For Mabel:

Amount of money Mabel saves in a year

$= \$(101+102+\cdots+112)$

$= \$\frac{12}{2}(101+112) = \$1278$

Amount of money Mabel saves in 5 year $= \$1278 \times 5 = \$6390$

For Janice:

| $n$th mth | Amount at start of month ($) | Amount at end of month ($) |
|---|---|---|
| 1 Jan26 | 100 | $100(1.003)$ |
| 2 | $100+100(1.003)$<br>$=100(1+1.003)$ | $100(1+1.003)(1.003)$<br>$=100\,(1.003+1.003^2)$ |
| ⋮ | | ⋮ |
| $n$ | | $100\,(1.003+1.003^2+\cdots+1.003^n)$<br>$=\frac{100(1.003)\,(1.003^n-1)}{1.003-1}$ |

At Dec 2030, $n=60$.

Therefore, Amount Janice saves in 5 years

$= \frac{100(1.003)\,(1.003^{60}-1)}{1.003-1} = \$6582.85$ (2 d.p.) [A1]

$\therefore$ Janice will have more money in her savings account than Mabel has in her piggy bank.

**(b)**
**[4]**

At end of 4th year,

Mabel will have \$1278 x 4 $= \$5112$

Janice will have $\frac{100(1.003)\,(1.003^{48}-1)}{1.003-1} = \$5169.97 > \$5112$

At end of 3rd year,

Mabel will have \$1278 x 3 $= \$3834$

Janice will have $\frac{100(1.003)\,(1.003^{36}-1)}{1.003-1} = \$3806.97 < \$3834$

Therefore, the year when Janice's saving is more than Mabel's saving happen in 4th year (i.e. 2029). Let it be the $k$th month.

Then set Janice's saving > Mabel's saving

$\Rightarrow \frac{100(1.003)\,(1.003^{36+k}-1)}{1.003-1} > 3834+\frac{k}{2}\,[2(100)+(k-1)(1)]$

$\Rightarrow \left(\frac{100003}{3}\right)(1.003^{36+k}-1) - 3834 - \frac{k}{2}\,[199+k] > 0$

Using GC, we have $k \geqslant 4$. Therefore, the month and year is Apr 2029.

**7.5** Sarah, a Singaporean Citizen, started a 4-year undergraduate Computing course at the National University of Singapore (NUS) in August 2021. The annual tuition fee, after the MOE Tuition Grant, is \$8,250. Sarah took a Tuition Fee Loan (TFL) covering 90% of her tuition fees. This loan is interest-free during her course of study.

Sarah decided to set aside \$$x$ per month from August 2021. This amount is increased by \$2 each subsequent month. (Sarah set aside \$$x$ in August 2021, \$$(x+2)$ in September 2021, ….) With the amount set aside, she repaid one-fifth of her total TFL at the end of July 2025.

**(a)** Find the value of $x$. [3]

Upon Sarah's graduation, the TFL's interest rate becomes effective at a fixed rate of 0.4% per month, calculated on the outstanding balance at the end of the month. The interest is first charged on 31st August 2025.

Sarah plans to make monthly repayments of \$$y$ to the bank on the 1st of every month, starting from 1st September 2025.

**(b)** Find the amount of outstanding loan after Sarah's first monthly repayment. [1]

**(c)** Show that the amount of outstanding loan after Sarah's $n$th monthly repayment is

$$23760(1.004)^n - 250y\,(1.004^n - 1)\,.$$

[3]

**(d)** Sarah wants to pay off her remaining loan by the end of 3 years. Find the value of $y$. [2]

Sarah decided on a different repayment strategy. For the first 24 months, she repays \$500 per month. After these 24 months, she doubles her monthly repayment to \$1000.

**(e)** Determine the month and year Sarah will clear her loan and calculate the amount of her final repayment. [4]

**[ACJC 2025 Prelim P1 Q12]**

| Solution | | Comments |
|---|---|---|
| **(a)**<br>**[3]** | Amount Sarah plans to repay (before graduation):<br>$\frac{1}{5}\times\frac{90}{100}\times 8250\times 4=\$5940$<br>Number of repayments made from August 2021 to July 2025 is 48 (i.e. $n=48$).<br>The repayments follow an arithmetic progression (AP) with first term $a=x$ and common difference $d=2$.<br>The sum of an arithmetic progression is<br>$S_n=\frac{n}{2}[2a+(n-1)d]$<br>$5940=\frac{48}{2}[2x+(48-1)2]$<br>$x=76.75$ | |
| **(b)**<br>**[1]** | Amount of outstanding loan after Sarah's graduation:<br>$\frac{4}{5}\times\frac{90}{100}\times 8250\times 4=\$23760$<br>On 31st August 2025, apply interest of 0.4% to outstanding loan to get $\$(23760(1.004))$.<br>On 1st September 2025, make 1st repayment of $\$y$ to the bank.<br>$\therefore$ Outstanding loan after 1st repayment<br>$=\$(23760(1.004)-y)$ or $\$(23855.04-y)$ | |
| **(c)**<br>**[3]** | On 30th September 2025, apply interest of 0.4%.<br>On 1st October 2025, make 2nd repayment of $\$y$.<br>Outstanding loan after 2nd repayment<br>$=(23760(1.004)-y)\,1.004-y$<br>$=23760(1.004)^2-1.004y-y$<br>Outstanding loan after 3rd repayment (on 1st Nov 2025)<br>$=(23760(1.004)^2-1.004y-y)\,1.004-y$<br>$=23760(1.004)^3-1.004^2y-1.004y-y$<br>Outstanding loan after $n^{\text{th}}$ repayment<br>$=23760(1.004)^n-1.004^{n-1}y-\cdots-1.004^2y-1.004y-y$<br>$=23760(1.004)^n-y\,(1+1.004+1.004^2+\cdots+1.004^{n-1})$<br>$=23760(1.004)^n-y\left(\frac{1.004^n-1}{1.004-1}\right)$<br>$=23760(1.004)^n-250y\,(1.004^n-1)$ (shown) | |
| **(d)**<br>**[2]** | To repay remaining loan by the end of 3 years (i.e. 36 monthly repayments of $\$y$), let $n=36$.<br>To fully repay the loan, we want outstanding loan amount after 36 repayments to be zero (or less). | |

| | | |
|---|---|---|
| | $23760(1.004)^{36} - 250y\,(1.004^{36} - 1) \leqslant 0$<br>$y \geqslant 709.9769$<br>$\therefore y = 709.98$ (nearest cent) | |
| **(e)**<br>**[4]** | After 24 monthly repayments of \$500 (i.e. $n = 24$, $y = 500$), outstanding loan amount<br>$= 23760(1.004)^{24} - 250(500)\,((1.004)^{24} - 1)$<br>$= 13580.49$<br>Outstanding loan amount as of 1st August 2027<br>$= 13580.49$<br>In a similar method to **(c)**, the outstanding loan amount after $m$ monthly repayments of \$1000 (i.e. $y = 1000$) is<br>$13580.49(1.004)^m - 250(1000)\,((1.004)^m - 1)$<br>To fully repay the loan, we need<br>$13580.49(1.004)^m - 250(1000)\,((1.004)^m - 1) \leqslant 0$<br>$m \geqslant 13.991185$<br>Least $m = 14$<br>Sarah would require another 14 monthly repayments to fully repay her loan, with her first repayment of \$1000 on 1st September 2027.<br>Hence, she would clear her loan on (1st) October 2028.<br>The outstanding loan amount after her 13th monthly repayment of \$1000 (on 1st Sep 2028) is<br>$13580.49(1.004)^{13} - 250(1000)\,((1.004)^{13} - 1)$<br>$= 987.2529$<br>Interest is applied on 30th Sep 2028. Hence her final repayment amount is<br>$987.2529 \times 1.004 =$ \$991.20 (nearest cent)<br>(Alternatively, outstanding loan amount after 14th monthly repayment is<br>$13580.49(1.004)^{14} - 250(1000)\,((1.004)^{14} - 1)$<br>$= -8.7981$<br>Hence the final repayment is $1000 - 8.7981 =$ \$991.20 (nearest cent). | |

**7.6** The dosage of medicine given to a patient needs to be carefully managed in order to achieve the desired result.

**(a)** To give the patient time to adapt to a particular medicine, the dosage is slowly increased over time. The patient is initially given a dosage of 0.75 units. The dosage is increased by 0.5 units each day until a total of at least 10 units of medicine has been taken. Find the least number of days it takes to achieve this, and the dosage on that day. [4]

**(b)** The doctor prescribed a new medication to another patient. After $n$ days, the total dosage of the new medicine administered to the patient is given by

$$\frac{3}{20}\left(\frac{5^n-4^n}{5^{n-1}}\right) \text{ units.}$$

**(i)** Show that the increase in dosage of the new medicine administered per day follows a geometric progression. [3]

**(ii)** Find the total dosage of the new medicine administered to the patient after a long period of treatment. [2]

**(c)** Doctors can manage the medication so that the daily dosage of medicine increases as an arithmetic progression. Explain in context why the arithmetic progression is not a preferred model for the dosage of medication. [1]

**[RVHS 2025 Prelim P2 Q4]**

| Solution | | Comments |
|---|---|---|
| **(a)**<br>**[4]** | Let $u_n$ be the dose on the $n^{th}$ day.<br>$u_n = 0.75 + 0.5(n-1)$<br>$S_n = \frac{n}{2}(2(0.75)+(n-1)(0.5))$<br>$10 \leqslant \frac{n}{2}\left(1+\frac{n}{2}\right)$ ——(∗)<br>$\frac{1}{4}n^2 + \frac{1}{2}n \geqslant 10$<br>$n^2 + 2n \geqslant 40$<br>$(n+1)^2 \geqslant 41$<br>$n \geqslant -1+\sqrt{41}$ or $n \leqslant -1-\sqrt{41}$<br>(rej as no. of days cannot be -ve)<br>$n \geqslant 5.40312$<br>Therefore, it takes 6 days.<br>Alternatively,<br>When $n=5$, $\frac{1}{4}n^2+\frac{1}{2}n = 8.75 < 10$<br>When $n=6$, $\frac{1}{4}n^2+\frac{1}{2}n = 12 \geqslant 10$<br>Therefore, it takes 6 days.<br>$u_6 = 0.75+0.5(5) = 3.25$<br>The dose on the 6th day is 3.25 units. | |
| **(b)(i)**<br>**[3]** | Increase on $n^{th}$ day<br>$= S_n - S_{n-1}$<br>$= \frac{3}{20}\left(\frac{5^n-4^n}{5^{n-1}}\right) - \frac{3}{20}\left(\frac{5^{n-1}-4^{n-1}}{5^{n-2}}\right)$<br>$= \frac{3}{20}\left(\frac{5^n-4^n}{5^{n-1}} - \frac{5^{n-1}-4^{n-1}}{5^{n-2}}\right)$<br>$= \frac{3}{20}\left(\frac{1}{5^{n-2}}\right)\left(5^{n-1} - \frac{4}{5}(4^{n-1}) - 5^{n-1} + 4^{n-1}\right)$<br>$= \frac{3}{20}\left(\frac{1}{5^{n-2}}\right)\left(\frac{1}{5}\right)(4^{n-1})$<br>$= \frac{3}{20}\left(\frac{4}{5}\right)^{n-1}$<br>which is of the form $ar^{n-1}$, where $a = \frac{3}{20}$ and $r = \frac{4}{5}$ which is a GP.<br>**Alternative:**<br>Consider $\frac{u_{n+1}}{u_n} = \frac{\frac{3}{20}\left(\frac{4}{5}\right)^{n+1-1}}{\frac{3}{20}\left(\frac{4}{5}\right)^{n-1}} = \frac{4}{5}$ which is a constant<br>Since consecutive terms have a common ratio $r = \frac{4}{5}$, the increase in dosage follows a GP. | |

**(b)(ii)**
**[2]**

**Method 1**

Long term dosage

$$= \lim_{n\to\infty} \frac{3}{20}\left(\frac{5^n - 4^n}{5^{n-1}}\right)$$

$$= \frac{3/20}{1-4/5}$$

$$= \frac{3}{4}$$

In the long term, the total dosage approaches 0.75 units.

**Note:**

After $n$ days, the total dosage of the medicine is given by $\frac{3}{20}\left(\frac{5^n - 4^n}{5^{n-1}}\right)$.

Let $S_n$ denote the total dosage of the new medicine after $n$ days.

Then $S_n = \frac{3}{20}\left(\frac{5^n - 4^n}{5^{n-1}}\right)$.

Let $v_n$ the change in dosage on $n$th day found in part (b)(i) where $v_n = S_n - S_{n-1}$

Since $v_n$ is a geometric progression, we can use $S_\infty = \frac{a}{1-r}$, the sum to infinity formula to find the long term total dosage.

**Method 2**

$$\frac{3}{20}\left(\frac{5^n - 4^n}{5^{n-1}}\right)$$

$$= \frac{3}{20}(5)\left(\frac{5^n - 4^n}{5^n}\right)$$

$$= \frac{3}{20}(5)\left(1 - \left(\frac{4}{5}\right)^n\right)$$

As $n \to \infty$, $\left(\frac{4}{5}\right)^n \to 0$

Total dosage in the long term

$$= \frac{3}{20}(5)(1-0)$$

$$= \frac{3}{4}$$

**(c)**
**[1]**

If the dosage increases as an AP, then it will continue to **increase to** (positive) **infinity**.

# 8 Differentiation

**8.1** A curve $C$ has equation $x^2+y^2-4y=xy$, where $x \geqslant 0$.

**(a)** Show that $(2y-x-4)\dfrac{\mathrm{d}y}{\mathrm{d}x}=y-2x$. [2]

**[RVHS 2025 Prelim P1 Q7(a)]**

| Solution | | Comments |
|---|---|---|
| **(a)**<br>**[2]** | $x^2+y^2-4y=xy$<br>$2x+2y\dfrac{\mathrm{d}y}{\mathrm{d}x}-4\dfrac{\mathrm{d}y}{\mathrm{d}x}=y+x\dfrac{\mathrm{d}y}{\mathrm{d}x}$<br>$(2y-x-4)\dfrac{\mathrm{d}y}{\mathrm{d}x}+2x-y=0$<br>$(2y-x-4)\dfrac{\mathrm{d}y}{\mathrm{d}x}=y-2x$ | |

**8.2** A curve has parametric equations $x=t+\sin t$, $y=1-\cos t$.

Show that $\dfrac{\mathrm{d}y}{\mathrm{d}x}=\tan\dfrac{t}{2}$. [3]

| Solution | | Comments |
|---|---|---|
| **[3]** | $x=t+\sin t \Rightarrow \dfrac{\mathrm{d}x}{\mathrm{d}t}=1+\cos t$<br>$y=1-\cos t \Rightarrow \dfrac{\mathrm{d}y}{\mathrm{d}t}=\sin t$<br>$\dfrac{\mathrm{d}t}{\mathrm{d}x}=\dfrac{1}{\frac{\mathrm{d}x}{\mathrm{d}t}}=\dfrac{1}{1+\cos t}$<br>$\dfrac{\mathrm{d}y}{\mathrm{d}x}=\dfrac{\mathrm{d}y}{\mathrm{d}t}\times\dfrac{\mathrm{d}t}{\mathrm{d}x}$<br>$=\dfrac{\sin t}{1+\cos t}$<br>$=\dfrac{2\sin\frac{t}{2}\cos\frac{t}{2}}{2\cos^2\frac{t}{2}}$<br>$=\dfrac{\sin\frac{t}{2}}{\cos\frac{t}{2}}$<br>$=\tan\dfrac{t}{2}$ (shown) | |

# 9 Applications of Differentiation

**9.1** The curve $C$ has equation $x^3+y^3-3xy=k$, where $k$ is a positive constant.

$C$ intersects the $x$-axis at the point $A$.

**(a)** Find the equation of the tangent to $C$ at $A$. Give your answer in the form $y=px+q$, where $p$ and $q$ are constants to be found in terms of $k$. [5]

**(b)** There are two points on $C$ where the tangents are parallel to the $y$-axis. Given that $k=3$, find the exact $y$-coordinates of these two points. [3]

**[EJC 2025 Prelim P2 Q1]**

| Solution | | Comments |
|---|---|---|
| **(a)**<br>**[5]** | Differentiating $C$ w.r.t. $x$,<br>$3x^2+3y^2\frac{dy}{dx}-3x\frac{dy}{dx}-3y=0$<br>$\frac{dy}{dx}=\frac{3y-3x^2}{3y^2-3x}=\frac{y-x^2}{y^2-x}$<br>When $y=0$,<br>$x^3=k$,<br>$x=k^{\frac{1}{3}}$<br>$\frac{dy}{dx}=x=k^{\frac{1}{3}}$<br>$y=k^{\frac{1}{3}}\left(x-k^{\frac{1}{3}}\right)=k^{\frac{1}{3}}x-k^{\frac{2}{3}}$ | |
| **(b)**<br>**[3]** | For tangent to be parallel to $y$-axis, $\frac{dy}{dx}$ must be undefined.<br>Hence,<br>$y^2-x=0$<br>$x=y^2$<br>Substituting into $C$ and $k=3$,<br>$y^6+y^3-3y^3-k=0$<br>$y^6-2y^3-3=0$<br>Factorising,<br>$(y^3-3)(y^3+1)=0$<br>$y^3=3$ or $y^3=-1$<br>$y=3^{\frac{1}{3}}$ or $y=-1$ | |

**9.2** A curve $C$ has parametric equations $x=\sqrt{t}, y=t^2+4$, where $0 \leqslant t \leqslant 16$.

**(i)** Find the equation of the tangent to $C$ at the point $P$ with parameter $p$. [3]

**(ii)** Given that the tangent in part **(i)** passes through the point $(0, 1)$, find the exact coordinates of the point $P$. [3]

**[MI 2025 Prelim P2 Q2(i), (ii)]**

| Solution | | Comments |
|---|---|---|
| **(i)** **[3]** | $x=\sqrt{t}, \quad y=t^2+4$<br>$\frac{\mathrm{d}x}{\mathrm{d}t}=\frac{1}{2\sqrt{t}}, \quad \frac{\mathrm{d}y}{\mathrm{d}t}=2t$<br>$\frac{\mathrm{d}y}{\mathrm{d}x}=\frac{2t}{\frac{1}{2\sqrt{t}}}=4t^{\frac{3}{2}}$<br>Note: $t=p$<br>The eqn of tangent to $C$ at $P\left(\sqrt{p}, p^2+4\right)$:<br>$y-(p^2+4)=4p^{\frac{3}{2}}\left(x-\sqrt{p}\right)$<br>$y=4p^{\frac{3}{2}}x-3p^2+4$ | |
| **(ii)** **[3]** | The eqn of tangent of $C$ at $P$ passes through $(0, 1)$<br>At $(0, 1)$:<br>$1=4p^{\frac{3}{2}}(0)-3p^2+4$<br>$3p^2=3$<br>$p=1 \quad$ (reject $-1 \because p \geqslant 0$)<br>When $p=1$<br>$x=1, \ y=5$<br>$P(1, 5)$ | |

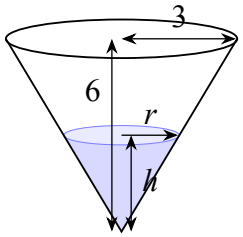
**9.3** A conical water tank is being filled with water at a constant rate of 0.2 $m^3s^{-1}$. The tank has a height of 6 m and a radius at the top of 3 m.

**(a)** Show that the volume of the water in the tank, $V$, can be expressed as $\frac{2}{3}\pi r^3$, where $r$ is the radius of the top water surface. [1]

**(b)** Find the exact rate of change of $r$ when the height of the water is 4 m. [3]

[The volume of a cone of base radius $r$ and height $h$ is given by $V=\frac{1}{3}\pi r^2 h$.]

**[RVHS 2025 Promo Q1]**

| **Solution** | | **Comments** |
|---|---|---|
| **(a)** **[1]** |  <br>$V=\frac{1}{3}\pi r^2 h,$ where $h$ is the height of the water.<br>$=\frac{1}{3}\pi r^2(2r),$ since $\frac{h}{r}=\frac{6}{3}=\frac{2}{1}$ by similar triangles.<br>$=\frac{2}{3}\pi r^3$. (Shown) | |
| **(b)** **[3]** | $\frac{dV}{dr}=2\pi r^2$<br>When $h=4$ (i.e. when $r=2$),<br>$\frac{dr}{dt}=\frac{dr}{dV}\times\frac{dV}{dt}$<br>$=\frac{1}{2\pi r^2}\times 0.2$<br>$=\frac{1}{8\pi}\times 0.2$<br>$=\frac{1}{40\pi}$ m $s^{-1}$ | |

**9.4** [The arc length $L$ of a sector of radius $a$ and angle $\theta$ is given by $L = a\theta$;
The volume $V$ of a cone of base radius $r$ and height $h$ is given by $V = \frac{1}{3}\pi r^2 h$ ]

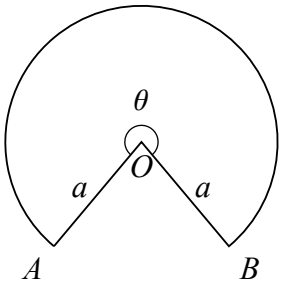


Figure 1

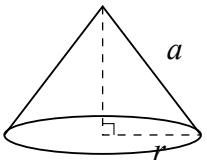


Figure 2

A metal sheet is shaped as a circular sector with angle $\theta$ and fixed radius $a$, as shown in Figure 1. This sector is then formed into a right circular cone with slant height $a$ by joining the two radii $OA$ and $OB$ together as shown in Figure 2.

**(a)** If $\theta$ is measured in radians, state the base radius of the cone, $r$, in term of $a$ and $\theta$. [1]

**(b)** Show that the volume of the cone, $V$, is given by

$$V^2 = \frac{a^6}{576\pi^4}\theta^4 (4\pi^2 - \theta^2) .$$

[3]

**(c)** Using differentiation, find the exact value of $\theta$ that will maximise $V$. [4]

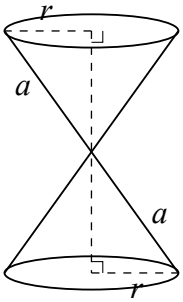


Figure 3

Two identical cones, as described in part **(c)**, are joined together tip-to-tip to form a new object, as shown in Figure 3. The upper cone is initially filled with water, which leaks out into the lower cone at a rate of $\frac{\pi}{15}$ cm$^3$ s$^{-1}$. At time $t$ seconds, the height of the water in the upper cone is $y$.

**(d)** Show that the volume of water in the cone, $W$ cm$^3$, is given by $W = \frac{2\pi y^3}{3}$. [2]

**(e)** Calculate the rate of change of $y$ when $y=2$. [2]

**[TJC 2025 Prelim P2 Q4]**

| Solution | | Comments |
|---|---|---|
| **(a)** **[1]** | Circumference of base = length of arc AB<br>$2\pi r = a\theta$<br>$r = \frac{a}{2\pi}\theta$ | |
| **(b)** **[3]** | (i) Let $h$ = height of cone<br>$h^2 = a^2 - r^2$<br>$= a^2 - \frac{a^2}{4\pi^2}\theta^2$<br>$= \frac{a^2}{4\pi^2}(4\pi^2 - \theta^2)$<br>$V = \frac{1}{3}\pi r^2 h$<br>$V^2 = \frac{1}{9}\pi^2 r^4 h^2$<br>$= \frac{1}{9}\pi^2 \left(\frac{a^4}{16\pi^4}\right) \cdot \frac{a^2}{4\pi^2}(4\pi^2 - \theta^2)$<br>$= \frac{a^6}{576\pi^4}\theta^4(4\pi^2 - \theta^2)$ | |

**(c)**
**[4]**

**Method 1**

$$\frac{\mathrm{d}V^2}{\mathrm{d}\theta} = \frac{a^6}{576\pi^4}\frac{\mathrm{d}}{\mathrm{d}\theta}\left(4\pi^2\theta^4 - \theta^6\right)$$
$$= \frac{a^6}{576\pi^4}\left(16\pi^2\theta^3 - 6\theta^5\right)$$
$$= \frac{a^6}{576\pi^4}\cdot 2\theta^3\left(8\pi^2 - 3\theta^2\right)$$

$\frac{\mathrm{d}V^2}{\mathrm{d}\theta} = 0$ when $\theta^3\left(8\pi^2 - 3\theta^2\right) = 0$

$$\theta^2 = \frac{8\pi^2}{3} \quad \Rightarrow \quad \theta = 2\pi\sqrt{\frac{2}{3}}$$

$$\frac{\mathrm{d}^2V^2}{\mathrm{d}\theta^2} = \frac{a^6}{576\pi^4}\left(48\pi^2\theta^2 - 30\theta^4\right)$$

When $\theta = 2\pi\sqrt{\frac{2}{3}}$,

$$\frac{\mathrm{d}^2V^2}{\mathrm{d}\theta^2} = \frac{a^6}{576\pi^4}\left(48\pi^2\left(\frac{8\pi^2}{3}\right) - 30\left(\frac{64\pi^4}{9}\right)\right) = -\frac{4a^6}{27} < 0$$

Therefore $V$ is max when $\theta = 2\pi\sqrt{\frac{2}{3}}$

**Method 2**

$$V^2 = \frac{a^6}{576\pi^4}\theta^4\left(4\pi^2 - \theta^2\right)$$

Differentiating w.r.t. $\theta$,

$$2V\frac{\mathrm{d}V}{\mathrm{d}\theta} = \frac{a^6}{576\pi^4}\frac{\mathrm{d}}{\mathrm{d}\theta}\left(4\pi^2\theta^4 - \theta^6\right)$$
$$= \frac{a^6}{576\pi^4}\left(16\pi^2\theta^3 - 6\theta^5\right)$$
$$= \frac{a^6}{576\pi^4}\cdot 2\theta^3\left(8\pi^2 - 3\theta^2\right)$$

$\frac{\mathrm{d}V}{\mathrm{d}\theta} = 0$ when $\theta^3\left(8\pi^2 - 3\theta^2\right) = 0$

$$\theta^2 = \frac{8\pi^2}{3} \quad \Rightarrow \quad \theta = 2\pi\sqrt{\frac{2}{3}} \quad (\because \theta > 0)$$

$$2V\frac{\mathrm{d}^2V}{\mathrm{d}\theta^2} + 2\left(\frac{\mathrm{d}V}{\mathrm{d}\theta}\right)^2 = \frac{a^6}{576\pi^4}\left(48\pi^2\theta^2 - 30\theta^4\right)$$

When $\theta = 2\pi\sqrt{\frac{2}{3}}$, $\frac{\mathrm{d}V}{\mathrm{d}\theta} = 0$

$$2V\frac{\mathrm{d}^2V}{\mathrm{d}\theta^2} = \frac{a^6}{576\pi^4}\left(48\pi^2\left(\frac{8\pi^2}{3}\right) - 30\left(\frac{64\pi^4}{9}\right)\right) = -\frac{4a^6}{27} < 0$$

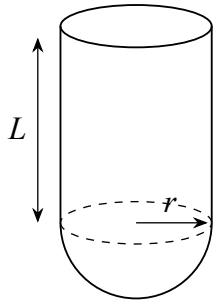
Since $V > 0$, $\frac{\mathrm{d}^2V}{\mathrm{d}\theta^2} < 0$

Therefore $V$ is max when $\theta = 2\pi\sqrt{\frac{2}{3}}$

| | | |
|---|---|---|
| **(d)**<br>**[2]** | Note that: $\theta = 2\pi\sqrt{\frac{2}{3}}$<br>$r = \frac{a}{2\pi}\theta = \frac{a}{2\pi}\left(2\pi\sqrt{\frac{2}{3}}\right) = a\sqrt{\frac{2}{3}}$<br>$h^2 = a^2 - r^2 = a^2 - \left(a\sqrt{\frac{2}{3}}\right)^2 = \frac{1}{3}a^2 \quad \Rightarrow h = \frac{1}{\sqrt{3}}a$<br>$\frac{r}{h} = a\sqrt{\frac{2}{3}} \times \frac{\sqrt{3}}{a} = \sqrt{2} \quad \Rightarrow r = \sqrt{2}h$<br>$W = \frac{1}{3}\pi r_w^2 h_w = \frac{1}{3}\pi\left(\sqrt{2}y\right)^2 y = \frac{2\pi y^3}{3}$ | |
| **(e)**<br>**[2]** | $\frac{\mathrm{d}W}{\mathrm{d}t} = \frac{\mathrm{d}W}{\mathrm{d}y} \times \frac{\mathrm{d}y}{\mathrm{d}t}$<br>$-\frac{\pi}{15} = 2\pi y^2 \times \frac{\mathrm{d}y}{\mathrm{d}t}$<br>$\frac{\mathrm{d}y}{\mathrm{d}t} = -\frac{1}{30y^2}$<br>When $y = 2$, $\frac{\mathrm{d}y}{\mathrm{d}t} = -\frac{1}{120}$ cm s$^{-1}$<br>Rate of change of $y$ is $-\frac{1}{120}$ cm s$^{-1}$. | |

**9.5** [The surface area and volume of a sphere are given by $4\pi r^2$ and $\frac{4}{3}\pi r^3$ respectively.]

An aquarist who has a fixed budget of \$$k$ wants to make a goldfish sanctuary. His design of the tank consists of a circular cover, a cylinder of length $L$ cm and a hemispherical base of radius $r$ cm neatly fitted together. The costs of the material per cm$^2$ used for the cover, the cylindrical surface and hemispherical base are \$20, \$10 and \$30 respectively. Assume that the tank is made of material with negligible thickness.

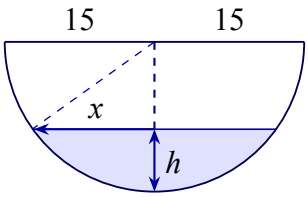



**(a)** Show that the volume of the tank is $V$ cm$^3$, where $V=\frac{k}{20}r-\frac{10}{3}\pi r^3$. [3]

**(b)** As $r$ varies, find the cost of the material used to make the hemispherical base in terms of $k$ when $V$ is at its maximum value. You need to show that $V$ is a maximum. [3]

The aquarist has fixed the radius of the cylinder to be 15 cm. Initially, there is some water in the hemispherical part of the tank. However, due to a defect in the tank, water is leaking at a constant rate of 20 cm$^3$ per minute.

The depth of the water at the hemispherical bottom is denoted by $h$ cm, and the radius of the water surface is $x$ cm. The volume of water in the hemispherical bottom is given by $W$ cm$^3$, where $W=\frac{\pi h\,(3x^2+h^2)}{6}$.




**(c)** By formulating a relationship between $x$ and $h$, find, when $h=3$, the rate of change of

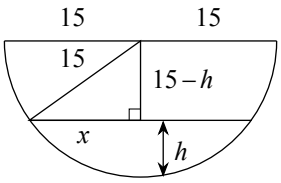
**(i)** the depth of water, [3]

**(ii)** the radius of the water surface. [3]

**[DHS 2025 Prelim P1 Q11]**

| Solution | | Comments |
|---|---|---|
| **(a)** **[3]** | $k=2\pi r^2(30)+2\pi rL(10)+\pi r^2(20)$<br>$k=80\pi r^2+20\pi rL$<br>$k=20\pi r(4r+L)$<br>$\Rightarrow L=\frac{k}{20\pi r}-4r$<br>$V=\frac{2}{3}\pi r^3+\pi r^2L$<br>$=\frac{2}{3}\pi r^3+\pi r^2\left(\frac{k}{20\pi r}-4r\right)$<br>$=\frac{2}{3}\pi r^3+\frac{kr}{20}-4\pi r^3$<br>$=\frac{k}{20}r-\frac{10}{3}\pi r^3$ (shown) | |
| **(b)** **[3]** | For maximum $V$ as $r$ varies,<br>$\frac{\mathrm{d}V}{\mathrm{d}r}=0$<br>$\frac{\mathrm{d}V}{\mathrm{d}r}=\frac{k}{20}-10\pi r^2$<br>$r=\sqrt{\frac{k}{200\pi}}$ (reject $r<0$)<br>$\frac{\mathrm{d}^2V}{\mathrm{d}r^2}=-20\pi r<0$<br>Therefore, $V$ is maximum.<br>Cost of material for the hemispherical base<br>$=2\pi r^2(30)=2\pi\left(\sqrt{\frac{k}{200\pi}}\right)^2(30)=\frac{3k}{10}$ dollars | |

**(c)(i)**
**[3]**




By Pythagoras' Theorem,

$x^2 = 15^2 - (15-h)^2$

$x^2 = (15-(15-h))\,(15+(15-h))$

$x^2 = h(30-h)$

$$W = \frac{\pi h\,(3x^2+h^2)}{6} = \frac{\pi h\,(3\,(h(30-h))+h^2)}{6}$$

$$W = \frac{\pi h\,(90h-2h^2)}{6} = \frac{\pi\,(45h^2-h^3)}{3}$$

$$\frac{\mathrm{d}W}{\mathrm{d}h} = \frac{\pi\,(90h-3h^2)}{3}$$

$$\frac{\mathrm{d}W}{\mathrm{d}t} = \frac{\mathrm{d}W}{\mathrm{d}h} \times \frac{\mathrm{d}h}{\mathrm{d}t} = \frac{\pi\,(90h-3h^2)}{3} \times \frac{\mathrm{d}h}{\mathrm{d}t}$$

At $h = 3$,

$$\frac{\mathrm{d}h}{\mathrm{d}t} = \frac{\frac{\mathrm{d}W}{\mathrm{d}t}}{\frac{\mathrm{d}W}{\mathrm{d}h}} = \frac{-20}{\frac{\pi(90(3)-3(3)^2)}{3}}$$

$= -0.078595 = -0.0786$ cm/min (3 s.f.)

**(c)(ii)**
**[3]**

$$\frac{\mathrm{d}x}{\mathrm{d}t} = \frac{\mathrm{d}x}{\mathrm{d}h} \times \frac{\mathrm{d}h}{\mathrm{d}t}$$

$x^2 = h(30-h) \Rightarrow x = \sqrt{h(30-h)} = \sqrt{30h-h^2} \quad (x>0)$

$$\frac{\mathrm{d}x}{\mathrm{d}h} = \frac{30-2h}{2\sqrt{30h-h^2}} = \frac{15-h}{\sqrt{30h-h^2}}$$

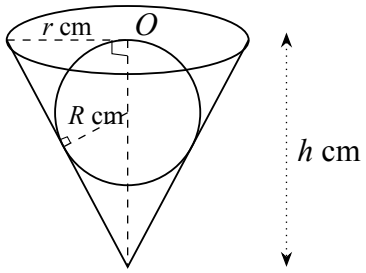
At $h = 3$,

$$\frac{\mathrm{d}x}{\mathrm{d}t} = \left(\frac{15-3}{\sqrt{30(3)-(3)^2}}\right) \times (-0.078595)$$

$= -0.10479 = -0.105$ cm/min (3sf)

**9.6** [It is given that the volume of cone with radius $r$ and height $h$ is $\frac{1}{3}\pi r^2 h$ ]

A water filter system takes the shape of an inverted right circular cone with a circular top of radius $r$ cm and height $h$ cm. It contains a spherical filter which the water passes through during filtration. The sphere has a constant radius $R$ cm and is inscribed in the cone such that the sphere is in contact with the center of the circular top, $O$, and the slant surface of the cone as shown in the diagram below.

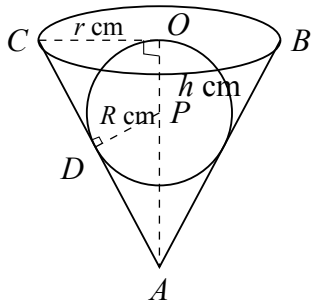



**(a)** Show that $r^2 = \dfrac{h^2R^2}{h^2 - 2hR}$. [3]

**(b)** As $h$ varies, find the minimum volume of the cone in terms of $R$. [5]

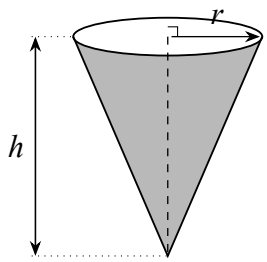
[You do not need to show that the volume is minimum]

**[TJC 2024 Promo Q12]**

| Solution | | Comments |
|---|---|---|
| **(a)** **[3]** |  <br>Let centre of the sphere be $P$.<br>Using similar triangles $APD$ and $ACO$,<br>$\frac{h-R}{AC}=\frac{R}{r}$<br>$AC=\frac{rh-rR}{R}$<br>By Pythagoras' Theorem, $AC^2=h^2+r^2$<br>$\left(\frac{rh-rR}{R}\right)^2=h^2+r^2$<br>$r^2h^2-2r^2hR+r^2R^2=h^2R^2+r^2R^2$<br>$r^2\left(h^2-2hR\right)=h^2R^2$<br>$\therefore r^2=\frac{h^2R^2}{h^2-2hR}$<br>**OR**<br>Let centre of the sphere be $P$.<br>$AD=\sqrt{(AP^2-PD^2)}$<br>$=\sqrt{((h-R)^2-R^2)}$<br>$=\sqrt{(h^2-2hR)}$<br>Using congruent triangles $PCO$ and $PCD$, $CO=CD=r$ cm.<br>$AC=AD+DC$<br>$=\sqrt{(h^2-2hR)}+r$<br>Since by Pythagoras' theorem, $AC^2=h^2+r^2$<br>$h^2+r^2=h^2-2hR+2r\sqrt{(h^2-2hR)}+r^2$<br>$\therefore r^2=\frac{h^2R^2}{h^2-2hR}$ | |
| **(b)** **[5]** | Volume of cone, $V$ | |

| | | |
|---|---|---|
| | $= \frac{1}{3}\pi r^2 h$<br>$= \frac{1}{3}\pi \frac{h^2R^2}{h^2-2hR}h$<br>$= \frac{1}{3}\pi R^2 \frac{h^3}{h^2-2hR}$<br>$= \frac{1}{3}\pi R^2 \frac{h^2}{h-2R}$<br>$\frac{\mathrm{d}V}{\mathrm{d}h} = \frac{1}{3}\pi R^2 \left[\frac{2h(h-2R)-h^2}{(h-2R)^2}\right]$<br>$= \frac{1}{3}\pi R^2 \left[\frac{h^2-4hR}{(h-2R)^2}\right]$<br>$= \frac{1}{3}\pi R^2 \frac{h(h-4R)}{(h-2R)^2}$<br>$\frac{\mathrm{d}V}{\mathrm{d}h} = 0$<br>$\frac{1}{3}\pi R^2 \frac{h(h-4R)}{(h-2R)^2} = 0$<br>$h = 4R$ or $h = 0$ (reject $\because h > 0$)<br>$\therefore$ minimum $V = \frac{1}{3}\pi R^2 \frac{(4R)^2}{4R-2R} = \frac{8}{3}\pi R^3$ cm$^3$ | |

**9.7**




A cone-shaped cup is made of paper of negligible thickness to hold $20\pi$ cm$^3$ of liquid. The open inverted cone has radius $r$ cm and height $h$ cm as shown in the diagram above.

The external surface area of the cup is denoted by $A$ cm$^2$. The manufacturer wants to reduce the cost of production by minimizing the value of $A$.

**(a)** Find $h$ in terms of $r$. [1]

**(b)** By considering $A^2$ or otherwise, show that $A\frac{\mathrm{d}A}{\mathrm{d}r} = \pi^2\left(2r^3 - \frac{3600}{r^3}\right)$. [3]

**(c)** Find the exact value of $r$ that gives the minimum value of $A$, proving that $A$ is a minimum. Find also the ratio of the radius to the height, $\frac{r}{h}$, giving your answer in terms of $k\sqrt{2}$, where $k$ is a constant to be determined. [5]

The manufacturer decides to make paper cups at minimum external surface area using the ratio $\frac{r}{h}$ found in part **(c)**.

**(d)** The cup is being filled completely with water. However, there is a small hole at the bottom of the cup that causes water to leak out at the rate of 3 cm$^3$ per second. Find the rate of decrease of the depth of the water at the instant when the depth is 2 cm. [3]

[It is given that the volume of a circular cone with base radius $r$ and height $h$ is $\frac{1}{3}\pi r^2 h$ and the curved surface area is $\pi rl$, where $l$ is the slant height of the cone.]

**[CJC 2024 Promo Q10]**

| **Solution** | | **Comments** |
|---|---|---|
| **(a)**<br>**[1]** | Volume of cone $= \frac{1}{3}\pi r^2 h = 20\pi$<br>$\frac{r^2 h}{3} = 20$<br>$h = \frac{60}{r^2}$ | |

**(b)**
**[3]**

**Method 1: Considering $A^2$**

$$A = \pi rl$$
$$= \pi r\sqrt{r^2+h^2}$$
$$= \pi r\sqrt{r^2+\left(\frac{60}{r^2}\right)^2}$$
$$= \pi r\sqrt{r^2+\frac{3600}{r^4}}$$
$$A^2 = \pi^2 r^2\left(r^2+\frac{3600}{r^4}\right)$$
$$= \pi^2\left(r^4+\frac{3600}{r^2}\right)$$
$$= \pi^2\left(r^4+3600r^{-2}\right)$$

Differentiating implicitly with respect to $r$:

$$2A\frac{\mathrm{d}A}{\mathrm{d}r} = \pi^2\left(4r^3-7200r^{-3}\right)$$
$$A\frac{\mathrm{d}A}{\mathrm{d}r} = \pi^2\left(2r^3-\frac{3600}{r^3}\right) \quad \text{(shown)}$$

**Method 2: "Otherwise" Method**

$$A = \pi rl$$
$$= \pi r\sqrt{r^2+h^2}$$
$$= \pi r\sqrt{r^2+\left(\frac{60}{r^2}\right)^2}$$
$$= \pi r\sqrt{r^2+\frac{3600}{r^4}}$$
$$= \pi r\left(r^2+3600r^{-4}\right)^{\frac{1}{2}}$$

Differentiating with respect to $r$:

$$\frac{\mathrm{d}A}{\mathrm{d}r} = \pi\left[r\cdot\frac{1}{2}\left(r^2+3600r^{-4}\right)^{-\frac{1}{2}}\left(2r-4\cdot3600r^{-5}\right)+\left(r^2+3600r^{-4}\right)^{\frac{1}{2}}\right]$$
$$= \pi\left(r^2+3600r^{-4}\right)^{-\frac{1}{2}}\left[r^2-7200r^{-4}+r^2+3600r^{-4}\right]$$
$$= \pi\left(r^2+3600r^{-4}\right)^{-\frac{1}{2}}\left[2r^2-3600r^{-4}\right]$$
$$= \frac{\pi}{\sqrt{r^2+3600r^{-4}}}\left(2r^2-\frac{3600}{r^4}\right)$$
$$A\frac{\mathrm{d}A}{\mathrm{d}r} = \pi r\sqrt{r^2+3600r^{-4}}\cdot\frac{\pi}{\sqrt{r^2+3600r^{-4}}}\left(2r^2-\frac{3600}{r^4}\right)$$
$$= \pi^2 r\left(2r^2-\frac{3600}{r^4}\right)$$
$$= \pi^2\left(2r^3-\frac{3600}{r^3}\right) \quad \text{(shown)}$$

**(c)**
**[5]**

At stationary point, $\frac{\mathrm{d}A}{\mathrm{d}r} = 0$

$$\pi^2\left(2r^3-\frac{3600}{r^3}\right)=0$$

$$2r^3-\frac{3600}{r^3}=0$$

$$2r^3=\frac{3600}{r^3}$$

$$r^6=1800$$

$$r=(1800)^{\frac{1}{6}}$$

To prove minimum:

**Method 1: Using 1st derivative test**

| $r$ | $\left((1800)^{\frac{1}{6}}\right)^-$ | $(1800)^{\frac{1}{6}}$ | $\left((1800)^{\frac{1}{6}}\right)^+$ |
|---|---|---|---|
| Sign of $\frac{\mathrm{d}A}{\mathrm{d}r}$ | $<0$ | 0 | $>0$ |
| Tangent | ＼ | — | ／ |

$r=(1800)^{\frac{1}{6}}$ gives the minimum $A$.

**Method 2: Using 2nd derivative test**

From the result $A\frac{\mathrm{d}A}{\mathrm{d}r}=\pi^2\left(2r^3-\frac{3600}{r^3}\right)$, using implicit differentiation to obtain:

$$A\frac{\mathrm{d}^2A}{\mathrm{d}r^2}+\left(\frac{\mathrm{d}A}{\mathrm{d}r}\right)^2=\pi^2\left(6r^2+\frac{10800}{r^4}\right)$$

At $\frac{\mathrm{d}A}{\mathrm{d}r}=0$ and $r=(1800)^{\frac{1}{6}}$, $\frac{\mathrm{d}^2A}{\mathrm{d}r^2}=\frac{\pi^2}{A}\left(6r^2+\frac{10800}{r^4}\right)>0$ (minimum)

$r=(1800)^{\frac{1}{6}}$ gives the minimum $A$.

**Method 3:**

$$\frac{r}{h}=\frac{r}{\left(\frac{60}{r^2}\right)}$$

$$=\frac{r^3}{60}$$

$$=\frac{\left(1800^{\frac{1}{6}}\right)^3}{60}$$

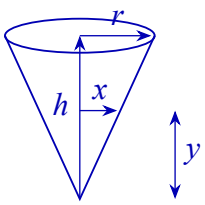
$$=\frac{\sqrt{1800}}{60}$$

$$=\frac{30\sqrt{2}}{60}$$

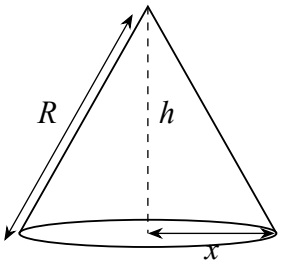
$$=\frac{\sqrt{2}}{2},\ \text{where } k=\frac{1}{2}$$

**Method 4:**

| | | |
|---|---|---|
| | $h = \frac{60}{r^2}$<br>$= \frac{60}{\left(1800^{\frac{1}{6}}\right)^2}$<br>$= \frac{60}{1800^{\frac{1}{3}}}$<br>$\frac{r}{h} = \frac{1800^{\frac{1}{6}}}{\left(\frac{60}{1800^{\frac{1}{3}}}\right)}$<br>$= \frac{\sqrt{1800}}{60}$<br>$= \frac{30\sqrt{2}}{60}$<br>$= \frac{\sqrt{2}}{2}$, where $k = \frac{1}{2}$ | |

| | | |
|---|---|---|
| **(d)**<br>**[3]** | Let $V$ be the volume of water in the cone when radius of water surface is $x$ cm and depth is $y$ cm.<br>By similar triangles,<br>  <br><br>$\frac{x}{y}=\frac{r}{h}=\frac{\sqrt{2}}{2}$<br>$x=\frac{\sqrt{2}}{2}y$<br>$V=\frac{1}{3}\pi x^2 y$<br>$=\frac{1}{3}\pi\left(\frac{\sqrt{2}}{2}y\right)^2 y$<br>$=\frac{1}{6}\pi y^3$<br>$\frac{\mathrm{d}V}{\mathrm{d}y}=\frac{1}{2}\pi y^2$<br>$\frac{\mathrm{d}V}{\mathrm{d}t}=\frac{\mathrm{d}V}{\mathrm{d}y}\times\frac{\mathrm{d}y}{\mathrm{d}t}$<br>$=\frac{1}{2}\pi y^2\times\frac{\mathrm{d}y}{\mathrm{d}t}$<br>When $y=2$,<br>$-3=\frac{1}{2}\pi(2)^2\times\frac{\mathrm{d}y}{\mathrm{d}t}$<br>$\frac{\mathrm{d}y}{\mathrm{d}t}=-\frac{3}{2\pi}$<br>Rate of decrease of depth of water is $\frac{3}{2\pi}$ cm/s or 0.477 cm/s (3 s.f.) | |

**9.8**




The diagram shows a right cone with a slant edge of fixed length $R$ cm, height $h$ cm and radius $x$ cm.

Show that $9V^2 = \pi^2 x^4 (R^2 - x^2)$, where $V$ is the volume of the cone.

Show further that, as $x$ varies, $V$ is a maximum when $x^2 = \frac{2}{3}R^2$. [7]

**[EJC 2018 Promo Q7]**

| Solution | | Comments |
|---|---|---|
| **[7]** | $h = \sqrt{R^2 - x^2}$<br>$V = \frac{1}{3}\pi x^2 h$<br>$= \frac{1}{3}\pi x^2 \sqrt{R^2 - x^2}$<br>$V^2 = \frac{1}{9}\pi^2 x^4 \left(\sqrt{R^2 - x^2}\right)^2$<br>$9V^2 = \pi^2 x^4 (R^2 - x^2)$ (shown)<br>Differentiate w.r.t. $x$,<br>$9V^2 = \pi^2 R^2 x^4 - \pi^2 x^6$<br>$18V\frac{\mathrm{d}V}{\mathrm{d}x} = 4\pi^2 R^2 x^3 - 6\pi^2 x^5$ — (*)<br>When the volume is maximum, $\frac{\mathrm{d}V}{\mathrm{d}x} = 0$<br>$-6\pi^2 x^5 + 4\pi^2 R^2 x^3 = 0$<br>$2\pi^2 x^3 (-3x^2 + 2R^2) = 0$<br>$-3x^2 + 2R^2 = 0$ or $x = 0$ (reject $\because x > 0$)<br>$x^2 = \frac{2}{3}R^2 \Rightarrow R^2 = \frac{3}{2}x^2$<br>**Check that $V$ is indeed a maximum:**<br><u>Second derivative method</u><br>Differentiating (*) wrt $x$<br>$18V\frac{\mathrm{d}^2V}{\mathrm{d}x^2} + 18\left(\frac{\mathrm{d}V}{\mathrm{d}x}\right)^2 = 12\pi^2 R^2 x^2 - 30\pi^2 x^4$<br>Substitute $x^2 = \frac{2}{3}R^2$, $\frac{\mathrm{d}V}{\mathrm{d}x} = 0$, | |

$$18V\frac{\mathrm{d}^2V}{\mathrm{d}x^2}+18(0)^2=12\pi^2R^2\left(\frac{2R^2}{3}\right)-30\pi^2\left(\frac{2R^2}{3}\right)^2$$

$$18V\frac{\mathrm{d}^2V}{\mathrm{d}x^2}=\frac{-40}{3}\pi^2R^4+8\pi^2R^4$$

$$18V\frac{\mathrm{d}^2V}{\mathrm{d}x^2}=-\frac{16}{3}\pi^2R^4$$

$$\frac{\mathrm{d}^2V}{\mathrm{d}x^2}=-\frac{16\pi^2R^4}{54V}<0$$

Therefore, $V$ is a maximum when $x^2=\frac{2}{3}R^2$.

<u>First derivative method</u>

From (*),

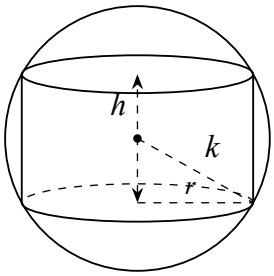
$$\frac{\mathrm{d}V}{\mathrm{d}x}=\frac{4\pi^2R^2x^3-6\pi^2x^5}{18V}=\frac{\pi^2x^3\,(2R^2-3x^2)}{9V}$$

(*Note to students*: Since $\frac{\pi^2x^3}{9V}>0$ for all $x$, we just need to consider $(2R^2-3x^2)$.)

| $x$ | $\left(\sqrt{\frac{2}{3}}R\right)^-$ | $\sqrt{\frac{2}{3}}R$ | $\left(\sqrt{\frac{2}{3}}R\right)^+$ |
|---|---|---|---|
| $\frac{\mathrm{d}V}{\mathrm{d}x}$ | $2R^2-3\left[\left(\sqrt{\frac{2}{3}}R\right)^-\right]^2$ $=2R^2-(2R^2)^->0$ | 0 | $2R^2-3\left[\left(\sqrt{\frac{2}{3}}R\right)^+\right]^2$ $=2R^2-(2R^2)^+<0$ |
| Slope | / | – | \ |

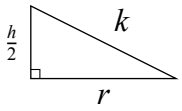
Therefore, $V$ is a maximum when $x^2=\frac{2}{3}R^2$.

**9.9**




A cylinder with height $h$ cm and base radius $r$ cm is inscribed within a sphere with fixed radius $k$ cm. The circumference of the circular top and bottom of the cylinder is in contact with the sphere (see figure above). By using differentiation, find, in terms of $k$, the exact value of $h$ for which the volume of the cylinder is maximum. [6]

**[TMJC 2024 Prelim P1 Q2]**

| **Solution** | | **Comments** |
|---|---|---|
| **[6]** | Let $V$ be the volume of the cylinder (in $\text{cm}^3$).<br>  <br><br>By Pythagoras Theorem,<br>$\left(\frac{h}{2}\right)^2 + r^2 = k^2$<br>$r^2 = k^2 - \frac{h^2}{4} \quad \text{— (1)}$<br>$V = \pi r^2 h$<br>$= \pi\left(k^2 - \frac{h^2}{4}\right)h \quad \text{,from (1)}$<br>$= \pi k^2 h - \frac{\pi}{4}h^3$<br>Differentiate w.r.t $h$,<br>$\frac{\mathrm{d}V}{\mathrm{d}h} = \pi k^2 - \frac{3\pi}{4}h^2$<br>For maximum $V$, $\frac{\mathrm{d}V}{\mathrm{d}h} = 0$<br>$\therefore \frac{\mathrm{d}V}{\mathrm{d}h} = \pi k^2 - \frac{3\pi}{4}h^2 = 0$<br>$k^2 - \frac{3}{4}h^2 = 0$<br>$h^2 = \frac{4k^2}{3}$<br>$h = \frac{2k}{\sqrt{3}} \quad (\because h > 0)$<br>**Testing $V$ is maximum**<br>**Method 1:** 1st derivative test | |

| $h$ | $\left(\frac{2k}{\sqrt{3}}\right)^-$ | $\frac{2k}{\sqrt{3}}$ | $\left(\frac{2k}{\sqrt{3}}\right)^+$ |
|---|---|---|---|
| $\frac{\mathrm{d}V}{\mathrm{d}h}$ | + | 0 | − |
| Slope | ／ | — | ＼ |

$\therefore V$ is maximum when $h=\frac{2k}{\sqrt{3}}$.

**Method 2:** 2nd derivative test

Differentiate w.r.t $h$,

$\frac{\mathrm{d}^2V}{\mathrm{d}h^2}=-\frac{3\pi}{2}h$

When $h=\frac{2k}{\sqrt{3}}$, $\frac{\mathrm{d}^2V}{\mathrm{d}h^2}=-\frac{3\pi}{2}\left(\frac{2k}{\sqrt{3}}\right)=-\sqrt{3}\pi k<0 \quad (\because k>0)$

$\therefore V$ is maximum when $h=\frac{2k}{\sqrt{3}}$.

# 10 Power Series

**10.1** It is given that $y = \ln \sqrt{1+e^{3x}}$.

**(a)** Show that $2e^{2y}\dfrac{dy}{dx} = 3e^{3x}$. [2]

**(b)** By further differentiation of the result in part **(a)**, find the Maclaurin series for $y$, up to and including the term in $x^2$, giving the coefficients in exact form. [4]

**(c)** Deduce the Maclaurin series for $\ln\left(\dfrac{\sqrt{1+e^{-3x}}}{2}\right)$, up to and including the term in $x^2$. [2]

**[TJC 2025 Prelim P2 Q1]**

| Solution | | Comments |
|---|---|---|
| **(a)** **[2]** | $y=\frac{1}{2}\ln(1+e^{3x})$<br>$e^{2y}=1+e^{3x}$<br>Diff. wrt $x$,<br>$2e^{2y}\frac{dy}{dx}=3e^{3x}$ (Shown) $-(1)$<br>Alternative solution:<br>$y=\ln\sqrt{1+e^{3x}}$<br>$\frac{dy}{dx}=\frac{3e^{3x}(1+e^{3x})^{-\frac{1}{2}}}{2(1+e^{3x})^{\frac{1}{2}}}=\frac{3e^{3x}}{2(1+e^{3x})}$<br>$2(1+e^{3x})\frac{dy}{dx}=3e^{3x}$<br>$2e^{2y}\frac{dy}{dx}=3e^{3x}$<br>$\left(\because y=\ln\sqrt{1+e^{3x}}=\frac{1}{2}\ln(1+e^{3x})\right.$ i.e. $\left.2y=\ln(1+e^{3x})\Rightarrow e^{2y}=1+e^{3x}\right)$ | |
| **(b)** **[4]** | Diff. wrt $x$,<br>$2e^{2y}\frac{d^2y}{dx^2}+4e^{2y}\left(\frac{dy}{dx}\right)^2=9e^{3x}$ —- (2)<br>When $x=0, y=\frac{1}{2}\ln(1+1)=\frac{1}{2}\ln 2$, $e^{2y}=e^{\ln 2}=2$<br>Sub into (1): $\frac{dy}{dx}=\frac{3}{4}$<br>Sub into (2): $2(2)\frac{d^2y}{dx^2}+8\left(\frac{3}{4}\right)^2=9 \Rightarrow \frac{d^2y}{dx^2}=\frac{9}{8}$<br>$y=\frac{1}{2}\ln 2+\frac{3}{4}x+\frac{9}{16}x^2+\ldots$ | |
| **(c)** **[2]** | $\ln\left(\frac{\sqrt{1+e^{-3x}}}{2}\right)=\ln\left(\sqrt{1+e^{3(-x)}}\right)-\ln 2$<br>$=-\ln 2+\frac{1}{2}\ln 2+\frac{3}{4}(-x)+\frac{9}{16}(-x)^2+\ldots$<br>$=-\frac{1}{2}\ln 2-\frac{3}{4}x+\frac{9}{16}x^2+\ldots$ | |

**10.2** It is given that $y = \mathrm{e}^{\sin^{-1}(3x)}$, where $-\frac{1}{3} < x < \frac{1}{3}$.

**(a)** Show that $(1-9x^2)\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - 9x\frac{\mathrm{d}y}{\mathrm{d}x} - 9y = 0$. [3]

**(b)** By differentiating the above result, find the Maclaurin series for $y$, up to and including the term in $x^3$. [4]

**(c)** Using your result in part **(b)** and the standard series in the List of Formulae (MF27), obtain the Maclaurin series for $\sin^{-1}(3x)$, up to and including the term in $x^3$. [3]

**[YIJC 2026 Prelim P1 Q10]**

| Solution | | Comments |
|---|---|---|
| **(a)** **[3]** | **EITHER** $y = e^{\sin^{-1}(3x)}$ $\frac{dy}{dx} = 3 \cdot \frac{1}{\sqrt{1-9x^2}} e^{\sin^{-1}(3x)}$ $= \frac{3y}{\sqrt{1-9x^2}}$ $\sqrt{1-9x^2}\frac{dy}{dx} = 3y$ **OR** $y = e^{\sin^{-1}(3x)}$ $\ln y = \sin^{-1}(3x)$ Differentiating w.r.t. $x$, $\frac{1}{y}\frac{dy}{dx} = \frac{1}{\sqrt{1-9x^2}} \cdot 3$ $\sqrt{1-9x^2}\frac{dy}{dx} = 3y$ **THEN** **EITHER** Differentiating with respect to $x$, $\sqrt{1-9x^2}\frac{d^2y}{dx^2} + \frac{1}{2}(1-9x^2)^{-\frac{1}{2}}(-18x)\frac{dy}{dx} = 3\frac{dy}{dx}$ Multiplying throughout by $\sqrt{1-9x^2}$, $(1-9x^2)\frac{d^2y}{dx^2} - 9x\frac{dy}{dx} = 3\sqrt{1-9x^2}\frac{dy}{dx}$ $= 3(3y)$ $(1-9x^2)\frac{d^2y}{dx^2} - 9x\frac{dy}{dx} - 9y = 0$ (shown) — (1) **OR** Squaring both sides, $(1-9x^2)\left(\frac{dy}{dx}\right)^2 = 9y^2$ Differentiating with respect to $x$, $(1-9x^2)\cdot 2\left(\frac{dy}{dx}\right)\frac{d^2y}{dx^2} - 18x\left(\frac{dy}{dx}\right)^2 = 18y\frac{dy}{dx}$ Dividing throughout by $2\frac{dy}{dx}$, $(1-9x^2)\frac{d^2y}{dx^2} - 9x\left(\frac{dy}{dx}\right) = 9y$ $(1-9x^2)\frac{d^2y}{dx^2} - 9x\frac{dy}{dx} - 9y = 0$ (shown) | |
| **(b)** **[4]** | Differentiating with respect to $x$, | |

| | | |
|---|---|---|
| **(c)**<br>**[3]** | $(1-9x^2)\dfrac{d^3y}{dx^3}-18x\dfrac{d^2y}{dx^2}-9x\dfrac{d^2y}{dx^2}-9\dfrac{dy}{dx}-9\dfrac{dy}{dx}=0$<br>$(1-9x^2)\dfrac{d^3y}{dx^3}-27x\dfrac{d^2y}{dx^2}-18\dfrac{dy}{dx}=0 \quad \text{— (2)}$<br>When $x=0, y=1.\ \dfrac{dy}{dx}=3$<br>From (1): $\dfrac{d^2y}{dx^2}-0-9=0\Rightarrow\dfrac{d^2y}{dx^2}=9$<br>From (2): $\dfrac{d^3y}{dx^3}-0-18(3)=0\Rightarrow\dfrac{d^3y}{dx^3}=54$<br>$\therefore y=1+3x+\dfrac{9}{2!}x^2+\dfrac{54}{3!}x^3+\ldots$<br>$=1+3x+\dfrac{9}{2}x^2+9x^3+\ldots$<br>$e^{\sin^{-1}(3x)}=1+3x+\dfrac{9}{2}x^2+9x^3+\ldots$<br>$\sin^{-1}(3x)\approx\ln\left(1+3x+\dfrac{9}{2}x^2+9x^3\right)$<br>$\approx\left(3x+\dfrac{9}{2}x^2+9x^3\right)-\dfrac{1}{2}\left(3x+\dfrac{9}{2}x^2\right)^2+\dfrac{1}{3}(3x)^3$<br>$\approx 3x+\dfrac{9}{2}x^2+9x^3-\dfrac{1}{2}(9x^2+27x^3)+\dfrac{1}{3}(27x^3)$<br>$=3x+\dfrac{9}{2}x^3$ | |

**10.3** It is given that $\mathrm{f}(x)=\dfrac{1}{\sqrt{1-4x^2}}$ where $-\dfrac{1}{2}<x<\dfrac{1}{2}$.

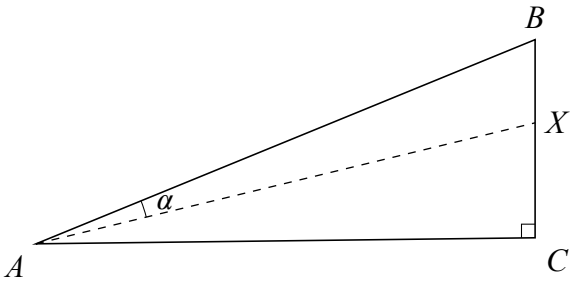
**(a)** Using standard series from the List of Formulae (MF27), find the Maclaurin's expansion of $\mathrm{f}(x)$, up to and including the term in $x^6$. [2]

**(b)** Hence find the first four non-zero terms of the Maclaurin series for $\sin^{-1}2x$. [4]

**[ACJC 2025 Prelim P2 Q2]**

| Solution | | Comments |
|---|---|---|
| **(a)** **[2]** | $\mathrm{f}(x)=\dfrac{1}{\sqrt{1-4x^2}}=(1-4x^2)^{-\frac{1}{2}}$<br>$=1-\dfrac{1}{2}(-4x^2)+\dfrac{-\frac{1}{2}\left(-\frac{3}{2}\right)}{2!}(-4x^2)^2+\dfrac{-\frac{1}{2}\left(-\frac{3}{2}\right)\left(-\frac{5}{2}\right)}{3!}(-4x^2)^3+\ldots$<br>$=1+2x^2+6x^4+20x^6+\ldots$ | |
| **(b)** **[4]** | $\displaystyle\int\frac{1}{\sqrt{1-4x^2}}\,\mathrm{d}x=\int 1+2x^2+6x^4+20x^6\,\mathrm{d}x$<br>$\dfrac{1}{2}\sin^{-1}2x=x+\dfrac{2}{3}x^3+\dfrac{6}{5}x^5+\dfrac{20}{7}x^7+c$<br>$\sin^{-1}2x=2x+\dfrac{4}{3}x^3+\dfrac{12}{5}x^5+\dfrac{40}{7}x^7+d$<br>$\sin^{-1}0=0 \quad \therefore d=0$<br>$\sin^{-1}2x=2x+\dfrac{4}{3}x^3+\dfrac{12}{5}x^5+\dfrac{40}{7}x^7$ | |

**10.4** In the right-angled triangle $ABC$, angle $C$ is a right angle. $AB = 13$ cm, $BC = 5$ cm and $X$ is a point on $BC$ such that angle $BAX$ is $\alpha$ radians.




**(a)** Find the exact value of $\cos \angle BAC$. [1]

**(b)** By considering triangle $AXC$, or otherwise, show that $AX = \dfrac{156}{12\cos\alpha + 5\sin\alpha}$. [3]

**(c)** Given that $\alpha$ is a sufficiently small angle, show that

$$AX \approx 13 + p\alpha + q\alpha^2,$$

where $p$ and $q$ are constants to be determined exactly. [4]

**[EJC 2025 Prelim P1 Q3]**

| Solution | | Comments |
|---|---|---|
| **(a)** **[1]** | $AC = \sqrt{13^2 - 5^2} = 12$<br>$\cos \angle BAC = \frac{AC}{AB} = \frac{12}{13}$ | |
| **(b)** **[3]** | $AX = \frac{12}{\cos(\angle BAC - \alpha)}$<br>$= \frac{12}{\cos \angle BAC \cos \alpha + \sin \angle BAC \sin \alpha}$<br>$= \frac{12}{\frac{12}{13}\cos \alpha + \frac{5}{13}\sin \alpha}$<br>$= \frac{156}{12\cos \alpha + 5\sin \alpha}$ (shown) | |
| **(c)** **[4]** | Using small angle approximation,<br>$AX \approx \frac{156}{12\left(1 - \frac{\alpha^2}{2}\right) + 5\alpha}$<br>$= 156\left[12\left(1 + \frac{5}{12}\alpha - \frac{\alpha^2}{2}\right)\right]^{-1}$<br>$= 156[12]^{-1}\left[1 + \left(\frac{5}{12}\alpha - \frac{\alpha^2}{2}\right)\right]^{-1}$<br>$= 13\left[1 - \left(\frac{5}{12}\alpha - \frac{\alpha^2}{2}\right) + \left(\frac{5}{12}\alpha - \frac{\alpha^2}{2}\right)^2 + \ldots\right]$<br>$= 13\left[1 - \frac{5}{12}\alpha + \frac{\alpha^2}{2} + \left(\frac{5}{12}\alpha\right)^2 + \ldots\right]$<br>$\approx 13 - \frac{65}{12}\alpha + \frac{1261}{144}\alpha^2$<br>$p = -\frac{65}{12}, \quad q = \frac{1261}{144}$ (shown) | |

**10.5** It is given that $\mathrm{f}(x) = \dfrac{1}{4+9x^2}$.

**(i)** Find $\int \mathrm{f}(x)\,\mathrm{d}x$. [2]

**(ii)** Find the binomial expansion for $\mathrm{f}(x)$, up to and including the term in $x^4$. Give the coefficients as exact fractions in their simplest form. [2]

**(iii)** Hence, find the Maclaurin series for $\tan^{-1}\dfrac{3x}{2}$. Give the coefficients as exact fractions in their simplest form. [3]

**(iv)** Use your series from part **(iii)** to estimate $\displaystyle\int_0^{0.5} \tan^{-1}\frac{3x}{2}\,\mathrm{d}x$, correct to 3 decimal places. [1]

**(v)** Use your calculator to find $\displaystyle\int_0^{0.5} \tan^{-1}\frac{3x}{2}\,\mathrm{d}x$, correct to 3 decimal places. [1]

**(vi)** Comparing your answers to parts **(iv)** and **(v)**, comment on the accuracy of your estimate in **(iv)** and how it can be improved. [2]

**[JPJC 2025 Prelim P1 Q9]**

| Solution | | Comments |
|---|---|---|
| **(i)** **[2]** | $\int \frac{1}{4+9x^2}\,\mathrm{d}x = \int \frac{1}{9\left(\frac{4}{9}+x^2\right)}\,\mathrm{d}x = \frac{1}{6}\tan^{-1}\frac{3x}{2}+K$ | |
| **(ii)** **[2]** | $\mathrm{f}(x) = (4+9x^2)^{-1} = \frac{1}{4}\left(1+\frac{9x^2}{4}\right)^{-1}$<br>$= \frac{1}{4}\left[1+(-1)\left(\frac{9x^2}{4}\right)+\frac{-1(-2)}{2}\left(\frac{9x^2}{4}\right)^2+\ldots\right]$<br>$= \frac{1}{4}\left(1-\frac{9x^2}{4}+\frac{81}{16}x^4+\ldots\right)$<br>$\approx \frac{1}{4}-\frac{9}{16}x^2+\frac{81}{64}x^4$ | |
| **(iii)** **[3]** | From **(i)** $\tan^{-1}\frac{3x}{2} = 6\int \frac{1}{4+9x^2}\,\mathrm{d}x + C, \quad$ where $C=-6K$<br>From **(ii)** $\tan^{-1}\frac{3x}{2} = 6\int\left(\frac{1}{4}-\frac{9}{16}x^2+\frac{81}{64}x^4\right)\mathrm{d}x+C$<br>$= 6\left(\frac{1}{4}x-\frac{3}{16}x^3+\frac{81}{320}x^5\right)+D$<br>$= \frac{3}{2}x-\frac{9}{8}x^3+\frac{243}{160}x^5+D$<br>$x=0,\ \tan^{-1}0=0\Rightarrow D=0$<br>$\therefore\ \tan^{-1}\frac{3x}{2} = \frac{3}{2}x-\frac{9}{8}x^3+\frac{243}{160}x^5$ | |
| **(iv)** **[1]** | $\int_0^{0.5}\tan^{-1}\frac{3x}{2}\,\mathrm{d}x = \int_0^{0.5}\left(\frac{3}{2}x-\frac{9}{8}x^3+\frac{243}{160}x^5\right)\mathrm{d}x$<br>$= 0.174$ (to 3 dp) (Using GC) | |
| **(v)** **[1]** | From GC, $\int_0^{0.5}\tan^{-1}\frac{3x}{2}\,\mathrm{d}x = 0.173$ (to 3 dp) | |
| **(vi)** **[2]** | The estimate in **(iv)** is accurate up to 2 decimal places but not to 3 decimal places.<br>To improve the estimate, we can include higher-order terms in the Maclaurin series expansion of $\tan^{-1}\frac{3x}{2}$. | |

**10.6** It is given that $y = \ln(\cos x)$.

**(i)** Prove that $\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + \left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2 + 1 = 0$. [2]

**(ii)** By differentiating further, find the first two non-zero terms of the expansion of $y$. [4]

**(iii)** Using the answer in **(ii)** and an appropriate expansion from MF26 up to and including the term in $x^4$, find an approximate value for $\int_0^{0.08} \ln\left(\frac{\cos^3 x}{1-x}\right) \mathrm{d}x$, leaving your answer correct to 7 decimal places. [2]

The calculated value of $\int_0^{0.08} \ln\left(\frac{\cos^3 x}{1-x}\right) \mathrm{d}x$ is given to be 0.0030328 correct to 7 decimal places.

**(iv)** Comparing the calculated value of $\int_0^{0.08} \ln\left(\frac{\cos^3 x}{1-x}\right) \mathrm{d}x$ given above to your answer in **(iii)**, and with reference to the value of $x$, comment on the accuracy of your approximation. [2]

**[SAJC 2024 Prelim P2 Q4]**

| Solution | | Comments |
|---|---|---|
| **(i)** **[2]** | $y=\ln(\cos x)$ —- (1)<br>Therefore, $e^y=\cos x$<br>Differentiate with respect to $x$,<br>$e^y\left(\frac{dy}{dx}\right)=-\sin x$ —- (2)<br>Differentiate with respect to $x$,<br>$e^y\left(\frac{d^2y}{dx^2}\right)+\left(\frac{dy}{dx}\right)^2 e^y=-\cos x$<br>$e^y\left(\frac{d^2y}{dx^2}\right)+\left(\frac{dy}{dx}\right)^2 e^y=-e^y$<br>$\frac{d^2y}{dx^2}+\left(\frac{dy}{dx}\right)^2=-1$ —- (3)<br>$\frac{d^2y}{dx^2}+\left(\frac{dy}{dx}\right)^2+1=0$ (shown)<br><u>Alternative Method</u><br>$y=\ln(\cos x)$<br>$\frac{dy}{dx}=\frac{1}{\cos x}(-\sin x)$<br>$=-\frac{\sin x}{\cos x}$<br>$\frac{d^2y}{dx^2}=-\frac{\cos x(\cos x)-\sin x(-\sin x)}{\cos^2 x}$<br>$=-\frac{\cos^2 x+\sin^2 x}{\cos^2 x}$<br>$=-1-\frac{\sin^2 x}{\cos^2 x}$<br>$=-1-\left(\frac{dy}{dx}\right)^2$<br>$\frac{d^2y}{dx^2}+\left(\frac{dy}{dx}\right)^2=-1$<br>$\frac{d^2y}{dx^2}+\left(\frac{dy}{dx}\right)^2+1=0$ (shown) | |
| **(ii)** **[4]** | Differentiate with respect to $x$,<br>$\frac{d^3y}{dx^3}+2\left(\frac{dy}{dx}\right)\left(\frac{d^2y}{dx^2}\right)=0$ (shown) —- (4)<br>Differentiate with respect to $x$,<br>$\frac{d^4y}{dx^4}+2\left[\frac{d^2y}{dx^2}\cdot\frac{d^2y}{dx^2}+\frac{d^3y}{dx^3}\cdot\frac{dy}{dx}\right]=0$<br>$\frac{d^4y}{dx^4}+2\left(\frac{dy}{dx}\right)\left(\frac{d^3y}{dx^3}\right)+2\left(\frac{d^2y}{dx^2}\right)^2=0$ —- (5)<br>Let $y=\ln(\cos x)=f(x)$ | |

When $x=0$,

From (1), $y=f(0)=\ln(\cos 0)=0$

From (2), $\frac{dy}{dx}=f'(0)=0$

From (3), $\frac{d^2y}{dx^2}=f''(0)=-(0)^2-1=-1$

From (4), $\frac{d^3y}{dx^3}=f'''(0)=-2(0)\,x\,(-1)=0$

From (5), $\frac{d^4y}{dx^4}=f^4(0)=-2\left[(-1)^2+(0)-(0)\right]=-2$

Using Maclaurin's expansion,

$$f(x)=f(0)+f'(0)x+\frac{f''(0)}{2!}x^2+\frac{f'''(0)}{3!}x^3+\frac{f^4(0)}{4!}x^4+\ldots$$

$$=-\frac{1}{2}x^2-\frac{1}{12}x^4+\ldots$$

$$\ln(\cos x)=-\frac{1}{2}x^2-\frac{1}{12}x^4+\ldots$$

**(iii)** **[2]**

$$\int_0^{0.08}\ln\left(\frac{\cos^3 x}{1-x}\right)dx$$

$$=\int_0^{0.08}\left[\ln(\cos x)^3-\ln(1-x)\right]dx$$

$$=\int_0^{0.08}\left[3\ln(\cos x)-\ln(1-x)\right]dx$$

$$\approx\int_0^{0.08}\left[3\left(-\frac{1}{2}x^2-\frac{1}{12}x^4\right)-\left(-x-\frac{x^2}{2}-\frac{x^3}{3}-\frac{x^4}{4}\right)\right]dx$$

$=0.0030327$ (to 7 decimal places)

**(iv)** **[2]**

The two values are the same up to 6 decimal places, and the $x$ value is between 0 and 0.08, which is relatively small and close to 0. Hence, the approximation is relatively accurate for the integral.

# 11 Integration

**11.1 (a)** Find $\int \frac{5-4x}{\sqrt{1+2x-x^2}}\,\mathrm{d}x$. [3]

**[VJC 2025 Promo Q7(a)]**

| Solution | | Comments |
|---|---|---|
| **(a)**<br>**[3]** | $\int \frac{5-4x}{\sqrt{1+2x-x^2}}\,\mathrm{d}x$<br>$=\int 2\,(2-2x)\,(1+2x-x^2)^{-\frac{1}{2}}\,\mathrm{d}x+\int \frac{1}{\sqrt{1+2x-x^2}}\,\mathrm{d}x$<br>$=2\left(\frac{1}{\frac{1}{2}}\right)(1+2x-x^2)^{\frac{1}{2}}+\int \frac{1}{\sqrt{2-(x-1)^2}}\,\mathrm{d}x$<br>$=4\sqrt{1+2x-x^2}+\sin^{-1}\left(\frac{x-1}{\sqrt{2}}\right)+C$ | Spilt the numerator into $kf'(x)+m$ allows us to use standard form $\int f'(x)\,[f(x)]^n\,\mathrm{d}x$ and from MF 27 $\int \frac{1}{\sqrt{a^2-x^2}}\,\mathrm{d}x$.<br><br>Must write $+C$ |

**11.2 (b) (i)** Use the substitution $x=\tan\theta$ to show that

$$\int \frac{x^5}{(x^2+1)^4}\,\mathrm{d}x=\int \sin^m\theta\cos^n\theta\,\mathrm{d}\theta,$$

where $m$ and $n$ are integers to be determined. [2]

**(ii)** Hence, find the exact value of $\int_0^{\sqrt{3}} \frac{x^5}{(x^2+1)^4}\,\mathrm{d}x$. [3]

**[VJC 2025 Promo Q7(b)]**

| Solution | | Comments |
|---|---|---|
| **(b)(i)** **[2]** | $x=\tan\theta \quad\Rightarrow\quad \frac{\mathrm{d}x}{\mathrm{d}\theta}=\sec^2\theta$<br>$\int \frac{x^5}{(x^2+1)^4}\,\mathrm{d}x=\int \frac{\tan^5\theta}{(\tan^2\theta+1)^4}\sec^2\theta\,\mathrm{d}\theta$<br>$=\int \frac{\tan^5\theta}{\sec^8\theta}\sec^2\theta\,\mathrm{d}\theta$<br>$=\int \frac{\left(\frac{\sin^5\theta}{\cos^5\theta}\right)}{\left(\frac{1}{\cos^6\theta}\right)}\mathrm{d}\theta$<br>$=\int \sin^5\theta\cos\theta\,\mathrm{d}\theta$ | |
| **(b)(ii)** **[3]** | When $x=\sqrt{3}$, $\sqrt{3}=\tan\theta \Rightarrow \theta=\frac{\pi}{3}$<br>When $x=0$, $0=\tan\theta \Rightarrow \theta=0$<br>$\int_0^{\sqrt{3}} \frac{x^5}{(x^2+1)^4}\,\mathrm{d}x=\int_0^{\frac{\pi}{3}} \sin^5\theta\cos\theta\,\mathrm{d}\theta$<br>$=\frac{1}{6}\left[\sin^6\theta\right]_0^{\frac{\pi}{3}}$<br>$=\frac{1}{6}\left(\frac{\sqrt{3}}{2}\right)^6$<br>$=\frac{9}{128}$ | Sufficient working must be shown to demonstrate that calculator is not used here since question ask for "exact" |

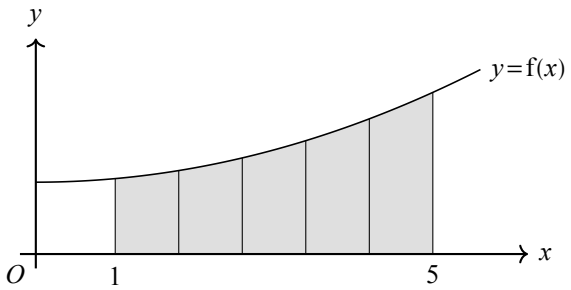
**11.3 (c)** Find $\int e^{-x}\cos 3x\,dx$. Hence, find the exact value of $\int_0^{\pi} e^{-x}\cos 3x\,dx$. [5]

**[YIJC 2024 Prelim P1 Q9(c)]**

| Solution | | Comments |
|---|---|---|
| **(c)**<br>**[5]** | $\int e^{-x}\cos 3x\,dx$<br>$= -e^{-x}\cos 3x - \int -e^{-x}(-3\sin 3x)\,dx$<br>$= -e^{-x}\cos 3x - 3\int e^{-x}\sin 3x\,dx$<br>$= -e^{-x}\cos 3x - 3\left[-e^{-x}\sin 3x - \int -e^{-x}(3\cos 3x)\,dx\right]$<br>$= -e^{-x}\cos 3x + 3e^{-x}\sin 3x - 9\int e^{-x}\cos 3x\,dx$<br>$\int e^{-x}\cos 3x\,dx = -e^{-x}\cos 3x + 3e^{-x}\sin 3x - 9\int e^{-x}\cos 3x\,dx$<br>$10\int e^{-x}\cos 3x\,dx = -e^{-x}\cos 3x + 3e^{-x}\sin 3x + C$<br>$\int e^{-x}\cos 3x\,dx = \frac{1}{10}e^{-x}(-\cos 3x + 3\sin 3x) + D$<br>$\int_0^{\pi} e^{-x}\cos 3x\,dx = \left[\frac{1}{10}e^{-x}(-\cos 3x + 3\sin 3x)\right]_0^{\pi}$<br>$= \frac{1}{10}e^{-\pi}(-\cos 3\pi + 3\sin 3\pi)$<br>$- \frac{1}{10}e^{0}(-\cos 3(0) + 3\sin 3(0))$<br>$= \frac{1}{10}e^{-\pi} + \frac{1}{10}$ | Remember to include the arbitrary constant once integration is completed |

# 12 Definite Integrals

**12.1** The diagram shows a sketch of the curve $y=\mathrm{f}(x)$. The region under the curve between $x=1$ and $x=5$, shown shaded in the diagram, is $A$. This region is split into 5 vertical stripes of equal width, $h$.




**(a)** State the value of $h$ and show, using a sketch, that $\sum_{n=0}^{4}(\mathrm{f}(1+nh))\,h$ is less than the area of $A$. [2]

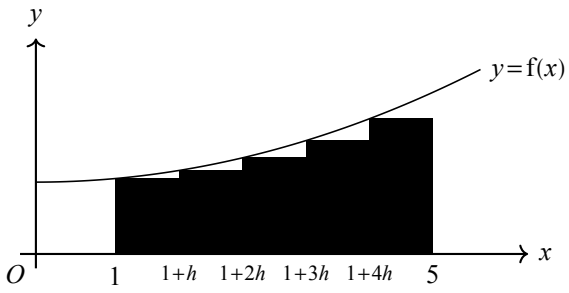
**(b)** Find a similar expression that is greater than the area of $A$. [1]

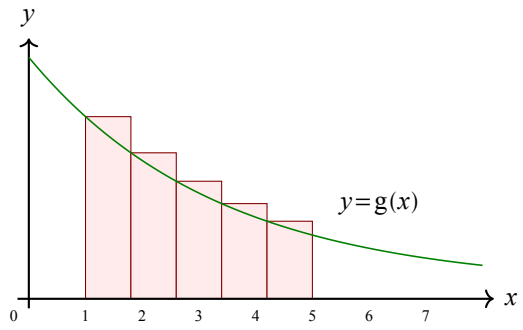
You are now given that $\mathrm{f}(x)=\frac{1}{20}x^2+1$.

**(c)** Use the expression given in part **(a)** and your expression from part **(b)** to find lower and upper bounds for the area of $A$. [2]

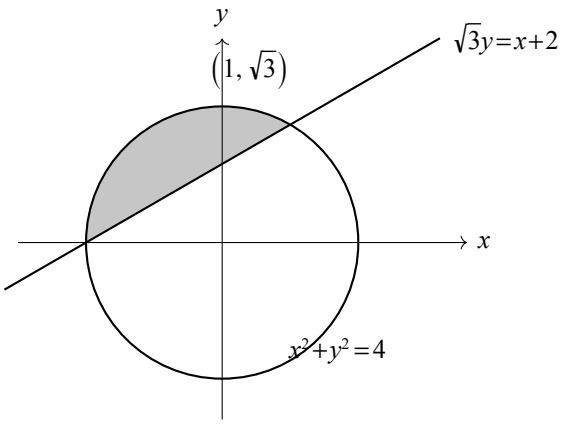
**(d)** Sketch the graph of a function $y=\mathrm{g}(x)$, between $x=1$ and $x=5$, for which the area between the curve, the $x$-axis and the lines $x=1$ and $x=5$ is less than $\sum_{n=0}^{4}(\mathrm{g}(1+nh))\,h$. [1]

**[2021 A-level P2 Q2]**

| Solution | | Comments |
|---|---|---|
| **(a)** **[2]** | The value of $h = \frac{5-1}{5} = \frac{4}{5}$.   [ Note that $\sum_{n=0}^{4} (\mathrm{f}(1+nh))\, h = h\mathrm{f}(1) + h\mathrm{f}(1+h) + h\mathrm{f}(1+2h) + h\mathrm{f}(1+3h) + h\mathrm{f}(1+4h)$ $h\mathrm{f}(1)$ represents the area of left-most rectangle in the diagram above. Similarly, $h\mathrm{f}(1+h)$ represents the area of second left-most rectangle in the diagram above, and $h\mathrm{f}(1+4h)$ represents the area of right-most rectangle in the diagram above. So, $\sum_{n=0}^{4} (\mathrm{f}(1+nh))\, h$ represents the sum of the area of the 5 shaded rectangles. Clearly, the total area of the rectangles is less than the area of $A$. ] | |
| **(b)** **[1]** | A similar expression is $\sum_{n=1}^{5} (\mathrm{f}(1+nh))\, h$ or $\sum_{n=0}^{4} (\mathrm{f}(1+(n+1)\, h))\, h$ | |
| **(c)** **[2]** | $\mathrm{f}(x) = \frac{1}{20}x^2 + 1$. Lower bound for area of $A = \sum_{n=0}^{4} (\mathrm{f}(1+nh))\, h$ $= \frac{4}{5}\sum_{n=0}^{4} \left( \frac{1}{20}\left(1+\frac{4}{5}n\right)^2 + 1 \right) = 5.608$ Upper bound for area of $A = \sum_{n=1}^{5} (\mathrm{f}(1+nh))\, h$ $= \frac{4}{5}\sum_{n=1}^{5} \left( \frac{1}{20}\left(1+\frac{4}{5}n\right)^2 + 1 \right)$ $= 6.568$ | Note that you can use the GC to evaluate the sum |

| | | |
|---|---|---|
| **(d)**<br>**[1]** |  | |

**12.2** Curve $C$ is a circle with radius 2 and center at the origin with equation $x^2+y^2=4$. Line $L$ has the equation $\sqrt{3}y=x+2$. The diagram below shows the shaded region $A$ which is enclosed between $C$ and $L$.

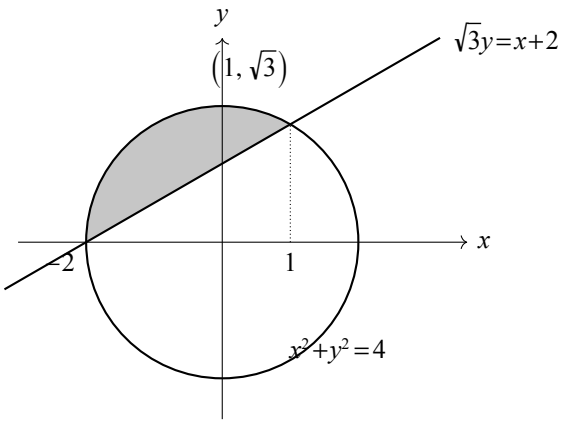



**(a)** Use the substitution $x=2\sin\theta$ to show that the area of region $A$ can be written in the expression $\int_b^a 4\cos^2\theta\,\mathrm{d}\theta - c$, where $a$, $b$ and $c$ are exact constants to be determined.

Hence evaluate this area exactly. [6]

**(b)** The region $B$ is bounded by the curve $C$, the line $L$ and the line $x=2$. Find the volume generated when region $B$ is rotated through $2\pi$ radians about the $y$-axis. Give your answer to two decimal places. [3]

**[ASRJC 2025 Prelim P1 Q7]**

| Solution | | Comments |
|---|---|---|
| **(a)**<br>**[6]** |  | |

The circle $x^2+y^2=4$ and line $\sqrt{3}y=x+2$ intersects at $x=-2$ and $x=1$

Area of region $A=\int_{-2}^{1}\sqrt{4-(x)^2}\,\mathrm{d}x-$ area of triangle

$$x=2\sin\theta \quad \Rightarrow \quad \frac{\mathrm{d}x}{\mathrm{d}\theta}=2\cos\theta$$

When $x=1 \Rightarrow \theta=\sin^{-1}\left(\frac{1}{2}\right)=\frac{\pi}{6}$

When $x=-2 \Rightarrow \theta=\sin^{-1}(-1)=-\frac{\pi}{2}$

$$=\int_{-\frac{\pi}{2}}^{\frac{\pi}{6}}\sqrt{4-(2\sin\theta)^2}\,(2\cos\theta)\,\mathrm{d}\theta-\frac{1}{2}\,(3)\,\sqrt{3}$$

$$=\int_{-\frac{\pi}{2}}^{\frac{\pi}{6}}(2\cos\theta)\,(2\cos\theta)\,\mathrm{d}\theta-\frac{3\sqrt{3}}{2}$$

$$=\int_{-\frac{\pi}{2}}^{\frac{\pi}{6}}4\cos^2\theta\,\mathrm{d}\theta-\frac{3\sqrt{3}}{2}\ \text{(shown)}$$

Where $a=\frac{\pi}{6}, b=-\frac{\pi}{2}, c=\frac{3\sqrt{3}}{2}$

$$4\int_{-\frac{\pi}{2}}^{\frac{\pi}{6}}\cos^2\theta\,\mathrm{d}\theta=4\int_{-\frac{\pi}{2}}^{\frac{\pi}{6}}\left(\frac{\cos 2\theta+1}{2}\right)\mathrm{d}\theta$$

$$=2\int_{-\frac{\pi}{2}}^{\frac{\pi}{6}}(\cos 2\theta+1)\,\mathrm{d}\theta$$

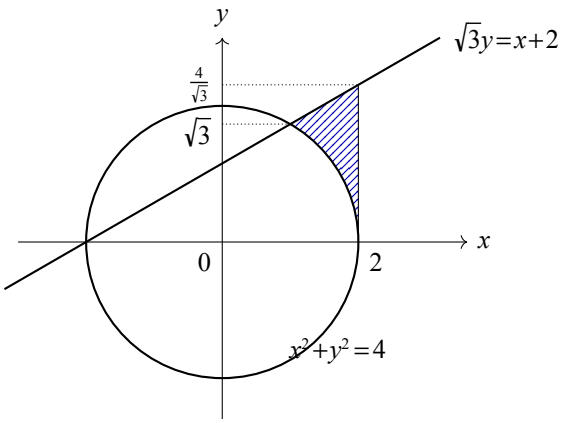
$$=2\left[\frac{\sin 2\theta}{2}+\theta\right]_{-\frac{\pi}{2}}^{\frac{\pi}{6}}$$

$$\text{Required area}=2\left[\left(\frac{1}{2}\sin 2\left(\frac{\pi}{6}\right)+\frac{\pi}{6}\right)-\left(\frac{1}{2}\sin 2\left(-\frac{\pi}{2}\right)-\frac{\pi}{2}\right)\right]-\frac{3\sqrt{3}}{2}$$

$$=\left(2\left(\frac{\sqrt{3}}{4}+\frac{\pi}{6}+\frac{\pi}{2}\right)\right)-\frac{3\sqrt{3}}{2}$$

$$=\frac{\sqrt{3}}{2}+\frac{4\pi}{3}-\frac{3\sqrt{3}}{2}=\frac{4\pi}{3}-\sqrt{3}$$

**(b)**
**[3]**




Volume obtained by rotating region $B$ through $2\pi$ radians about $y$-axis

$= \text{volume of cylinder} - \pi\int_{\sqrt{3}}^{\frac{4}{\sqrt{3}}} \left(\sqrt{3}y-2\right)^2 \mathrm{d}y - \pi\int_0^{\sqrt{3}} (4-y^2)\,\mathrm{d}y$

$= \pi(2)^2\left(\frac{4}{\sqrt{3}}\right) - \pi\int_{\sqrt{3}}^{\frac{4}{\sqrt{3}}} \left(\sqrt{3}y-2\right)^2 \mathrm{d}y - \pi\int_0^{\sqrt{3}} (4-y^2)\,\mathrm{d}y$

$\approx 8.4644$

$\approx 8.46$ (2 d.p.)

**12.3** **(a)** Differentiate $\sqrt{1-x^2}$ with respect to $x$.

Hence evaluate $\int_{-1}^{1} \cos^{-1} x \, \mathrm{d}x$, giving your answer in exact form. [5]

**(b)** The finite region $R$ is bounded by the curve $y = \cos^{-1}(x-3)$, where $2 \leqslant x \leqslant 4$, the line $y = \pi$ and the axes.

**(i)** Describe a geometrical transformation which will map the graph of $y = \cos^{-1}(x-3)$ onto the graph of $y = \cos^{-1} x$. [1]

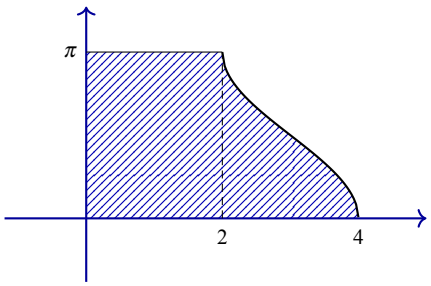
**(ii)** Hence or otherwise, find the area of $R$ in exact form. [2]

**(c)** Find the exact value of the volume generated when $R$ is rotated completely about the $y$-axis. [4]

**[TJC 2025 Prelim P1 Q10]**

| **Solution** | | **Comments** |
|---|---|---|
| **(a)** **[5]** | $\dfrac{\mathrm{d}}{\mathrm{d}x}\sqrt{1-x^2} = \dfrac{-2x}{2\sqrt{1-x^2}} = -\dfrac{x}{\sqrt{1-x^2}}$<br>$\int_{-1}^{1} \cos^{-1} x \, \mathrm{d}x = \left[x \cos^{-1} x\right]_{-1}^{1} - \int_{-1}^{1} x\left[-\dfrac{1}{\sqrt{1-x^2}}\right] \mathrm{d}x$<br>$= 1(0) - (-1)(\pi) - \int_{-1}^{1} \dfrac{1}{2} \cdot \dfrac{-2x}{\sqrt{1-x^2}} \, \mathrm{d}x$<br>$= \pi - \left[\sqrt{1-x^2}\right]_{-1}^{1}$<br>$= \pi - 0$<br>$= \pi$ | |
| **(b)(i)** **[1]** | Translate the graph of $y = \cos^{-1}(x-3)$ three units in the negative direction of $x$-axis to obtain the graph of $y = \cos^{-1} x$ | |

**(b)(ii)**
**[2]**




Area of the region $R = (2)\ (\pi) + \int_2^4 \cos^{-1}(x-3)\,\mathrm{d}x$

$= 2\pi + \int_{-1}^{1} \cos^{-1} u\,\mathrm{d}u$

$= 3\pi$

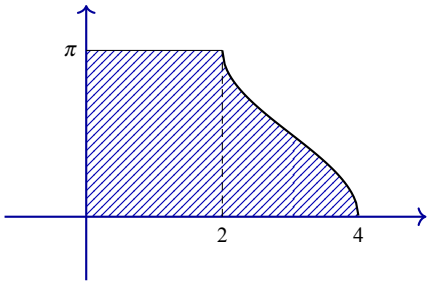
Alternative Method:

Area of the region $R = \int_0^{\pi} 3 + \cos y\,\mathrm{d}y$

$= \left[3x + \sin y\right]_0^{\pi}$

$= 3\pi$

**(c)**
**[4]**




Volume generated when rotated completely about $y$-axis

$= \pi \int_0^{\pi} x^2\,\mathrm{d}y$

$= \pi \int_0^{\pi} (\cos y + 3)^2\,\mathrm{d}y$

$= \pi \int_0^{\pi} (\cos^2 y + 6\cos y + 9)\,\mathrm{d}y$

$= \pi \int_0^{\pi} \left[\frac{1}{2}(\cos 2y + 1) + 6\cos y + 9\right]\mathrm{d}y$

$= \pi \left[\frac{1}{4}\sin 2y + 6\sin y + \frac{19}{2}y\right]_0^{\pi}$

$= \frac{19}{2}\pi^2$

# 13 Differential Equations

**13.1** **(a)** Express $\dfrac{1}{Q(500-Q)}$ in partial fractions. [2]

**(b)** A group of scientists is monitoring the population of a particular species of birds on an island. At time $t$ years after the start of the monitoring, the number of birds on the island is $Q$. The scientists observe that the birth rate is proportional to the bird's population and the birds are dying at a rate proportional to the square of the bird's population. There were 1000 birds on the island when the scientists first started monitoring the population. They discover that the population remains unchanged when there are 500 birds.

**(i)** Show that the differential equation relating $Q$ and $t$ is given by

$$\frac{\mathrm{d}Q}{\mathrm{d}t} = kQ(500-Q),$$

where $k$ is a positive constant. [2]

**(ii)** Hence, solve the differential equation, expressing $Q$ in terms of $k$ and $t$. [5]

**[HCI 2025 Prelim P1 Q10]**

| Solution | | Comments |
|---|---|---|
| **(a)** **[2]** | $\dfrac{1}{Q(500-Q)} = \dfrac{A}{Q} + \dfrac{B}{500-Q} = \dfrac{(500-Q)A+BQ}{Q(500-Q)}$<br>$\Rightarrow 1 = A(500-Q)+BQ$<br>For $Q=0$, $1=500A \Rightarrow A=\dfrac{1}{500}$.<br>For $Q=500$, $1=500B \Rightarrow B=\dfrac{1}{500}$.<br>$\therefore \dfrac{1}{Q(500-Q)} = \dfrac{1}{500Q} + \dfrac{1}{500(500-Q)}$ | |

**(b)(i)**
**[2]**

Increase rate $\propto Q$

$\Rightarrow$ Increase rate $= aQ$ ($a$ is the constant of proportionality for increase rate)

Decrease rate $\propto Q^2$

$\Rightarrow$ Increase rate $= bQ^2$ ($b$ is the constant of proportionality for decrease rate)

$$\therefore \frac{\mathrm{d}Q}{\mathrm{d}t} = aQ - bQ^2 \quad \text{(Increase rate – Decrease rate)}$$

$$\frac{\mathrm{d}Q}{\mathrm{d}t} = 0 \text{ when } Q = 500$$

$$\Rightarrow 500a - 250000b = 0$$

$$\Rightarrow a = 500b$$

$$\therefore \frac{\mathrm{d}Q}{\mathrm{d}t} = 500bQ - bQ^2$$

$$\Rightarrow \frac{\mathrm{d}Q}{\mathrm{d}t} = kQ\,(500 - Q) \quad \text{where } k = b$$

**(b)(ii)**
**[5]**

$$\int \frac{1}{Q\,(500-Q)}\,\mathrm{d}Q = \int k\,\mathrm{d}t$$

$$\frac{1}{500}\int \frac{1}{Q} + \frac{1}{500-Q}\,\mathrm{d}Q = \int k\,\mathrm{d}t \quad \text{(from part (a))}$$

$$\frac{1}{500}\left[\ \ln Q - \ln|500-Q|\ \right] = kt + C$$

$$\ln\left|\frac{Q}{500-Q}\right| = 500kt + 500C$$

**Resolving the modulus**:

**Method 1 (using arbitrary constant):**

$$\ln\left|\frac{Q}{500-Q}\right| = 500kt + 500C$$

$$\Rightarrow \left|\frac{Q}{500-Q}\right| = \mathrm{e}^{500kt+500C}$$

$$\Rightarrow \frac{Q}{500-Q} = \pm\mathrm{e}^{500C}\mathrm{e}^{500kt}$$

$$\Rightarrow \frac{Q}{500-Q} = H\mathrm{e}^{500kt}, \quad \text{where } H = \pm\mathrm{e}^{500C} \text{ is an arbitrary constant}$$

Therefore, we have

$$Q = 500H\mathrm{e}^{500kt} - QH\mathrm{e}^{500kt}$$

$$Q = \frac{500H\mathrm{e}^{500kt}}{1 + H\mathrm{e}^{500kt}}$$

When $t = 0$, $Q = 1000$

$$\frac{1000}{500-1000}=H \Rightarrow H=-2$$

$$\therefore Q=\frac{-1000\mathrm{e}^{500kt}}{1-2\mathrm{e}^{500kt}}$$

$$\Rightarrow Q=\frac{1000\mathrm{e}^{500kt}}{2\mathrm{e}^{500kt}-1}$$

**Method 2 (using the initial condition):**

When $t=0$, $Q=1000$, and $Q$ stabilized at 500 given in the question.

Therefore $Q \geq 500 \Rightarrow Q-500 \geq 0$

$$\Rightarrow \ln\left|\frac{Q}{500-Q}\right|=\ln\left(\frac{Q}{Q-500}\right)$$

Hence we have

$$\ln\left(\frac{Q}{Q-500}\right)=500kt+500C=500kt+D$$

When $t=0$, $Q=1000$

$$\ln\left(\frac{1000}{1000-500}\right)=0+D$$

$$\Rightarrow D=\ln 2$$

$$\ln\left(\frac{Q}{Q-500}\right)=500kt+\ln 2$$

$$\frac{Q}{Q-500}=\mathrm{e}^{500kt+\ln 2}$$

$$\frac{Q}{Q-500}=\mathrm{e}^{\ln 2}\mathrm{e}^{500kt}=2\mathrm{e}^{500kt}$$

$$Q=2\mathrm{e}^{500kt}(Q-500)$$

$$Q-2\mathrm{e}^{500kt}Q=-1000\mathrm{e}^{500kt}$$

$$Q=\frac{-1000\mathrm{e}^{500kt}}{1-2\mathrm{e}^{500kt}}$$

$$Q=\frac{1000\mathrm{e}^{500kt}}{2\mathrm{e}^{500kt}-1}$$

**13.2** It is given that $\left(2xy\frac{\mathrm{d}y}{\mathrm{d}x}+y^2\right)\cos x=\frac{1}{xy^2}$, where $0<x<\frac{\pi}{2}$ and $y\neq 0$.

**(a)** Using the substitution $u=xy^2$, show that the differential equation can be reduced to $\frac{\mathrm{d}u}{\mathrm{d}x}=\frac{\sec x}{u}$. [3]

**(b)** Hence find the general solution to the differential equation. [3]

**[HCI 2025 Prelim P1 Q3]**

| Solution | | Comments |
|---|---|---|
| **(a)** **[3]** | $u=xy^2$<br>$\frac{\mathrm{d}u}{\mathrm{d}x}=y^2+x\left(2y\frac{\mathrm{d}y}{\mathrm{d}x}\right)$<br>$\cos x\left(y^2+2xy\frac{\mathrm{d}y}{\mathrm{d}x}\right)=\frac{1}{xy^2}$<br>$\cos x\left(\frac{\mathrm{d}u}{\mathrm{d}x}\right)=\frac{1}{u}$<br>$\frac{\mathrm{d}u}{\mathrm{d}x}=\frac{\sec x}{u}$ | |
| **(b)** **[3]** | $\int u\,\mathrm{d}u=\int \sec x\,\mathrm{d}x$<br>$\frac{1}{2}u^2=\ln(\sec x+\tan x)+C \quad \left(\because \sec x, \tan x>0 \text{ for } 0<x<\frac{\pi}{2}\right)$<br>$x^2y^4=2\ln(\sec x+\tan x)+C$ | |

**13.3** An electrical circuit comprises a power source of $V$ volts in series with a resistance of $R$ ohms, a capacitance of $C$ farads and an inductance of $L$ henries. The current in the circuit, $t$ seconds after turning on the power, is $I$ amps and the charge on the capacitor is $q$ coulombs. The circuit can be used by scientists to investigate resonance, to model heavily damped motion and to tune into radio stations on a stereo tuner. It is given that $R$, $C$ and $L$ are constants, and that $I=0$ when $t=0$.

A differential equation for the circuit is $L\frac{\mathrm{d}I}{\mathrm{d}t}+RI+\frac{q}{C}=V$, where $I=\frac{\mathrm{d}q}{\mathrm{d}t}$.

**(i)** Show that, under certain conditions on $V$ which should be stated,

$$L\frac{\mathrm{d}^2I}{\mathrm{d}t^2}+R\frac{\mathrm{d}I}{\mathrm{d}t}+\frac{I}{C}=0.$$

[2]

It is now given that the differential equation in part (i) holds for the rest of the question.

**(ii)** Given that $I=At\mathrm{e}^{-\frac{Rt}{2L}}$ is a solution of the differential equation, where $A$ is a positive constant, show

[5]

**(iii)** In a particular circuit, $R=4$, $L=3$ and $C=0.75$. Find the maximum value of $I$ in terms of $A$, showi

[4]

**(iv)** Sketch the graph of $I$ against $t$. [2]

**[2018 A-level P1 Q10]**

| Solution | | Comments |
|---|---|---|
| **(i)** **[2]** | $L\frac{\mathrm{d}I}{\mathrm{d}t}+RI+\frac{q}{C}=V$, where $I=\frac{\mathrm{d}q}{\mathrm{d}t}$<br>Differentiate with respect to $t$,<br>$L\frac{\mathrm{d}^2I}{\mathrm{d}t^2}+R\frac{\mathrm{d}I}{\mathrm{d}t}+\frac{1}{C}\frac{\mathrm{d}q}{\mathrm{d}t}=\frac{\mathrm{d}V}{\mathrm{d}t}$<br>$L\frac{\mathrm{d}^2I}{\mathrm{d}t^2}+R\frac{\mathrm{d}I}{\mathrm{d}t}+\frac{I}{C}=\frac{\mathrm{d}V}{\mathrm{d}t}$, since $I=\frac{\mathrm{d}q}{\mathrm{d}t}$<br>Therefore,<br>$L\frac{\mathrm{d}^2I}{\mathrm{d}t^2}+R\frac{\mathrm{d}I}{\mathrm{d}t}+\frac{I}{C}=0$, when $\frac{\mathrm{d}V}{\mathrm{d}t}=0$ that is, $V$ is a constant. | |

**(ii)**
**[5]**

$I = At\mathrm{e}^{-\frac{Rt}{2L}}$

$$\frac{\mathrm{d}I}{\mathrm{d}t} = A\mathrm{e}^{-\frac{Rt}{2L}} + At\mathrm{e}^{-\frac{Rt}{2L}}\left(-\frac{R}{2L}\right) = A\mathrm{e}^{-\frac{Rt}{2L}} - \frac{AR}{2L}t\mathrm{e}^{-\frac{Rt}{2L}}$$

$$\frac{\mathrm{d}^2I}{\mathrm{d}t^2} = -\frac{AR}{2L}\mathrm{e}^{-\frac{Rt}{2L}} - \frac{AR}{2L}\mathrm{e}^{-\frac{Rt}{2L}} + \frac{AR^2}{4L^2}t\mathrm{e}^{-\frac{Rt}{2L}} = -\frac{AR}{L}\mathrm{e}^{-\frac{Rt}{2L}} + \frac{AR^2}{4L^2}t\mathrm{e}^{-\frac{Rt}{2L}}$$

Sub into $L\frac{\mathrm{d}^2I}{\mathrm{d}t^2} + R\frac{\mathrm{d}I}{\mathrm{d}t} + \frac{I}{C} = 0$,

$$-AR\mathrm{e}^{-\frac{Rt}{2L}} + \frac{AR^2}{4L}t\mathrm{e}^{-\frac{Rt}{2L}} + AR\mathrm{e}^{-\frac{Rt}{2L}} - \frac{AR^2}{2L}t\mathrm{e}^{-\frac{Rt}{2L}} + \frac{A}{C}t\mathrm{e}^{-\frac{Rt}{2L}} = 0$$

$$-AR\mathrm{e}^{-\frac{Rt}{2L}} + AR\mathrm{e}^{-\frac{Rt}{2L}} + \frac{AR^2}{4L}t\mathrm{e}^{-\frac{Rt}{2L}} - \frac{AR^2}{2L}t\mathrm{e}^{-\frac{Rt}{2L}} + \frac{A}{C}t\mathrm{e}^{-\frac{Rt}{2L}} = 0$$

$$-\frac{R^2}{4L} + \frac{1}{C} = 0 \Rightarrow C = \frac{4L}{R^2}$$

**(iii)**
**[4]**

Note that the given values satisfy (ii),

when $R = 4$ and $L = 3$, from (ii), $C = \frac{4L}{R^2} = \frac{3}{4}$

$I = At\mathrm{e}^{-\frac{2}{3}t}$

$$\frac{\mathrm{d}I}{\mathrm{d}t} = 0 \Rightarrow A\mathrm{e}^{-\frac{2}{3}t} - \frac{2A}{3}t\mathrm{e}^{-\frac{2}{3}t} = 0$$

$$\Rightarrow t = \frac{3}{2}$$

$$I = \frac{3A}{2\mathrm{e}}$$

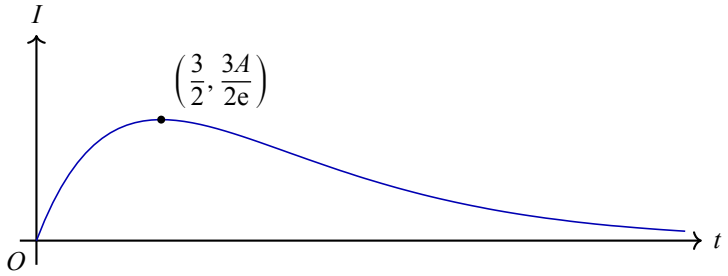
Sub into $L\frac{\mathrm{d}^2I}{\mathrm{d}t^2} + R\frac{\mathrm{d}I}{\mathrm{d}t} + \frac{I}{C} = 0$,

$$3\frac{\mathrm{d}^2I}{\mathrm{d}t^2} + 4\,(0) + \frac{1}{0.75}\cdot\frac{3A}{2\mathrm{e}} = 0$$

$$\frac{\mathrm{d}^2I}{\mathrm{d}t^2} = -\frac{A}{1.5\mathrm{e}} < 0$$

By right, we should sub $R = 4$, $L = 3$ and $C = 0.75$ into $I$ and check that it still a solution of the DE.

**(iv)**
**[2]**




Remember to label the max point $\left(\frac{3}{2}, \frac{3A}{2\mathrm{e}}\right)$

**13.4** The rate of decrease of atmospheric pressure $P$ Pascals (Pa) with respect to altitude $h$ kilometres above sea level, is proportional to $\frac{P}{T}$, where $T$ is the temperature measured in Kelvin (K). The temperature $T$ decreases linearly by 6.5 K for every one-kilometre increase in the altitude above sea level. It is given that the temperature is 293 K at sea level.

**(i)** Find a differential equation for $\frac{\mathrm{d}P}{\mathrm{d}h}$ in terms of $P$ and $h$. [2]

**(ii)** Given that $P>0$, solve this model in part (i) to show that

$$P=A\,|293-6.5h|^{b}, \text{ where } A \text{ and } b \text{ are constants.}$$

Explain whether $b$ is positive or negative. [4]

It is given that the atmospheric pressures are 101 300 Pa at sea level and 80 000 Pa at $h=2$.

**(iii)** Find the atmospheric pressure at the top of Mount Everest with an altitude of 8848 metres. [4]

**(iv)** For $h\geq 0$, sketch the graph of $P$ against $h$. Explain a limitation of this model. [3]

**[NJC 2025 Prelim P2 Q4]**

| Solution | | Comments |
|---|---|---|
| **(i)** **[2]** | $\frac{\mathrm{d}P}{\mathrm{d}h}=-k\left(\frac{P}{T}\right)$<br>$T=293-6.5h$<br>$\frac{\mathrm{d}P}{\mathrm{d}h}=\frac{-kP}{293-6.5h}$, where $k>0$<br>or $\frac{\mathrm{d}P}{\mathrm{d}h}=\frac{kP}{293-6.5h}$, where $k<0$ | |
| **(ii)** **[4]** | $\frac{1}{P}\frac{\mathrm{d}P}{\mathrm{d}h}=-\frac{k}{293-6.5h}$<br>$\int\frac{1}{P}\,\mathrm{d}P=\int-\frac{k}{293-6.5h}\,\mathrm{d}h$<br>Since $P>0$,<br>$\ln P=\frac{k}{6.5}\ln\lvert 293-6.5h\rvert+C$<br>$P=\mathrm{e}^{\frac{k}{6.5}\ln\lvert 293-6.5h\rvert+C}$<br>$=A\left[\mathrm{e}^{\ln\lvert 293-6.5h\rvert}\right]^{\frac{k}{6.5}}$, where $A=\mathrm{e}^{C}$<br>$=A\,\lvert 293-6.5h\rvert^{\frac{k}{6.5}}$<br>$=A\,\lvert 293-6.5h\rvert^{b}$<br>Since $k>0$, $b=\frac{k}{6.5}>0$. | |
| **(iii)** **[4]** | When $h=0$, $P=101300$, | |

$101300 = A(293^b)$ … (1)

When $h = 2$, $P = 80000$,

$80000 = A(293 - 6.5 \times 2)^b$

$80000 = A(280^b)$ … (2)

(1)/(2),

$$\frac{101300}{80000} = \left(\frac{293}{280}\right)^b$$

$$b = \frac{\ln \frac{101300}{80000}}{\ln \frac{293}{280}}$$

$$= 5.2015 \text{ (5 s.f.)}$$

$$A = \frac{101300}{293^{5.2015}}$$

$$= 1.4935 \times 10^{-8}$$

$$P = 1.4935 \times 10^{-8} |293 - 6.5h|^{5.2015}$$

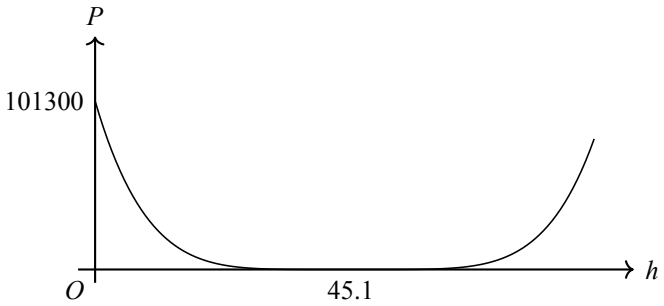
When $h = 8.848$,

$P = 1.4935 \times 10^{-8} |293 - 6.5(8.848)|^{5.2015}$

$= 32500$ (to 3 s.f.)

The atmospheric pressure is 32 500 Pascals.

**(iv)**
**[3]**




**Either**

The model suggests that when altitude is greater than 45.1 km, atompheric pressure increases (to infinity) as altitude increases, which is not possible.

**Or**

As the temperature $T$ measured in Kelvin can only be positive, the model is invalid if $T = 293 - 6.5h < 0$.

# 14 Complex Numbers

**14.1** **(a)** The complex numbers $z$ and $w$ satisfy the following equations.

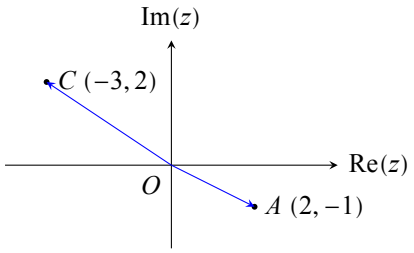
$$2z+|w|=2-4\mathrm{i} \qquad \mathrm{i}z-w=2\mathrm{i}$$

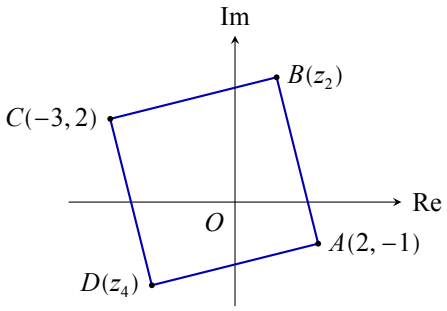
Find $z$ and $w$, giving your answers in the form $a+\mathrm{i}b$, where $a$ and $b$ are real numbers. [5]

**(b)** It is given that $z_1=2-\mathrm{i}$ and $z_3=-3+2\mathrm{i}$. On an Argand diagram, mark the points $A$ and $C$ representing $z_1$ and $z_3$ respectively. [1]

The points $B$ and $D$ on the Argand diagram represent complex numbers $z_2$ and $z_4$ respectively. Given that $ABCD$ is a square, labelled in an anti-clockwise direction, find $z_2$ and $z_4$. [4]

**[TJC 2025 Prelim P1 Q8]**

| Solution | | Comments |
|---|---|---|
| **(a)**<br>**[5]** | $2z+\|w\|=2-4\text{i}$ — (1)<br>$\text{i}z-w=2\text{i}$ — (2)<br>From (1):<br>$z=\dfrac{2-4\text{i}-\|w\|}{2}$<br>Substitute into (2):<br>$\text{i}\left(\dfrac{2-4\text{i}-\|w\|}{2}\right)-w=2\text{i}$<br>$2\text{i}+4-\text{i}\|w\|-2w=4\text{i}$<br>Let $w=a+\text{i}b,\ a,b\in\mathbb{R}$,<br>$2\text{i}+4-\text{i}\sqrt{a^2+b^2}-2(a+\text{i}b)=4\text{i}$<br>$4-2a+\text{i}\left(2-\sqrt{a^2+b^2}-2b\right)=4\text{i}$<br>Compare real and imaginary parts:<br>$4-2a=0\ \Rightarrow\ a=2$ — (3)<br>$2-\sqrt{a^2+b^2}-2b=4$ — (4)<br>Substitute (3) into (4):<br>$-2-2b=\sqrt{4+b^2}$ — (5)<br>$\Rightarrow(-2-2b)^2=4+b^2$<br>$4+8b+4b^2=4+b^2$<br>$b(3b+8)=0$<br>$\Rightarrow b=0$ (rejected since $\sqrt{4+b^2}\neq-2$ from (5)) or $b=-\dfrac{8}{3}$<br>$\therefore w=2-\dfrac{8}{3}\text{i}$<br>$z=\dfrac{2-4\text{i}-(-2-2b)}{2}=2-2\text{i}+\left(-\dfrac{8}{3}\right)=-\dfrac{2}{3}-2\text{i}$ | |
| **(b)**<br>**[4]** |  <br><br>$\overrightarrow{BA}$ represents the complex number $2-\text{i}-z_2$ and $\overrightarrow{BC}$ represents the complex number $-3+2\text{i}-z_2$.<br>Since $\angle ABC=\dfrac{\pi}{2}$, $A$ is obtained by rotating $C$ through $90^\circ$ in anti-clockwise direction about $B$. i.e. $\overrightarrow{BC}$ rotated by $90^\circ$ anticlockwise gives $\overrightarrow{BA}$. | |

| | | |
|---|---|---|
| | $\overrightarrow{BC}\times \mathrm{i}=\overrightarrow{BA}$<br>  <br><br>$\mathrm{i}(-3+2\mathrm{i}-z_2)=2-\mathrm{i}-z_2$<br>$-3\mathrm{i}-2-\mathrm{i}z_2=2-\mathrm{i}-z_2$<br>$(1-\mathrm{i})z_2=4+2\mathrm{i}$<br>$z_2=\dfrac{4+2\mathrm{i}}{1-\mathrm{i}}\cdot\dfrac{1+\mathrm{i}}{1+\mathrm{i}}=\dfrac{2+6\mathrm{i}}{2}=1+3\mathrm{i}$<br>$\overrightarrow{AB}=\overrightarrow{DC}$<br>$1+3\mathrm{i}-(2-\mathrm{i})=-3+2\mathrm{i}-z_4$<br>$z_4=-3+2\mathrm{i}+1-4\mathrm{i}=-2-2\mathrm{i}$ | |

**14.2 Do not use a calculator in answering this question.**

It is given that $1+2\text{i}$ is a complex root of the equation $z^3+pz^2+qz-10=0$, where $p$ and $q$ are real numbers.

**(a)** State another complex root of the equation in the form $a+\text{i}b$, where $a$ and $b$ are real numbers and $b\neq 0$. Justify your answer. [2]

**(b)** Find the values of $p$ and $q$, and find the third root. [6]

**(c)** Hence find the roots of the equation $1+pw+qw^2-10w^3=0$. [2]

**[2025 A-level P1 Q12]**

| Solution | | Comments |
|---|---|---|
| **(a)** **[2]** | All the coefficients of the equation are real numbers and thus by Conjugate Root Theorem, $1-2\text{i}$ is a root of the equation. | |
| **(b)** **[6]** | $(1+2\text{i})^2=1+4\text{i}-4=-3+4\text{i}$<br>$(1+2\text{i})^3=(1+2\text{i})(-3+4\text{i})=-11-2\text{i}$<br>$-11-2\text{i}+p(-3+4\text{i})+q(1+2\text{i})-10=0$<br>$-11-2\text{i}-3p+4p\text{i}+q+2q\text{i}-10=0$<br>$(-11-3p+q-10)+(-2+4p+2q)\,\text{i}=0$<br>Comparing real parts, Comparing imaginary parts,<br>$-3p+q-21=0,\quad 4p+2q-2=0$<br>$-3p+q=21$ — (1), $\quad 2p+q=1$ — (2)<br>(1) − (2)<br>$-5p=20\ \Rightarrow\ p=-4$<br>Substitute $p=-4$ into (2)<br>$q=1-2(-4)=9$<br>Attempt to long divide or compare coefficients<br>The third root is $z=2$. | |
| **(c)** **[2]** | Let $z=\dfrac{1}{w}\ \Rightarrow\ w=\dfrac{1}{z}$.<br>$w=\dfrac{1}{2}$ or $w=\dfrac{1}{1+2\text{i}}$ or $w=\dfrac{1}{1-2\text{i}}$<br>$w=\dfrac{1}{2}$ or $w=\dfrac{1\pm 2\text{i}}{5}$ | |

**14.3** **(a)** It is given that the roots to the equation $-\mathrm{i}x^3+5\mathrm{i}x^2+ax+b=0$, where $a$ and $b$ are purely imaginary, are $2+\mathrm{i}$, $2-\mathrm{i}$ and 1. Explain why the complex roots occur in conjugate pairs. [2]

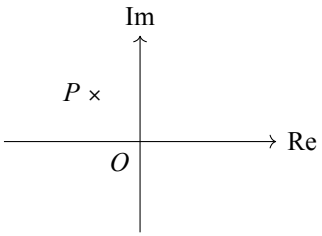
**(b)** By using (a) and an appropriate substitution, find the roots of the equation $\mathrm{i}x^3+5\mathrm{i}x^2+a^*x+b=0$, where the complex conjugate of $a$ is denoted by $a^*$. [4]

**[NYJC 2025 Prelim P2 Q2]**

| **Solution** | | **Comments** |
|---|---|---|
| **(a)** **[2]** | $-\mathrm{i}x^3+5\mathrm{i}x^2+ax+b=0$<br>Replacing $a=\lambda\mathrm{i}$ and $b=\mu\mathrm{i}$ where $\lambda, \mu \in \mathbb{R}$:<br>$-\mathrm{i}x^3+5\mathrm{i}x^2+\lambda\mathrm{i}x+\mu\mathrm{i}=0$<br>$x^3-5x^2-\lambda x-\mu=0$<br>Since the coefficients of the equation are all real, the complex roots occur in conjugate pair. | |
| **(b)** **[4]** | $\mathrm{i}x^3+5\mathrm{i}x^2+a^*x+b=0$<br>Since $a$ is purely imaginary, $a^*=-a$:<br>$-\mathrm{i}(-x)^3+5\mathrm{i}(-x)^2+a(-x)+b=0$<br>$-x=2-\mathrm{i},\ 2+\mathrm{i}$ and $1$<br>$\therefore x=-2+\mathrm{i},\ -2-\mathrm{i}$ and $-1$ | |

**14.4 Do not use a calculator in answering this question.**

The complex number $z$ has modulus 1 and argument $\theta$, where $\frac{\pi}{2} < \theta < \pi$, and the complex number $w$ is given by $w = \mathrm{i}\sqrt{3}z$. The point $P$ on the Argand diagram represents $z$.




**(a)** On the copy of the Argand diagram with origin $O$ in the Printed Answer Booklet, plot the points $Q$ and $R$ to represent $w$ and $z - w$ respectively. Show clearly the geometrical relationship between the points $P$, $Q$ and $R$. [3]

**(b)** Find the area of the quadrilateral $ORPQ$. [1]

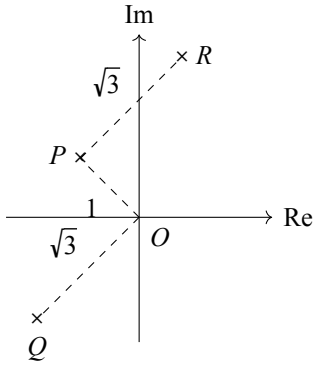
It is given that $\theta = \frac{3\pi}{4}$.

**(c)** Find $z$ in the form $x+y\mathrm{i}$, where $x$ and $y$ are real numbers. [2]

**(d)** Show that $z - w = k\left[\left(\sqrt{3}-1\right)+\left(\sqrt{3}+1\right)\mathrm{i}\right]$, where $k$ is a constant to be determined. [2]

**(e)** Hence show that $\tan\frac{5\pi}{12} = \frac{\sqrt{3}+1}{\sqrt{3}-1}$. [1]

**[VJC 2025 Prelim P2 Q4]**

| **Solution** | | **Comments** |
|---|---|---|
| **(a)**<br>**[3]** |  | |
| **(b)**<br>**[1]** | Area $ORPQ = \frac{1}{2}(1)\left(\sqrt{3}+\sqrt{3}\right) = \sqrt{3}$ | |

**(c)** **[2]**

$$\text{Length } x = 1\cos\frac{\pi}{4} = \frac{1}{\sqrt{2}}$$

$$\text{Length } y = 1\sin\frac{\pi}{4} = \frac{1}{\sqrt{2}}$$

$$\therefore z = -\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}\text{i}$$

**(d)** **[2]**

$$z - w = \left(z - \sqrt{3}\text{i}z\right)$$

$$= \left(-\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}\text{i}\right)\left(1 - \sqrt{3}\text{i}\right)$$

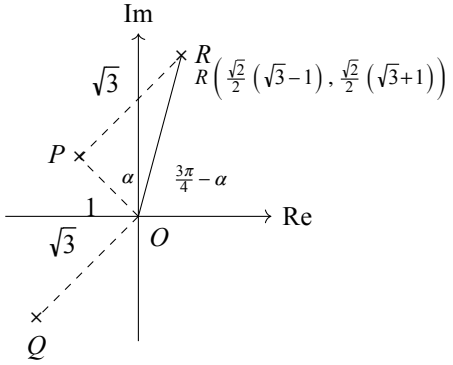
$$= \frac{1}{\sqrt{2}}(-1+\text{i})\left(1 - \sqrt{3}\text{i}\right)$$

$$= \frac{1}{\sqrt{2}}\left(-1 + \sqrt{3} + \text{i} + \sqrt{3}\text{i}\right)$$

$$= \frac{1}{\sqrt{2}}\left(\left(\sqrt{3}-1\right) + \left(\sqrt{3}+1\right)\text{i}\right)$$

$$k = \frac{1}{\sqrt{2}}$$

**(e)** **[1]**




$$\alpha = \tan^{-1}\frac{\sqrt{3}}{1} = \frac{\pi}{3}$$

$$\tan\left(\frac{3\pi}{4} - \frac{\pi}{3}\right) = \frac{\frac{\sqrt{2}}{2}\left(\sqrt{3}+1\right)}{\frac{\sqrt{2}}{2}\left(\sqrt{3}-1\right)}$$

$$\tan\frac{5\pi}{12} = \frac{\sqrt{3}+1}{\sqrt{3}-1}$$

**14.5 (i)** The complex numbers $p$ and $q$ satisfy the simultaneous equations

$$p^* + 10\mathrm{i} = q\mathrm{i} + 5 \qquad |p|^2 - q - 5 + 2\mathrm{i} = 0.$$

Given that $\mathrm{Im}(p) < 0$, find $p$ in the Cartesian form. [3]

Hence find the values of $n$ for which $p^{2n}$ is purely imaginary. [2]

**(ii)** The complex number $w$ is such that $\arg\left(\frac{w}{p} - p^*\right) = -\frac{\pi}{2}$ and $w + w^* = -2$.

Find $w$. [4]

**[AJC 2011 P1 Q13]**

| Solution | | Comments |
|---|---|---|
| **(i)**<br>**[2]** | $q=\lvert p\rvert^2-5+2\text{i}$<br>$p^*+10\text{i}=\text{i}\left(\lvert p\rvert^2-5+2\text{i}\right)+5$<br>$p^*=3+\left(\lvert p\rvert^2-15\right)\text{i}$<br>Let $p=a+b\text{i},\ a,b\in\mathbb{R}$,<br>$a-b\text{i}=3+(a^2+b^2-15)\,\text{i}$<br>By comparing real & imaginary parts,<br>$a=3,\quad -b=a^2+b^2-15$<br>$b^2+b-6=0$<br>$(b+3)(b-2)=0$<br>$b=-3\quad(\because \text{Im}(p)<0)$<br>$\therefore p=3-3\text{i}$<br>$p^2=9-18\text{i}-9=-18\text{i}$<br>$p^{2n}=(-18\text{i})^n=(-18)^n\,\text{i}^n$<br>$\text{i}^n$ is purely imaginary when $n$ is odd.<br>$\therefore n=2k-1,\ k\in\mathbb{Z}^+$ | |
| **(ii)**<br>**[4]** | Let $w=a+b\text{i},\ a,b\in\mathbb{R}$,<br>$w+w^*=2a=-2$<br>$a=-1$<br>$\dfrac{w}{p}-p^*=\dfrac{-1+b\text{i}}{3-3\text{i}}-(3+3\text{i})$<br>$=\dfrac{(-1+b\text{i})(3+3\text{i})}{18}-(3+3\text{i})$<br>$=\left(\dfrac{-3-3b}{18}-3\right)+\left(\dfrac{3b-3}{18}-3\right)\text{i}$<br>$\arg=-\dfrac{\pi}{2}$: real part $=0$, imaginary part $<0$.<br>$\dfrac{-3-3b}{18}-3=0$<br>$b=-19$<br>$\dfrac{3(-19)-3}{18}-3=-\dfrac{19}{3}<0$<br>$\therefore w=-1-19\text{i}$ | |